

Robust and efficient methods for single-cell CRISPR screens: PerturbPlan and spaCRT

McGill Statistics Seminar
April 2026

Ziang Niu
University of Pennsylvania
<https://ziangniu6.github.io>

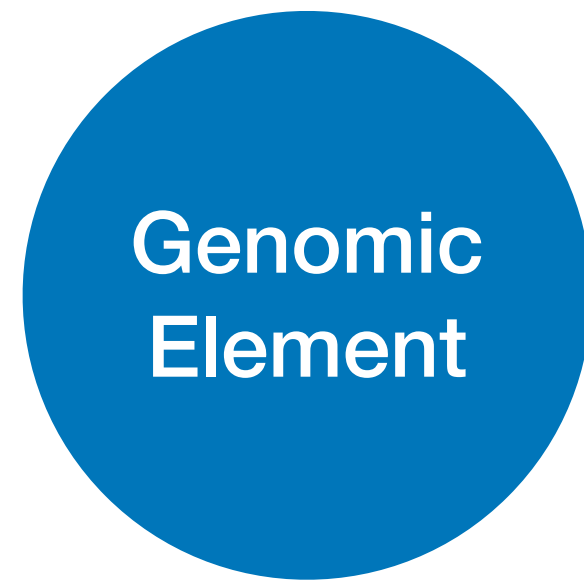
A model for genetic origins of human diseases

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Normal

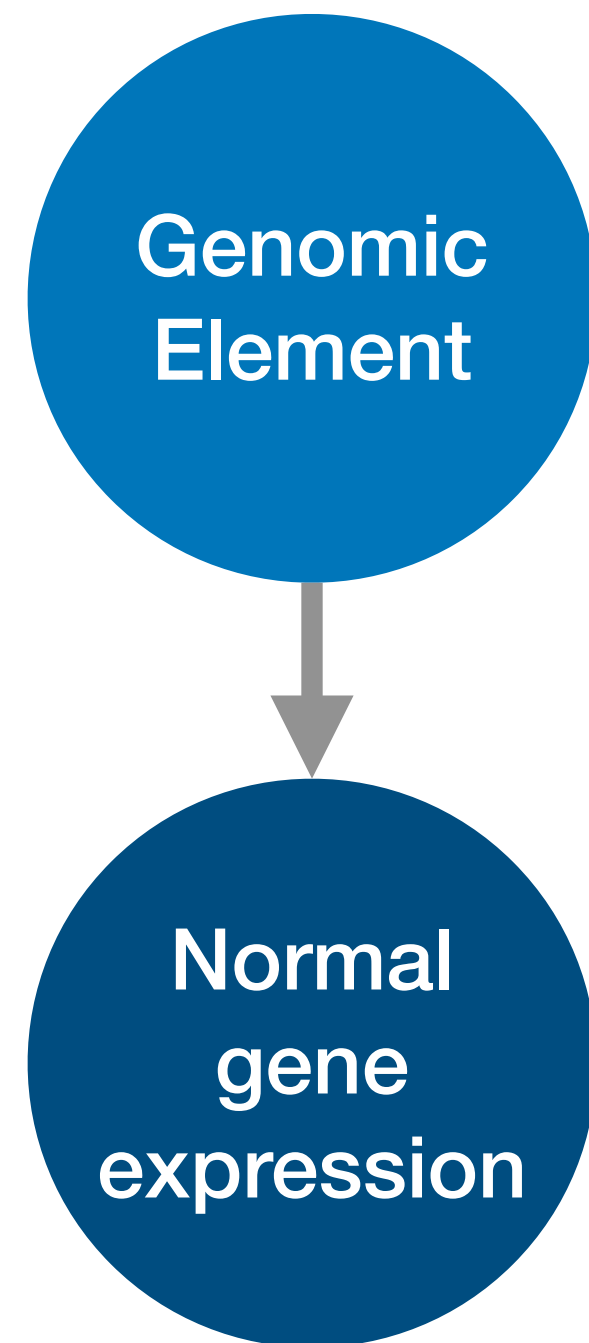
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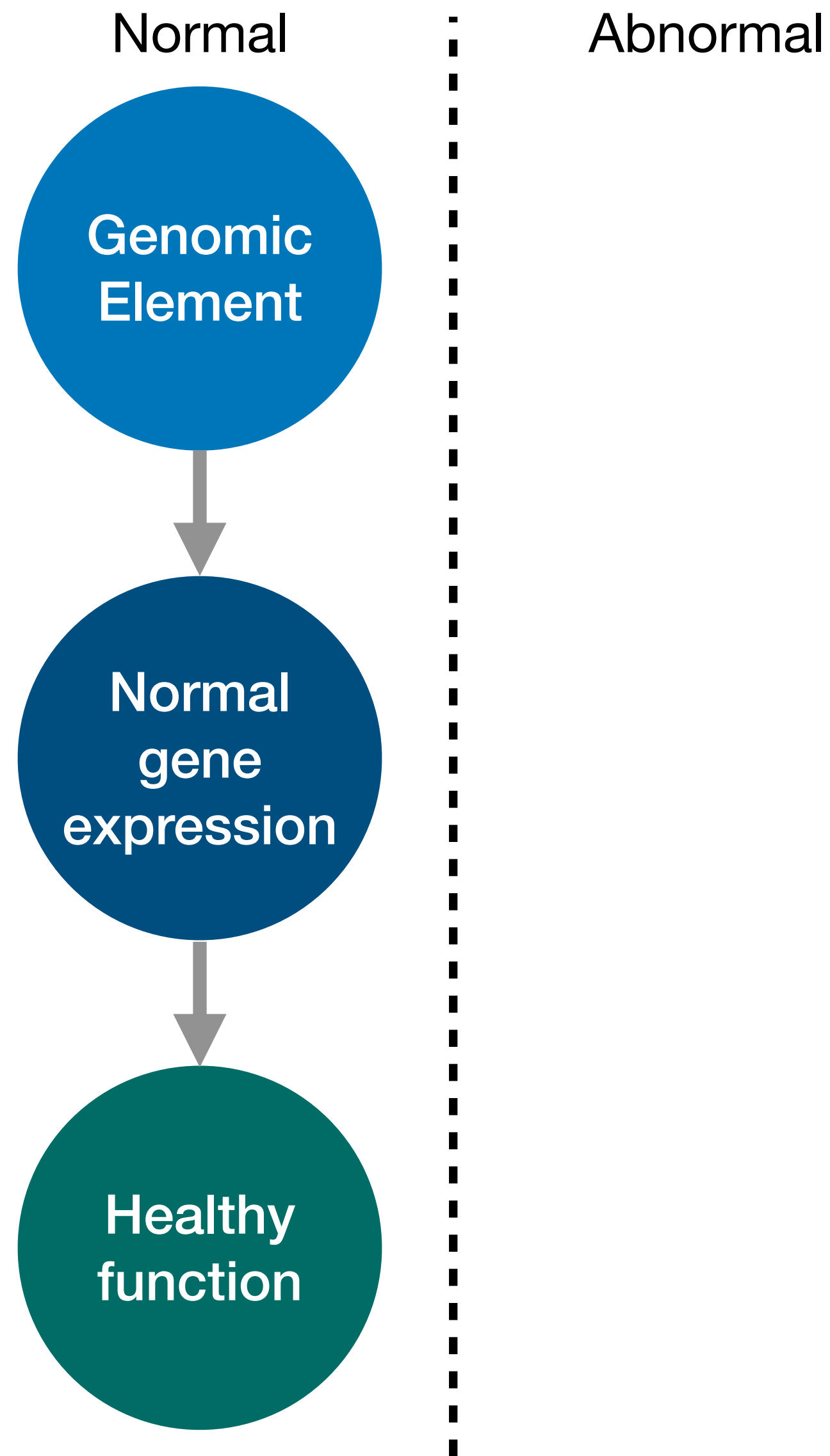


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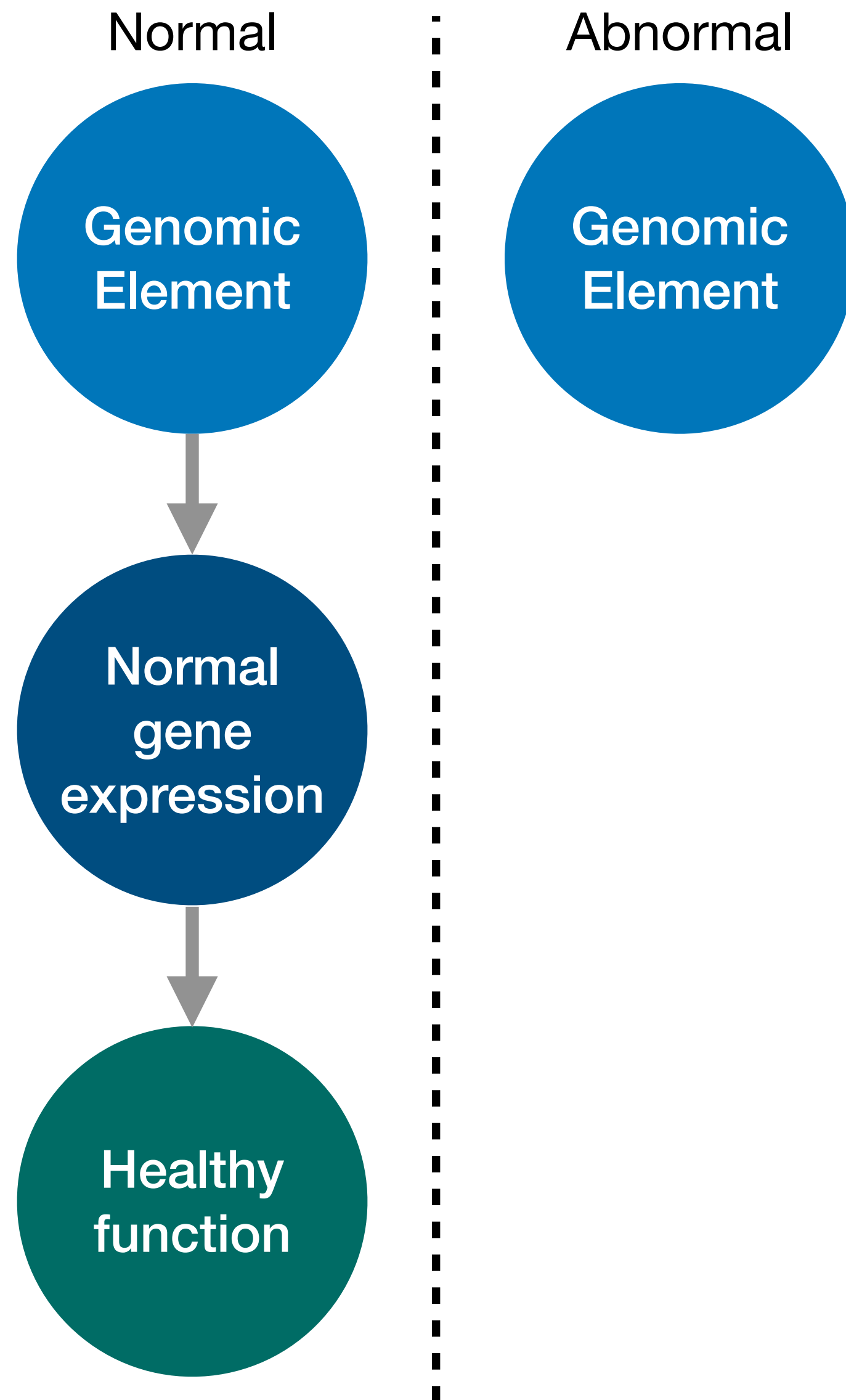
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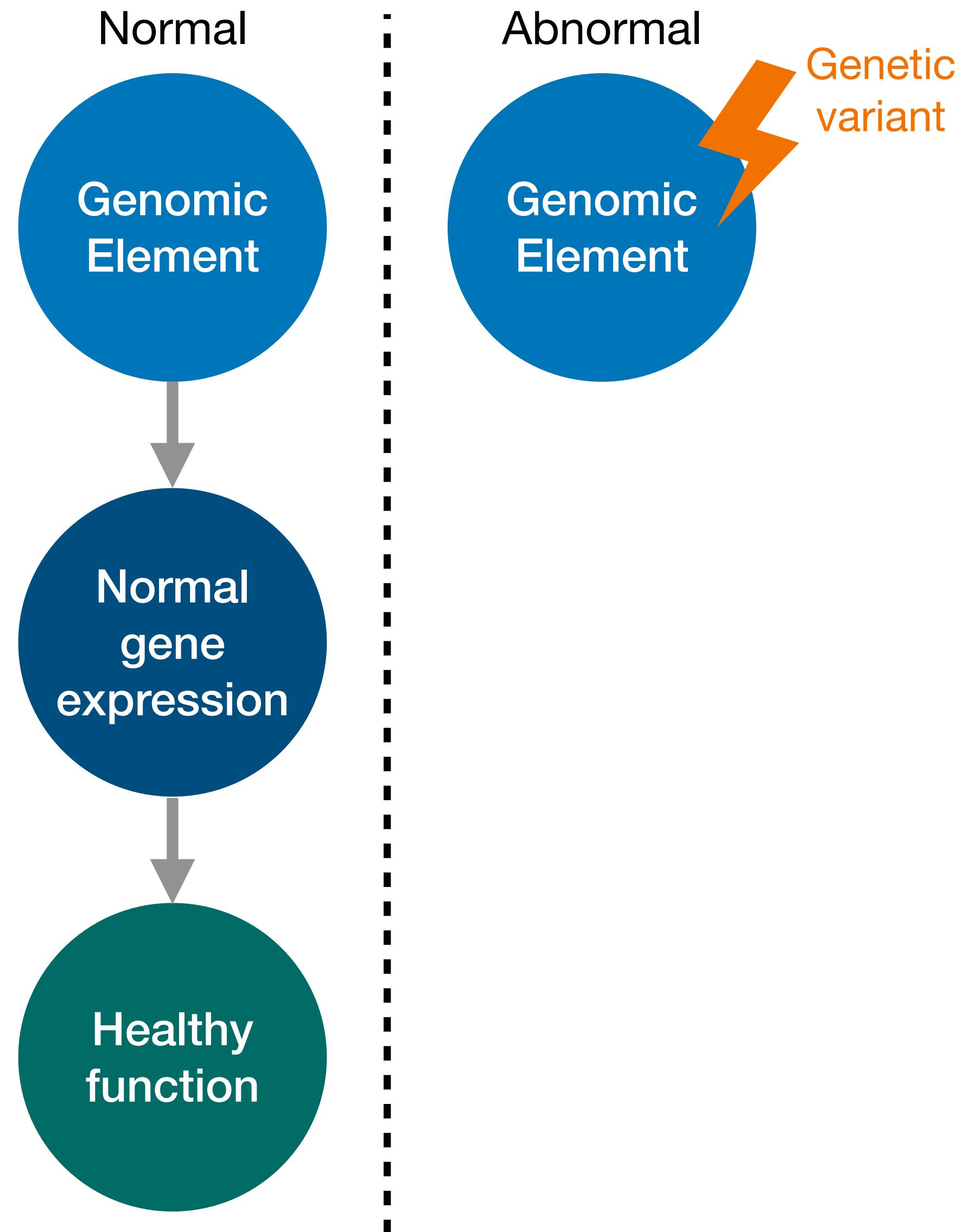
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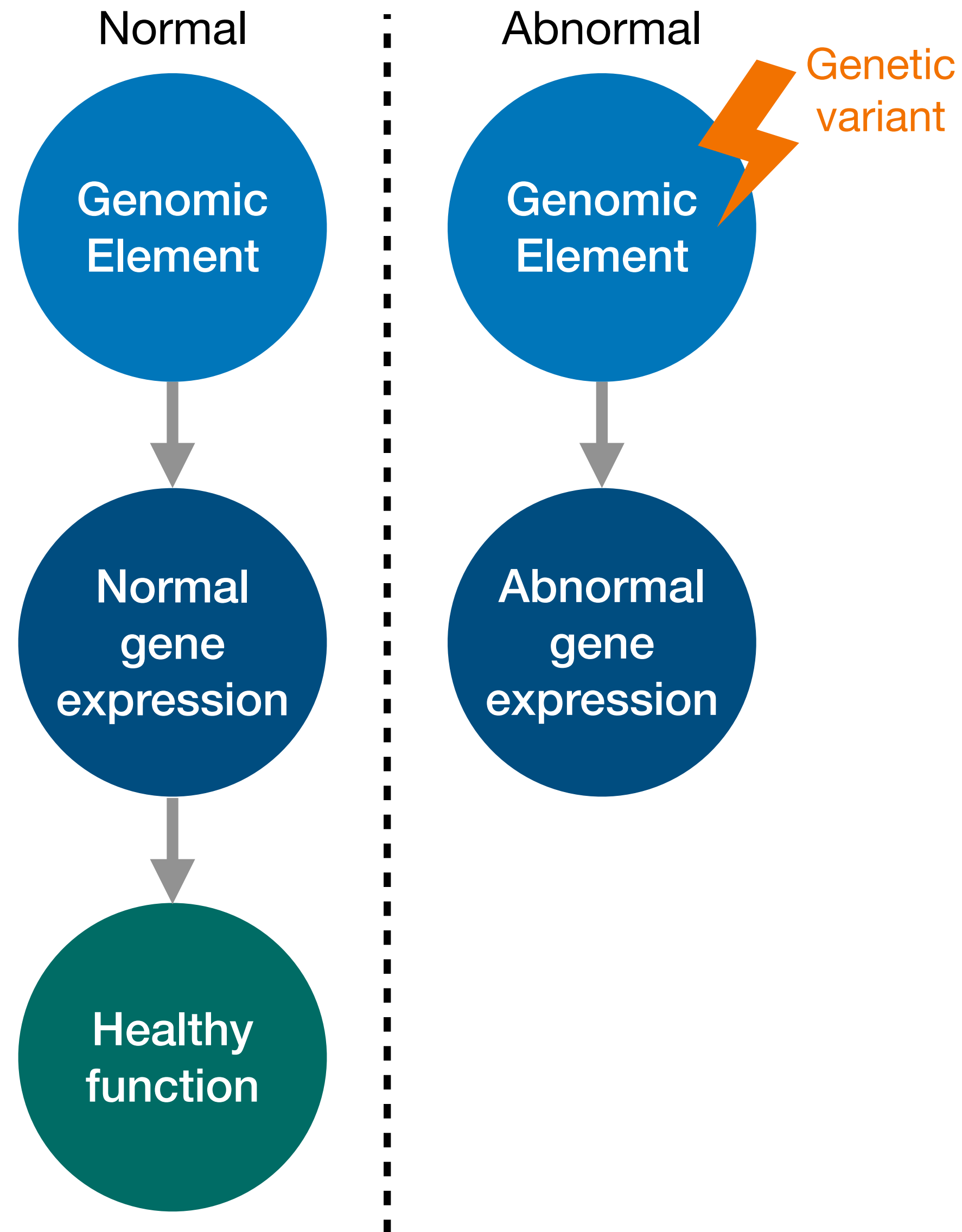
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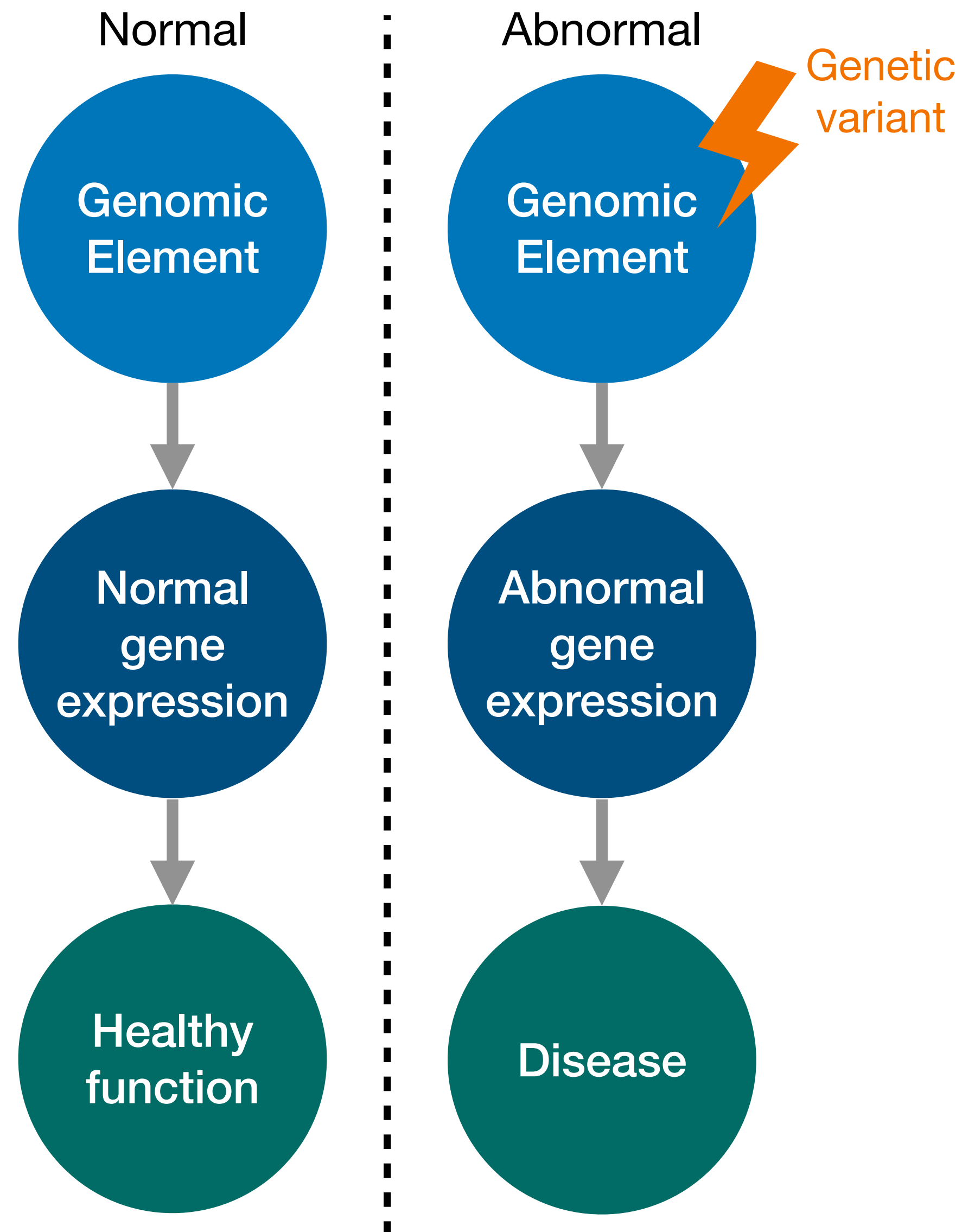
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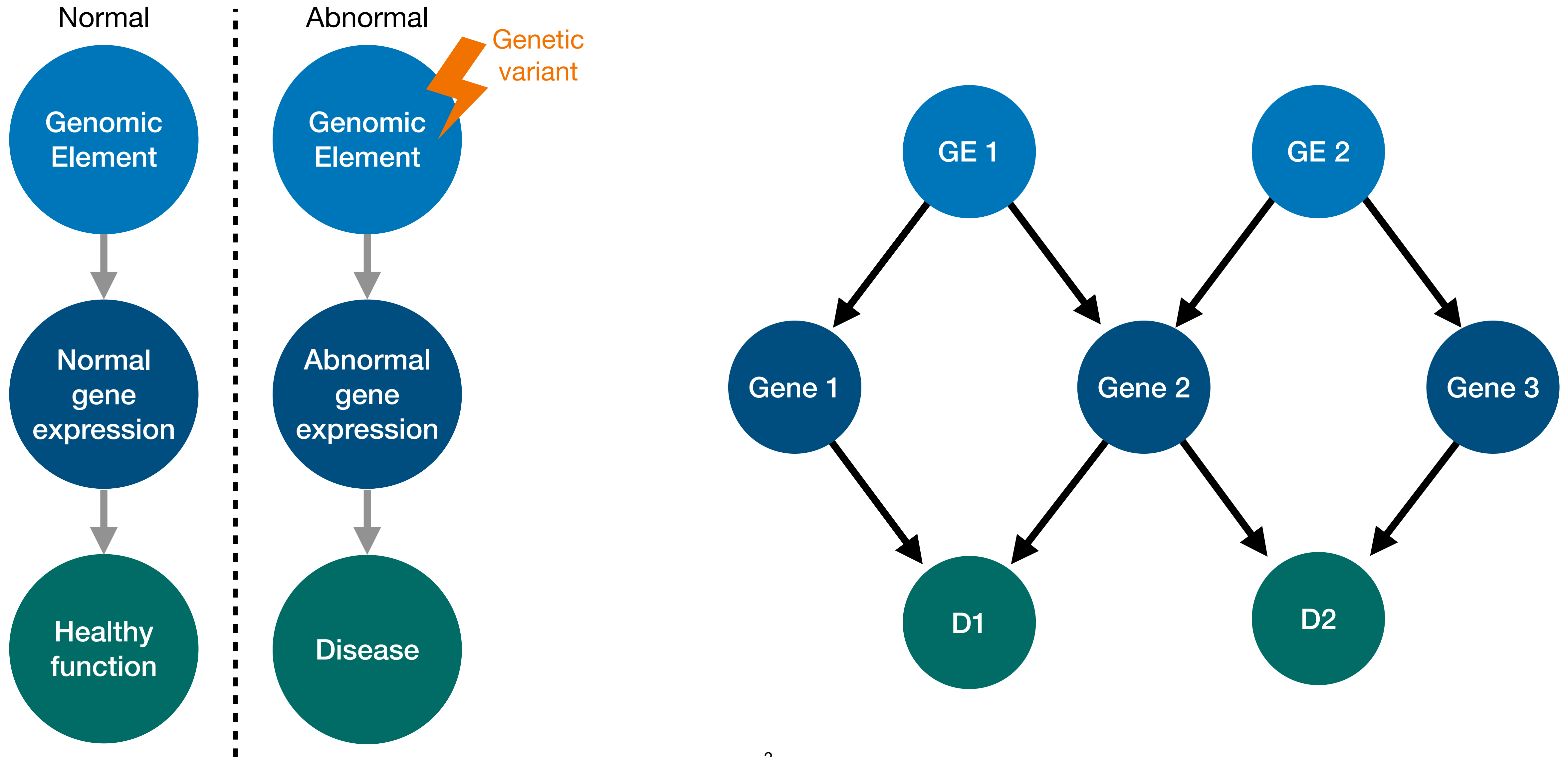
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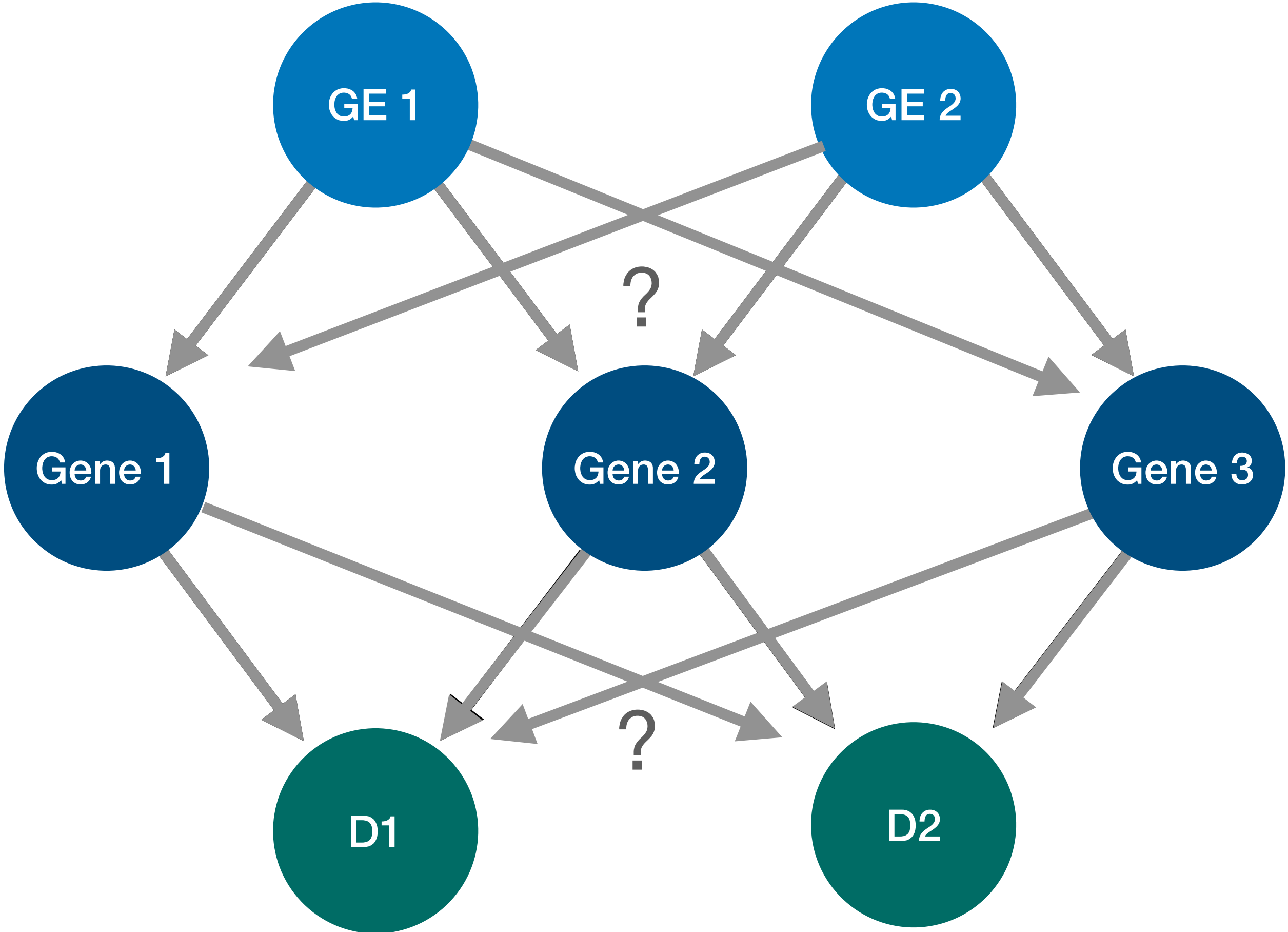
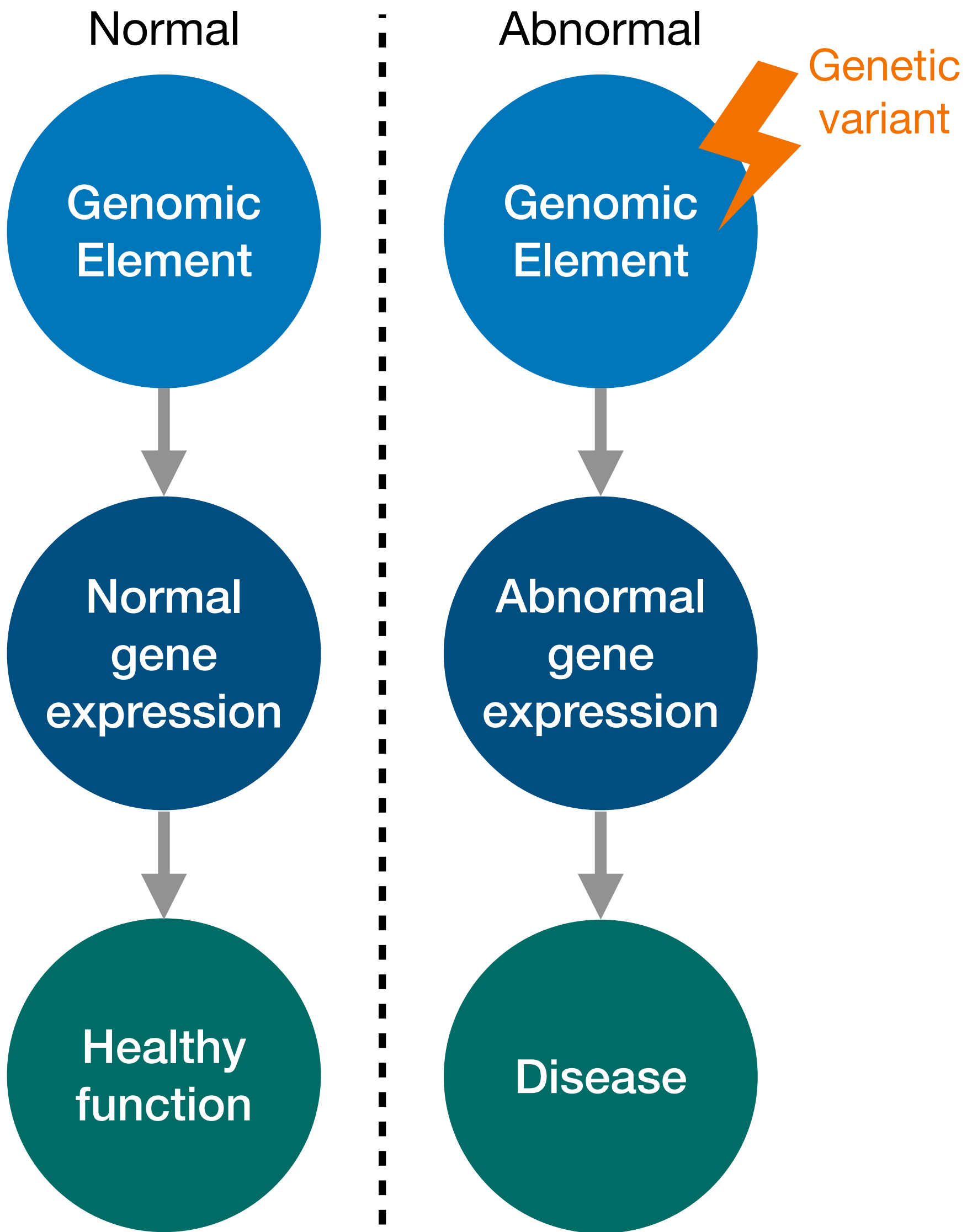
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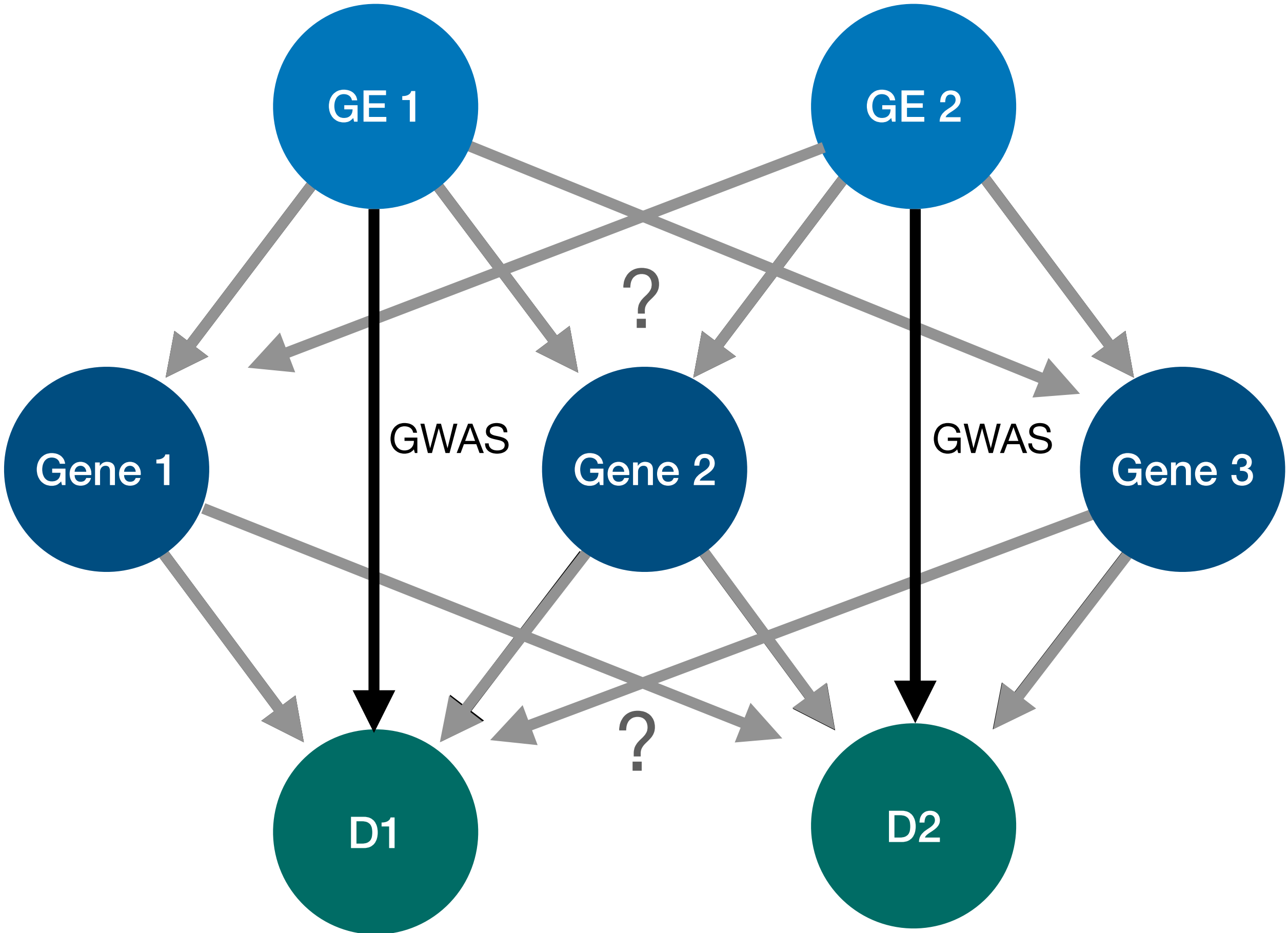
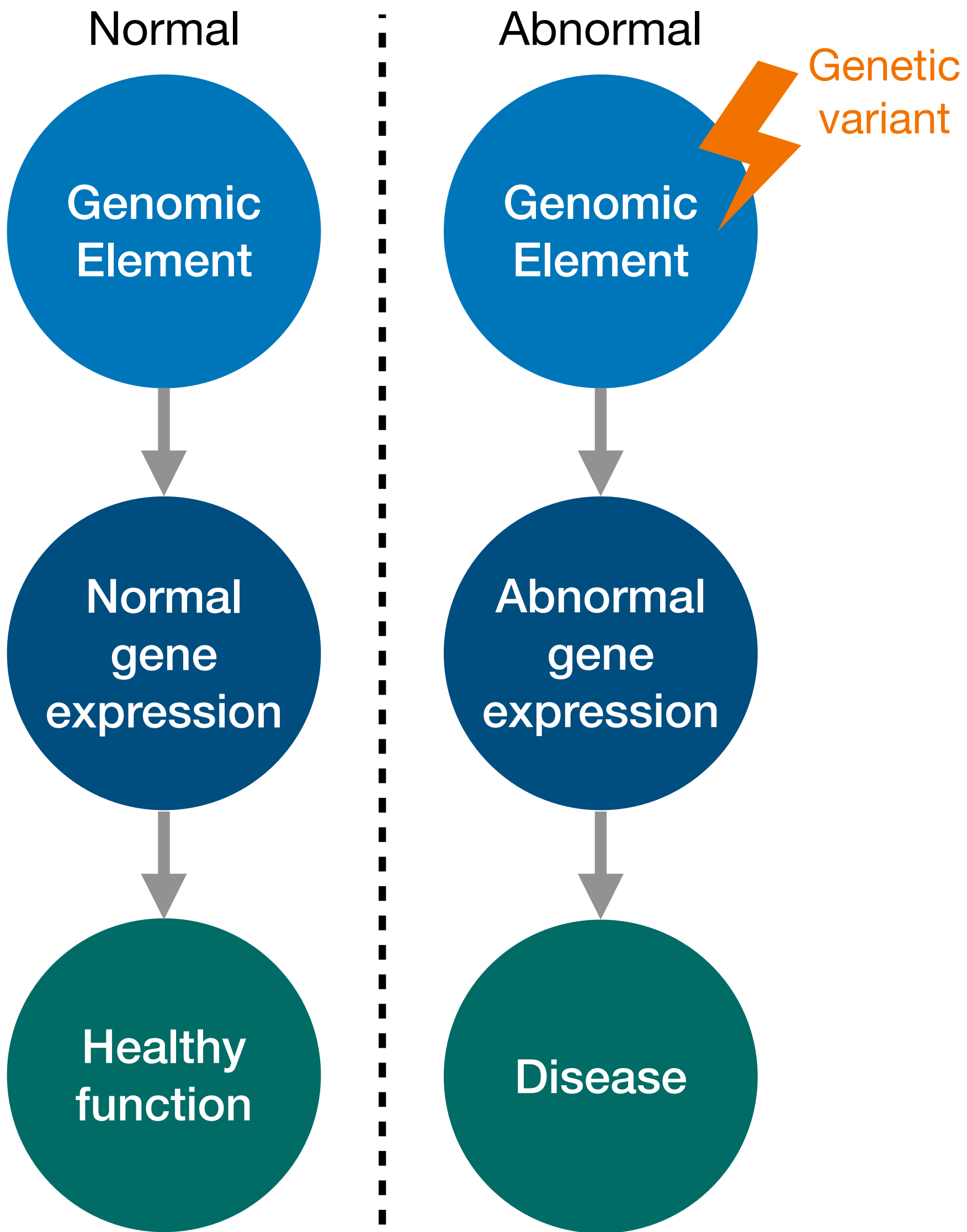
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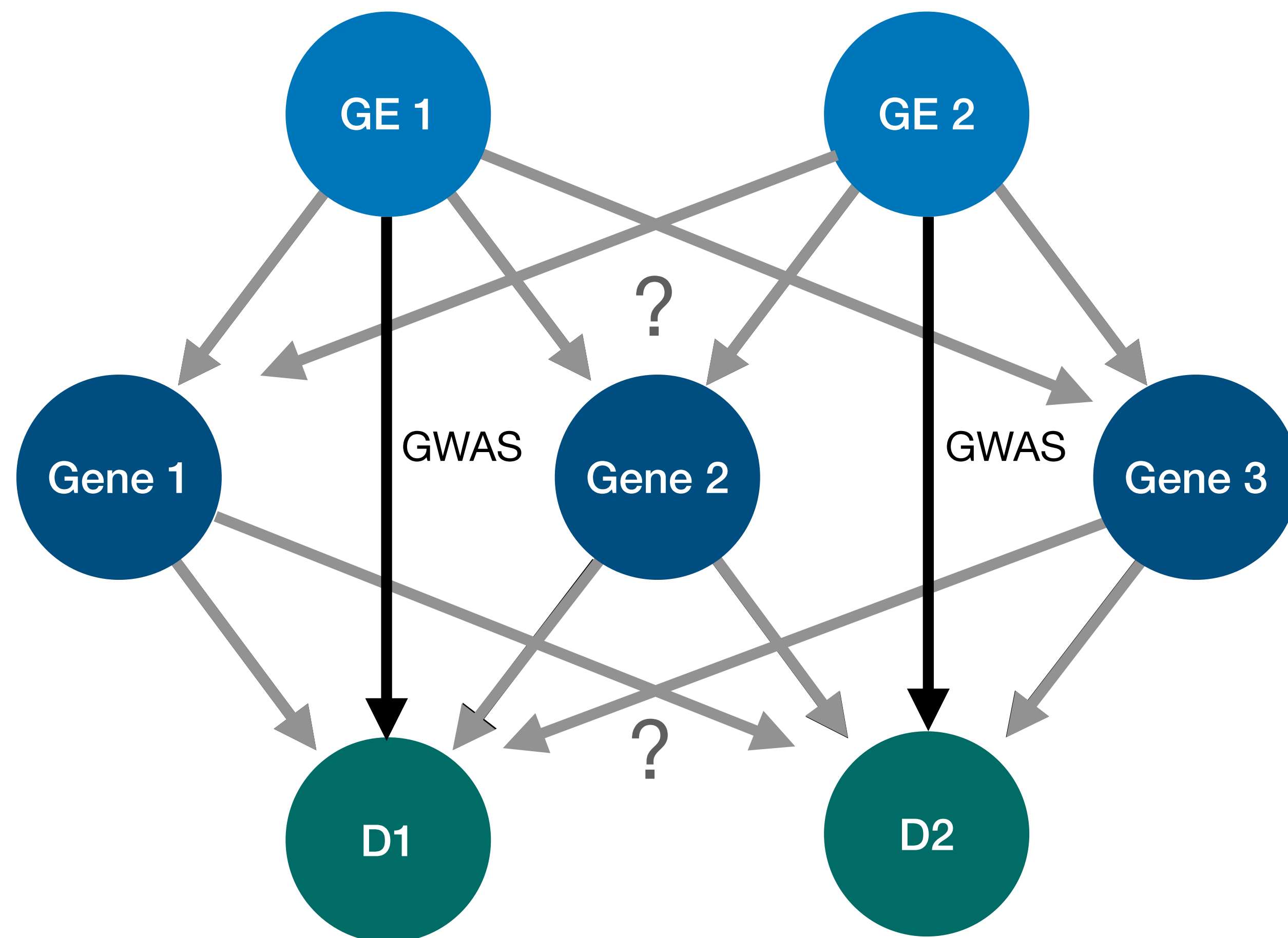
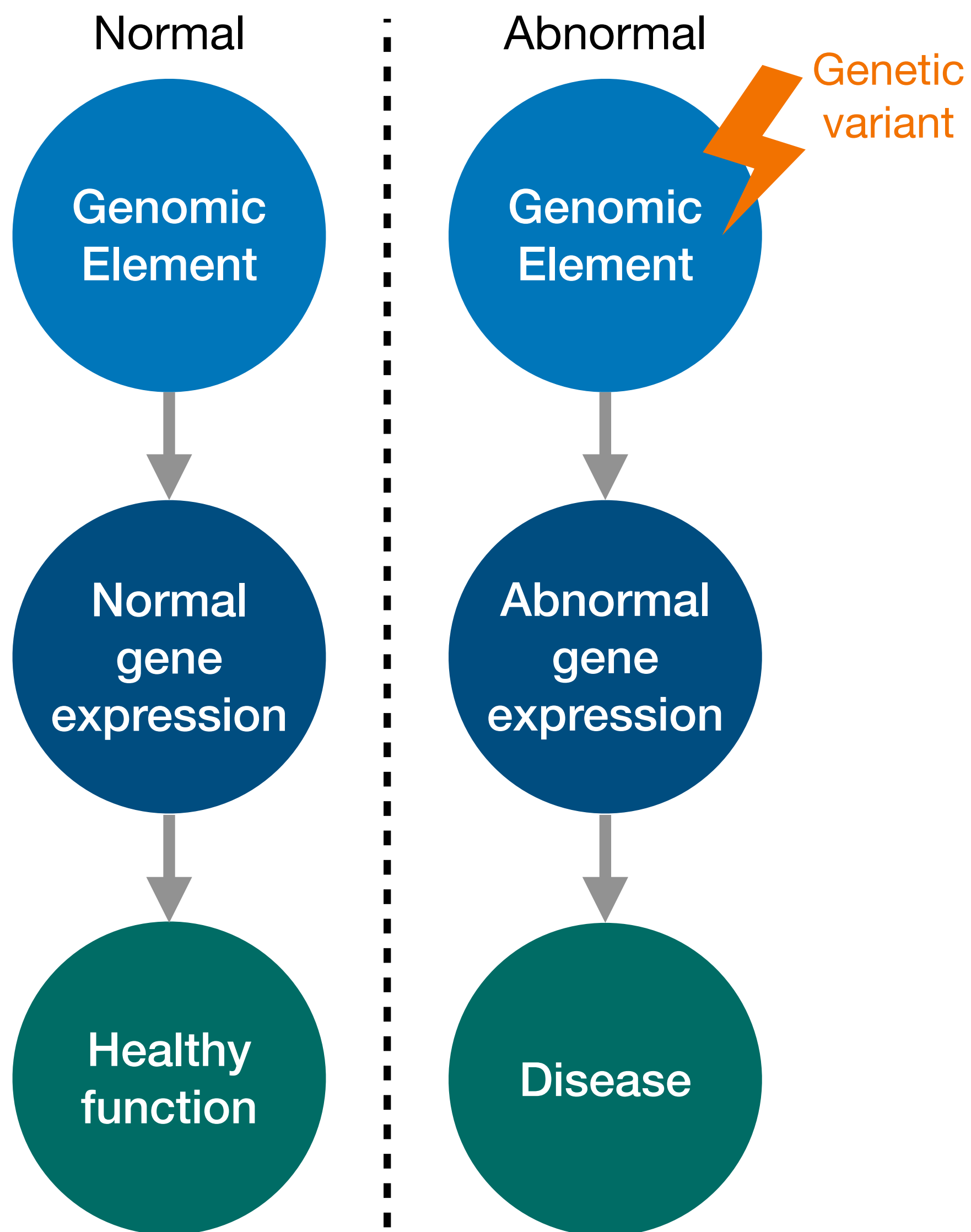
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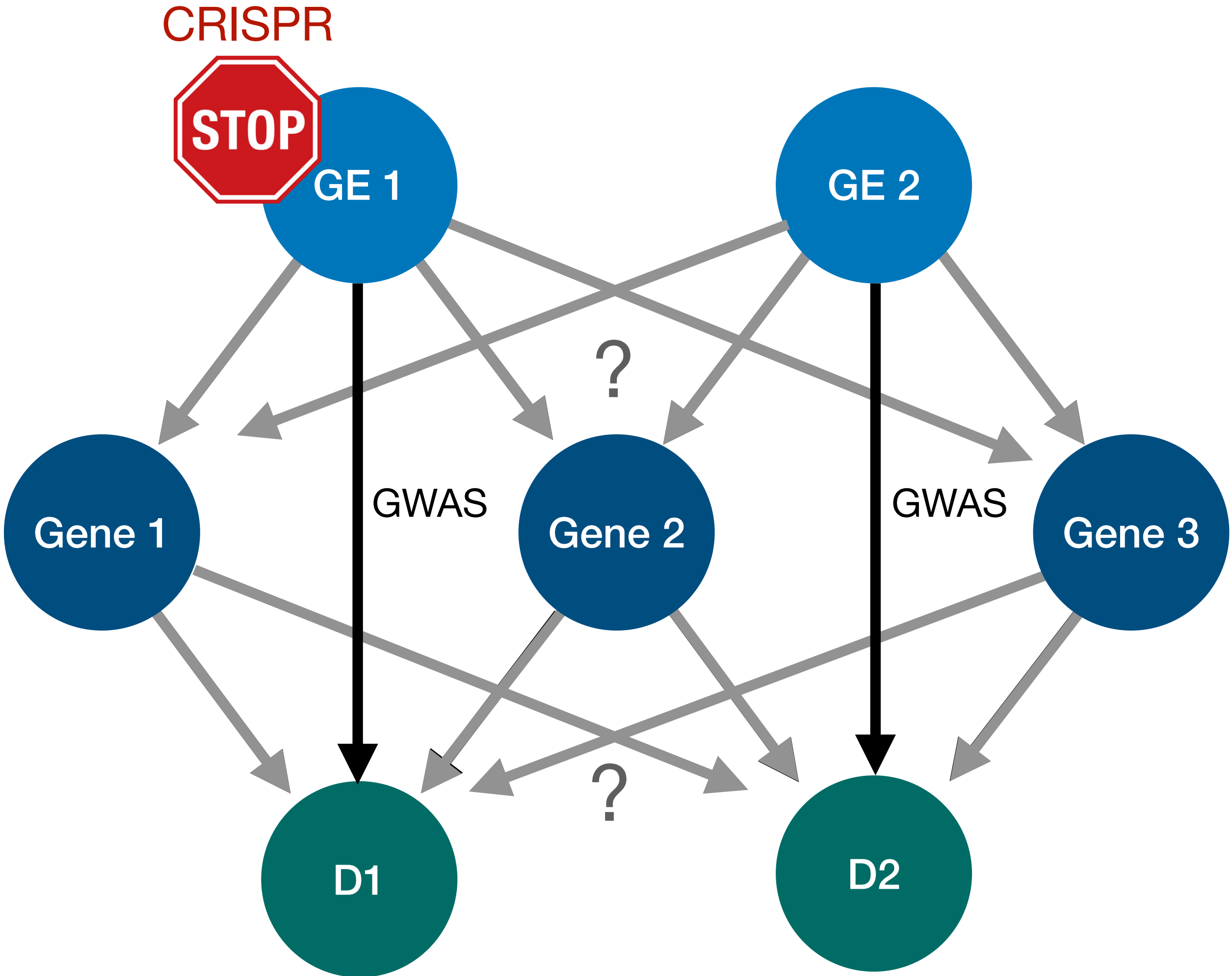
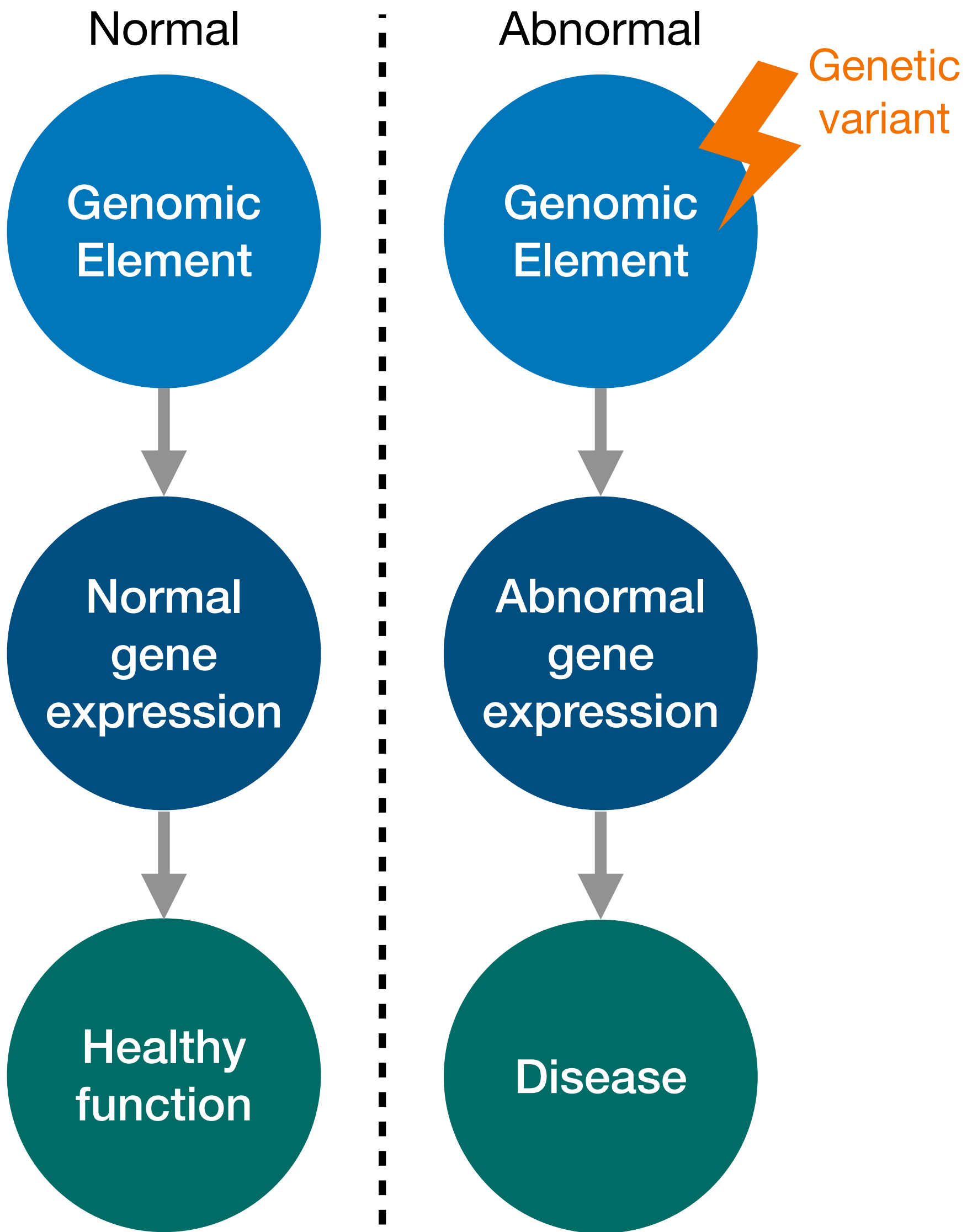


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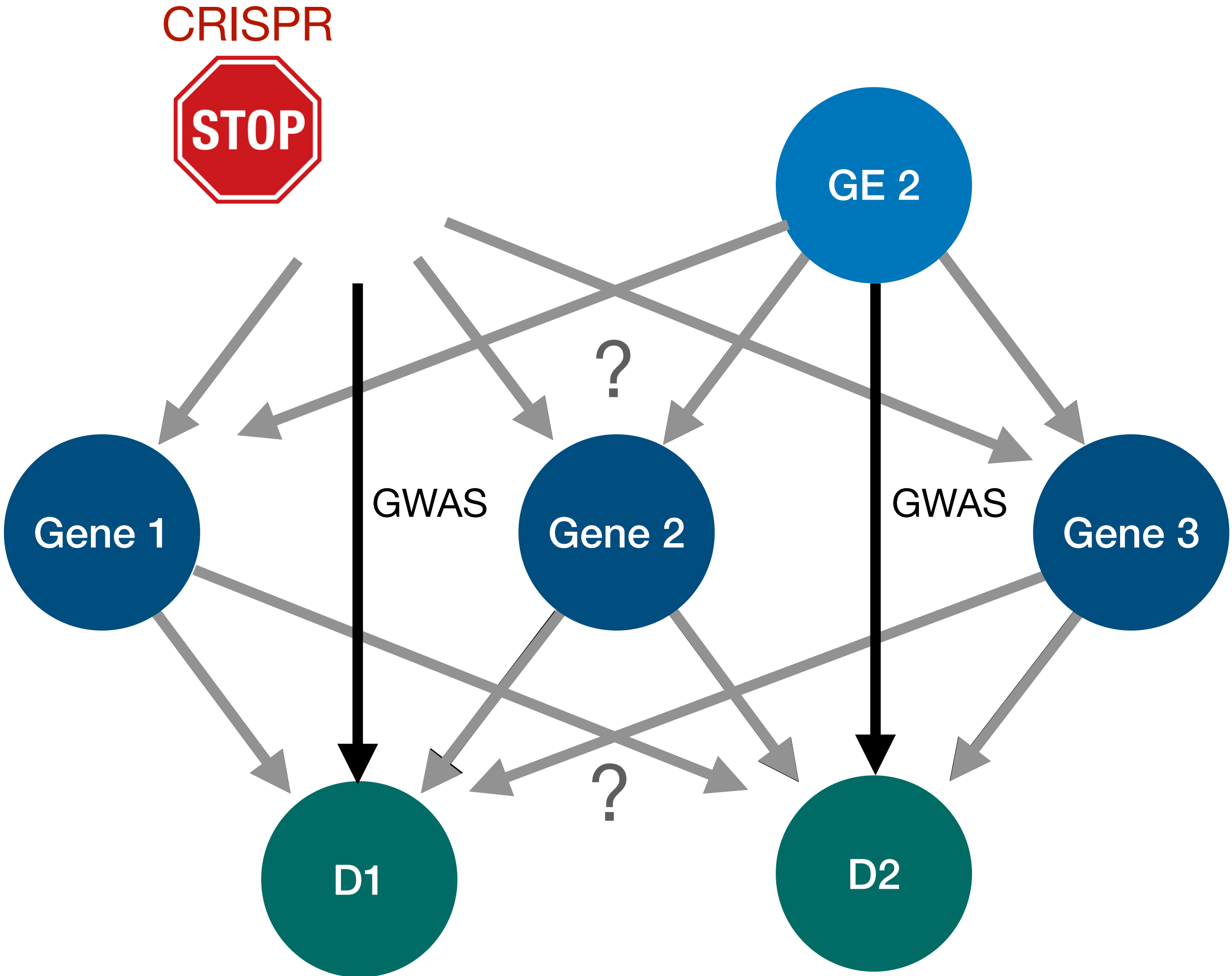
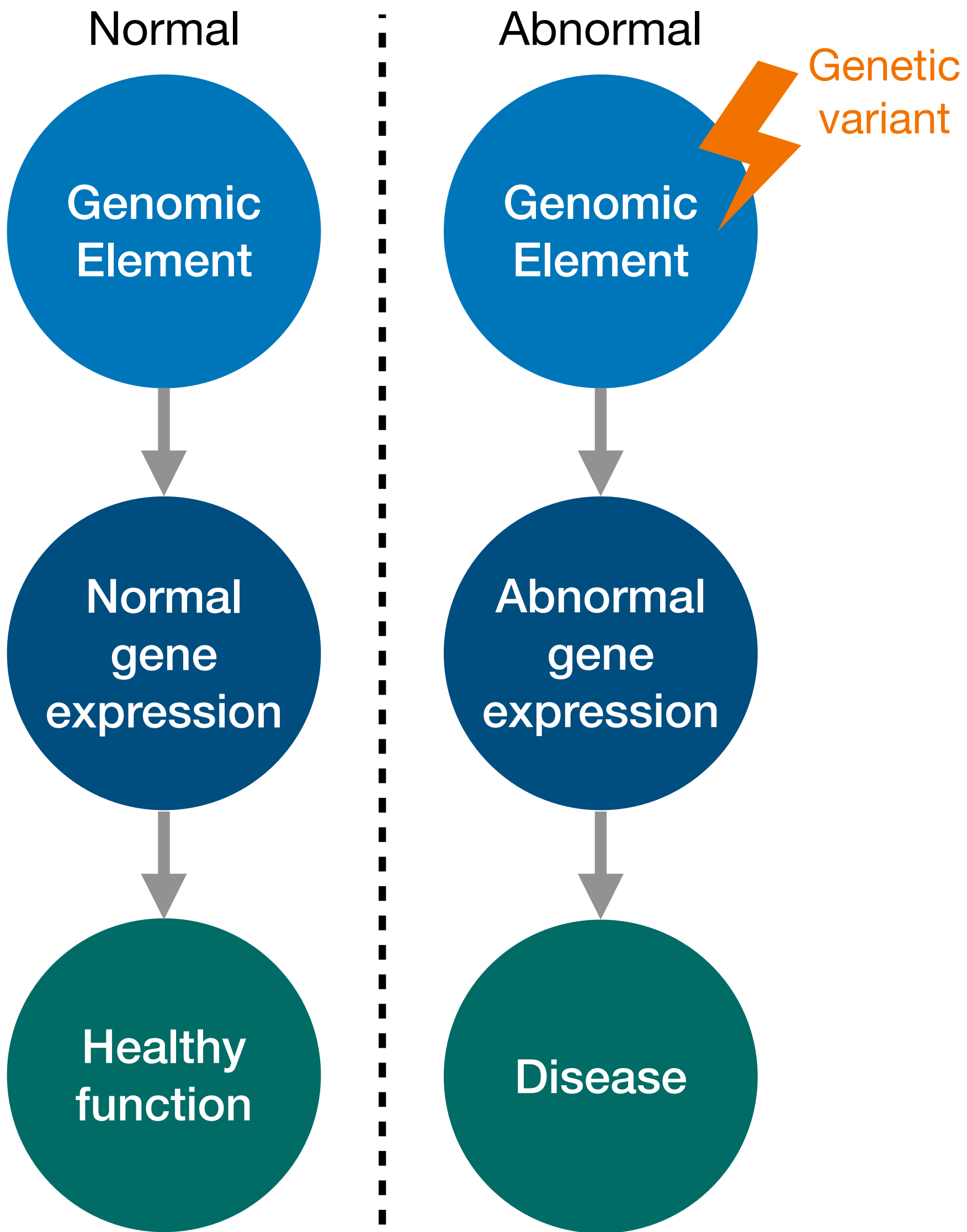
Single-cell CRISPR screens

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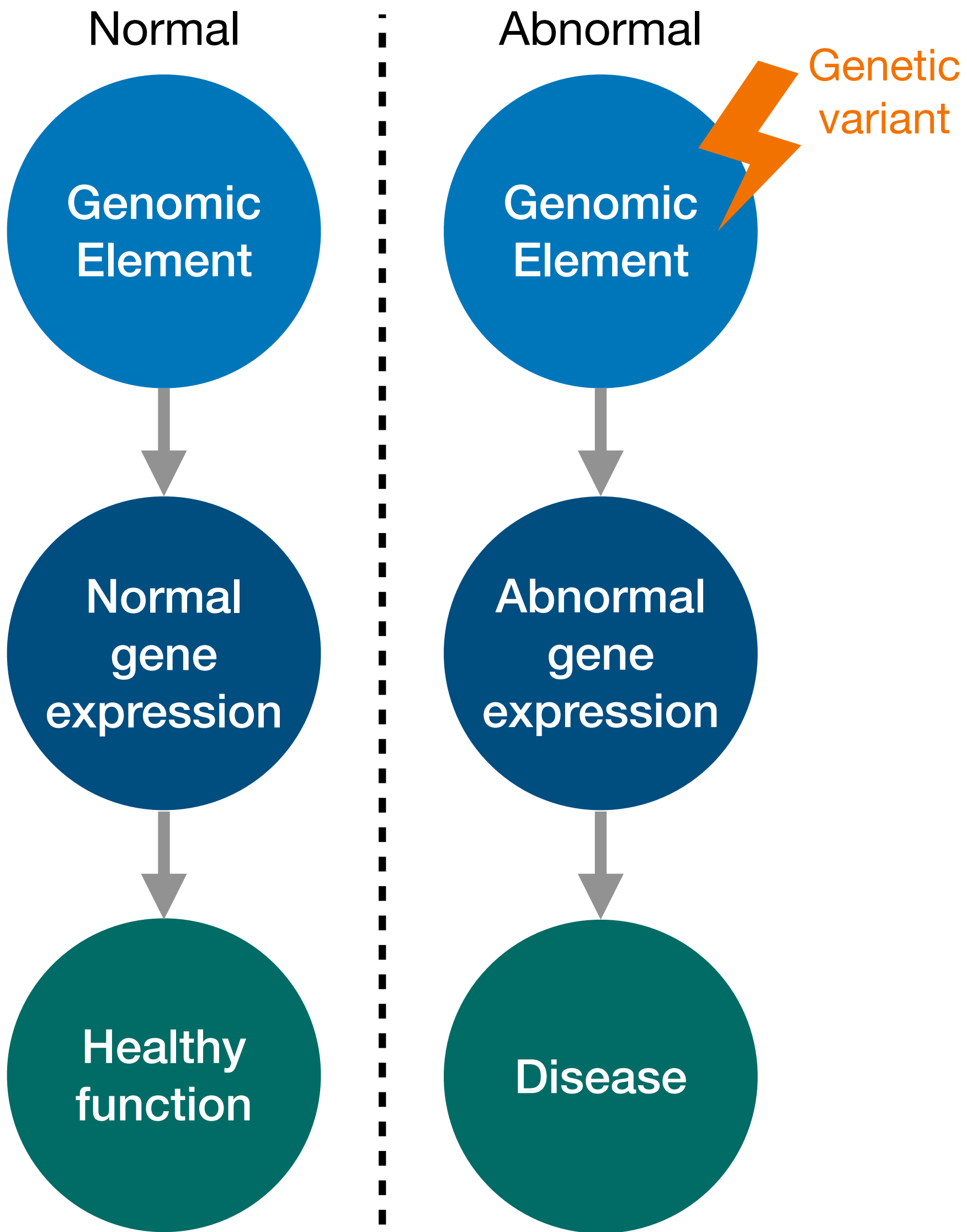
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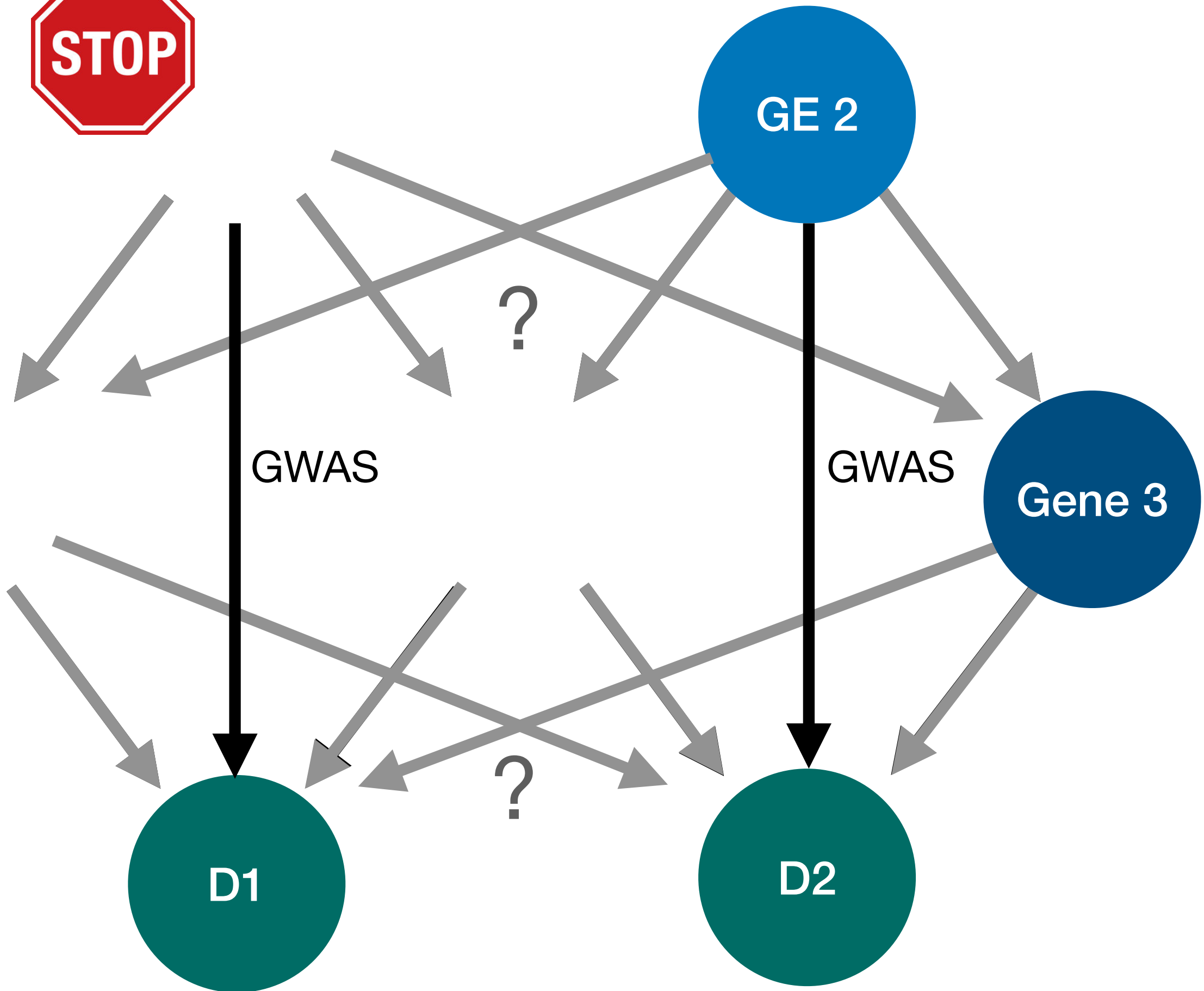


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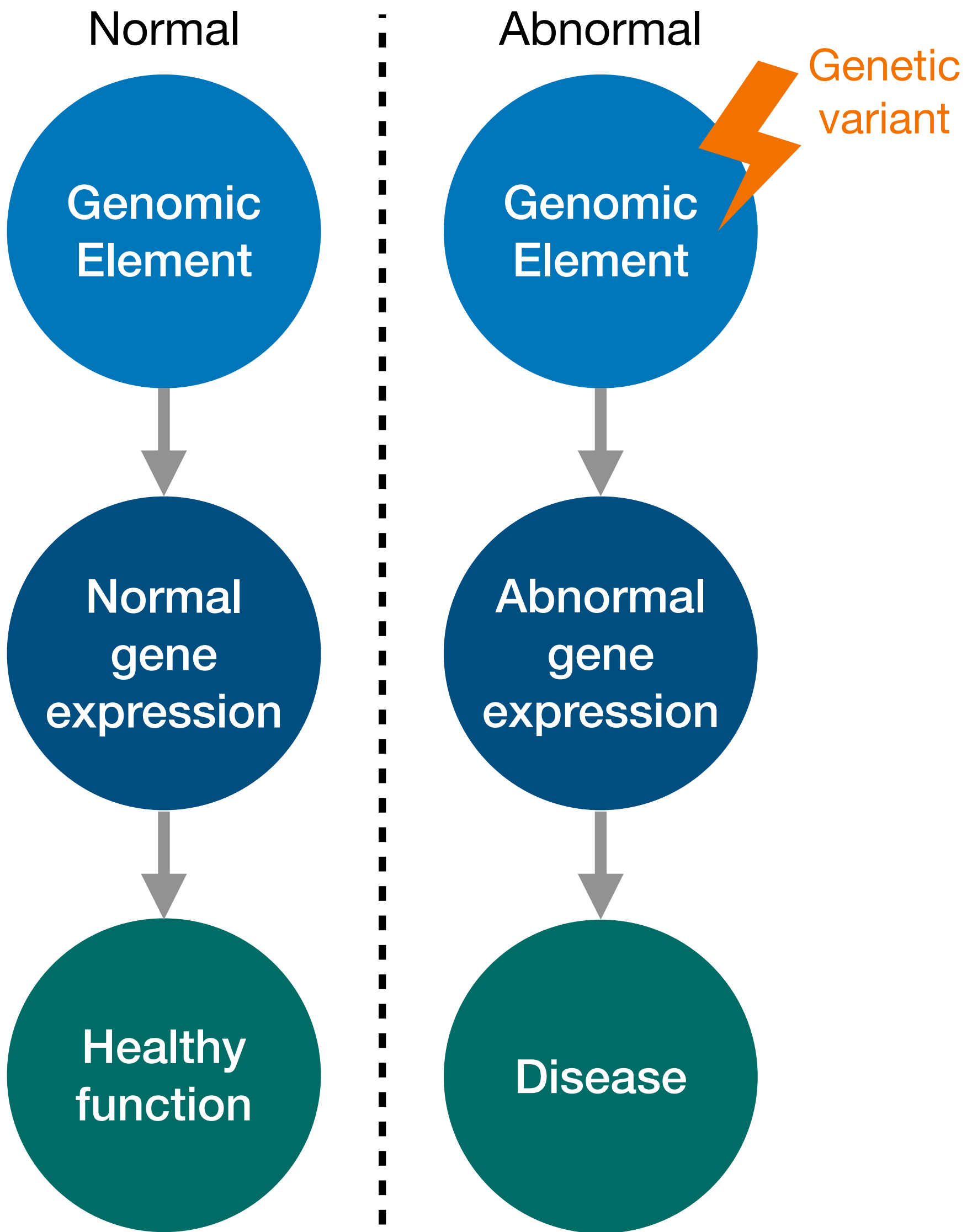


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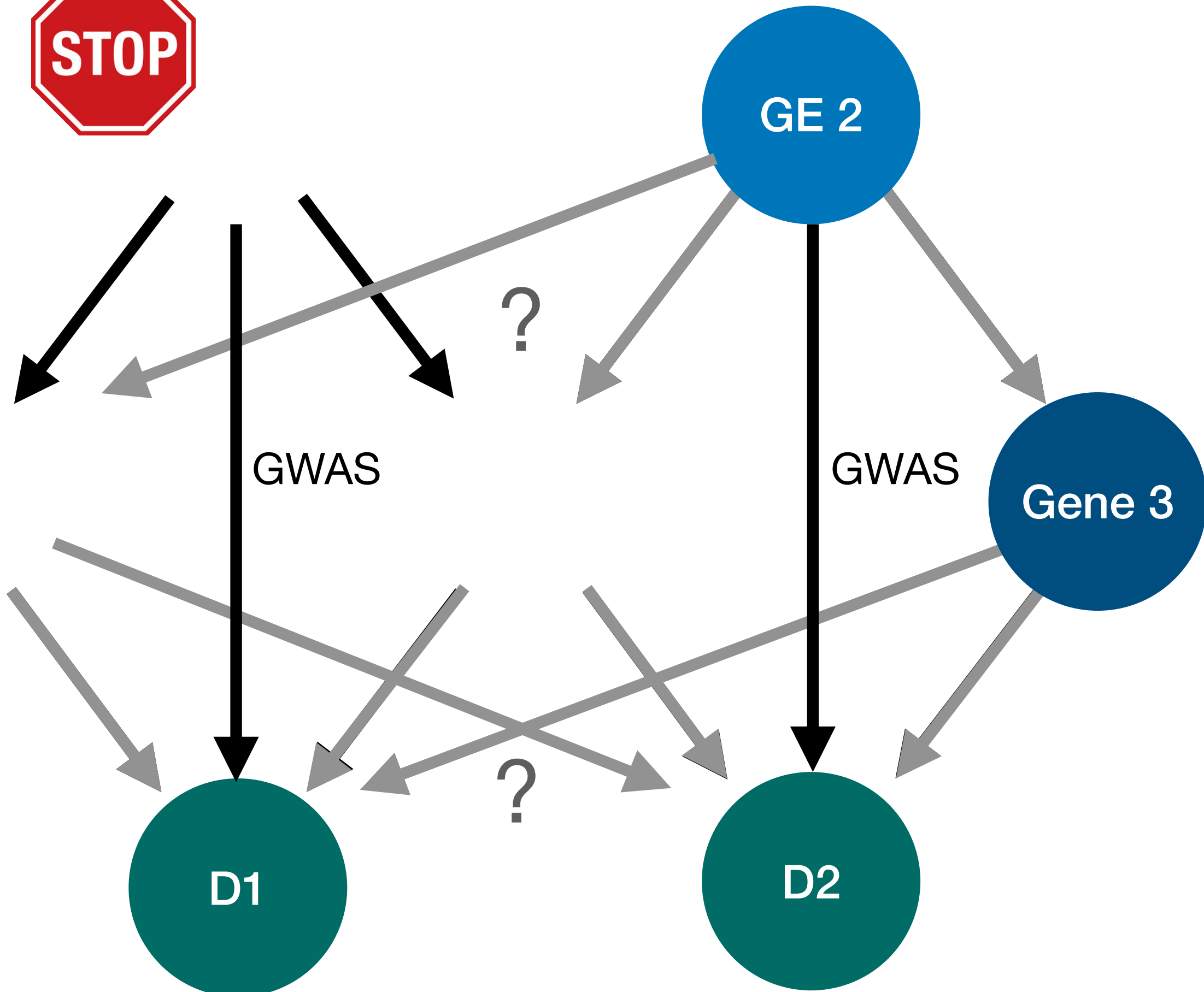


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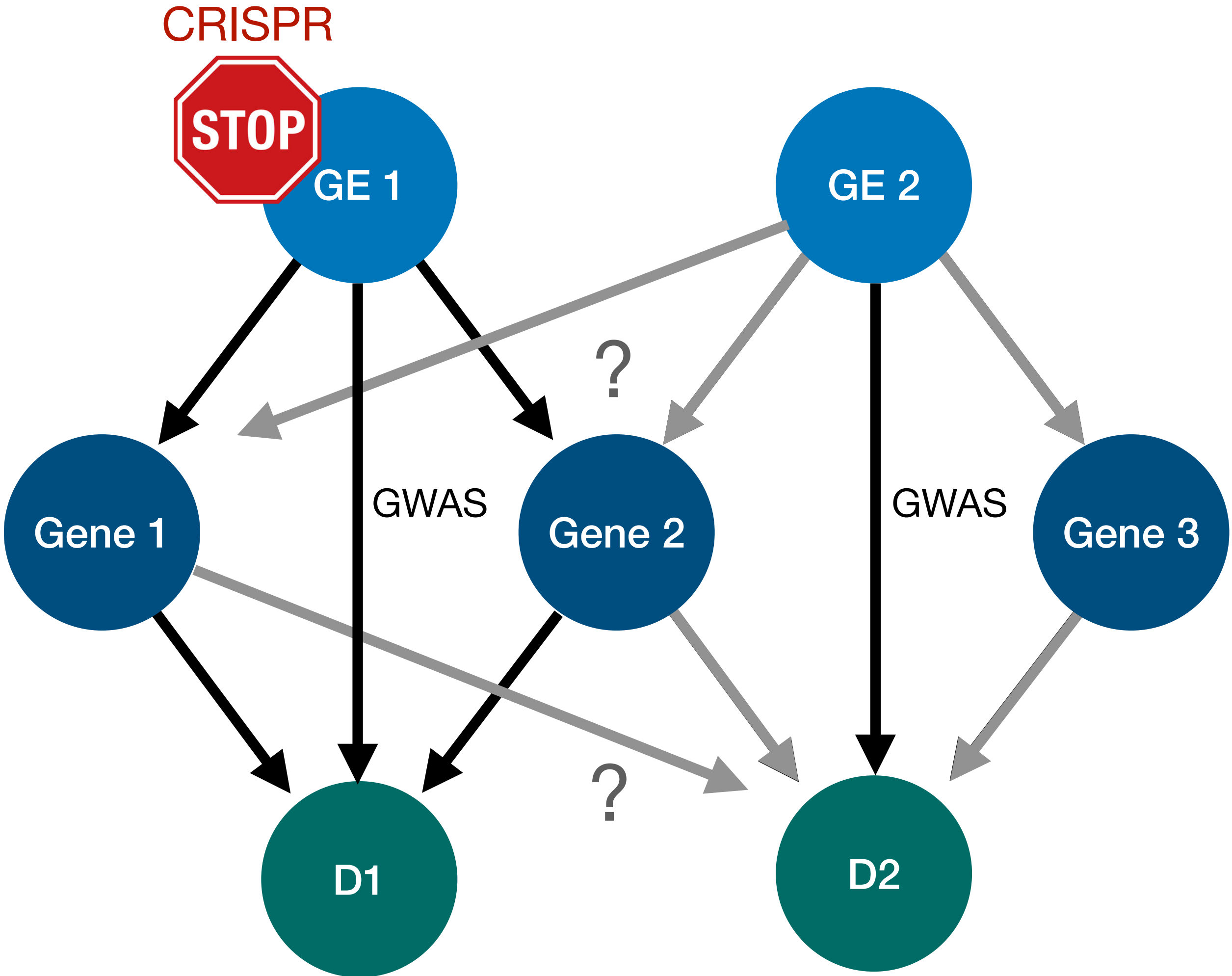
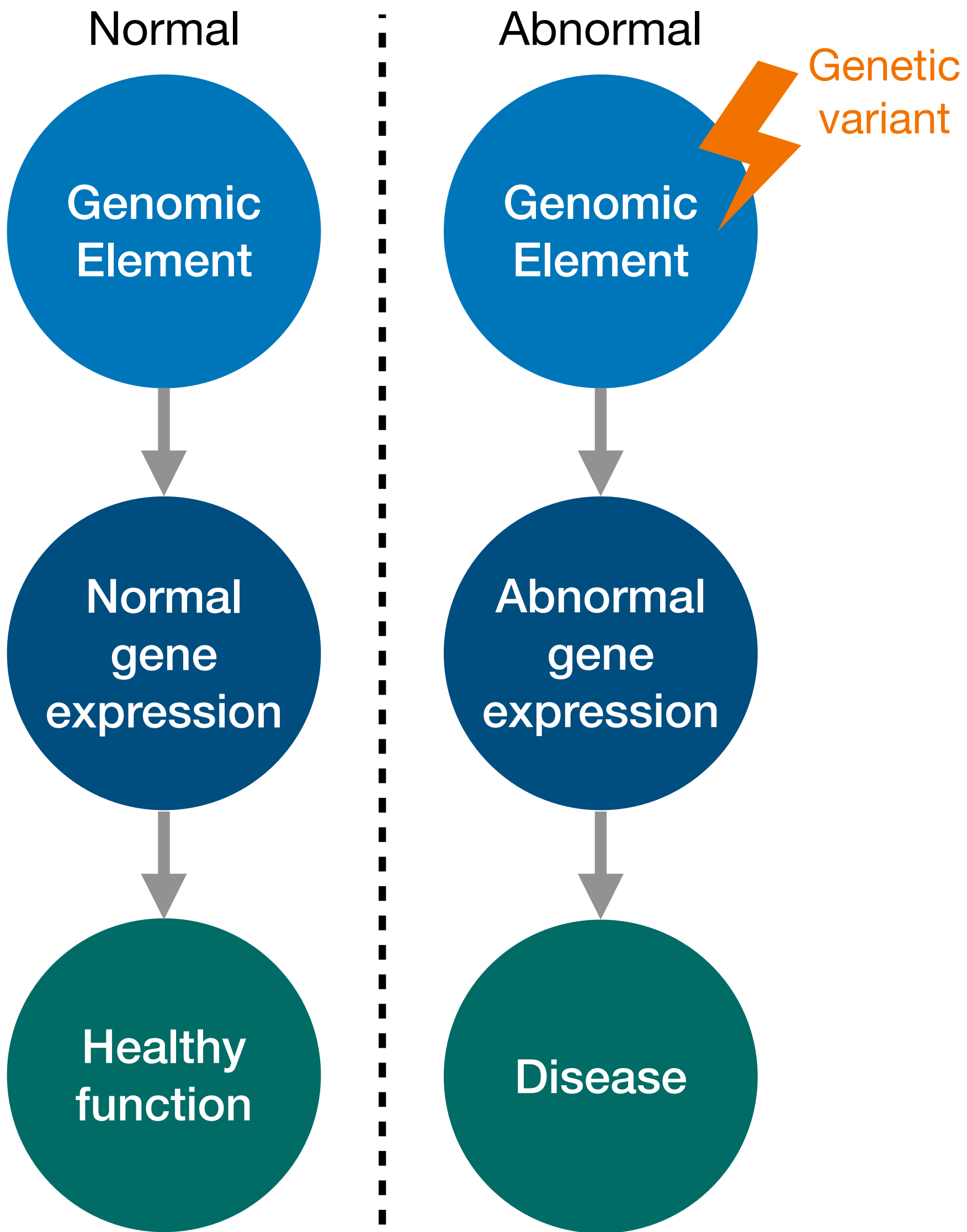


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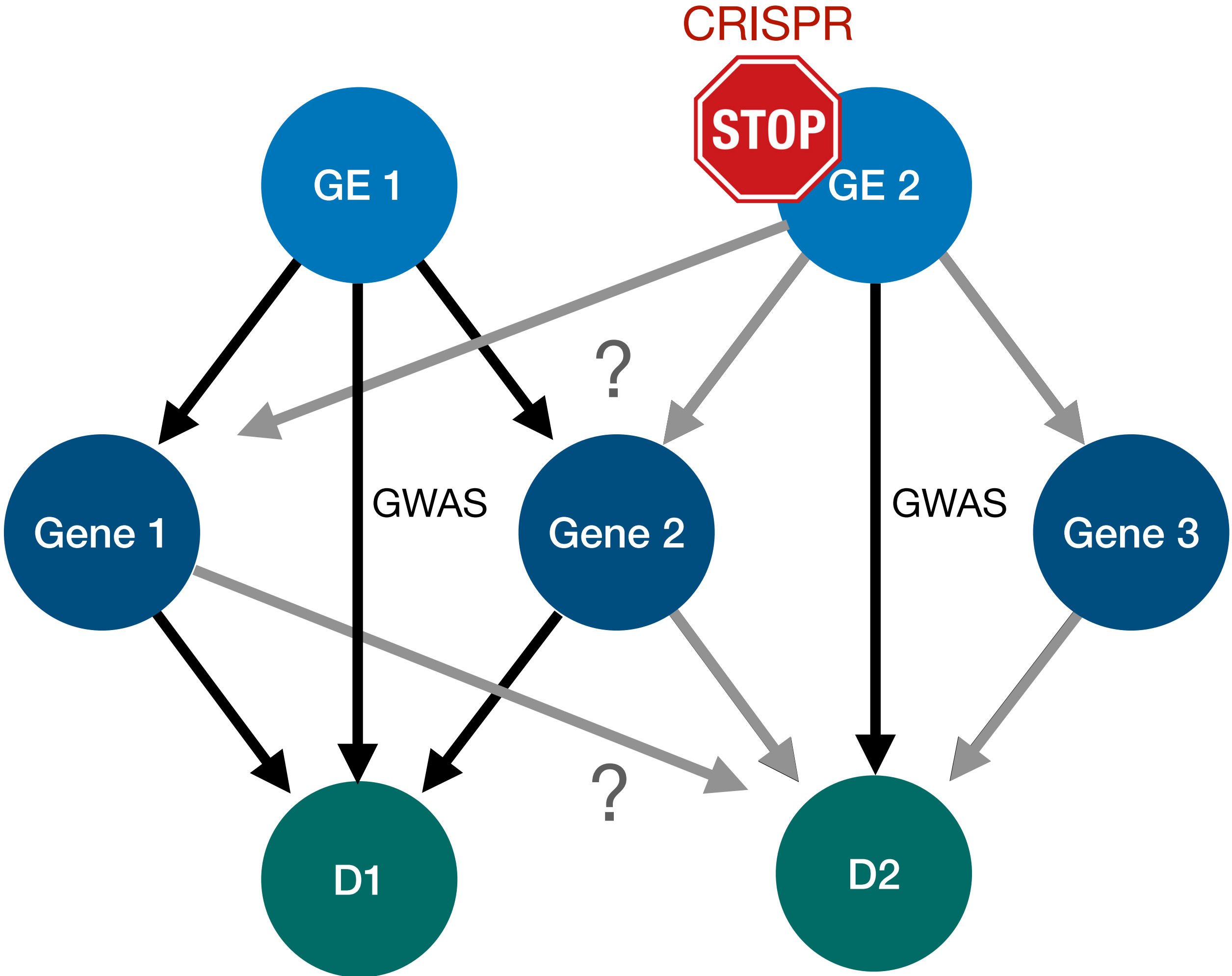
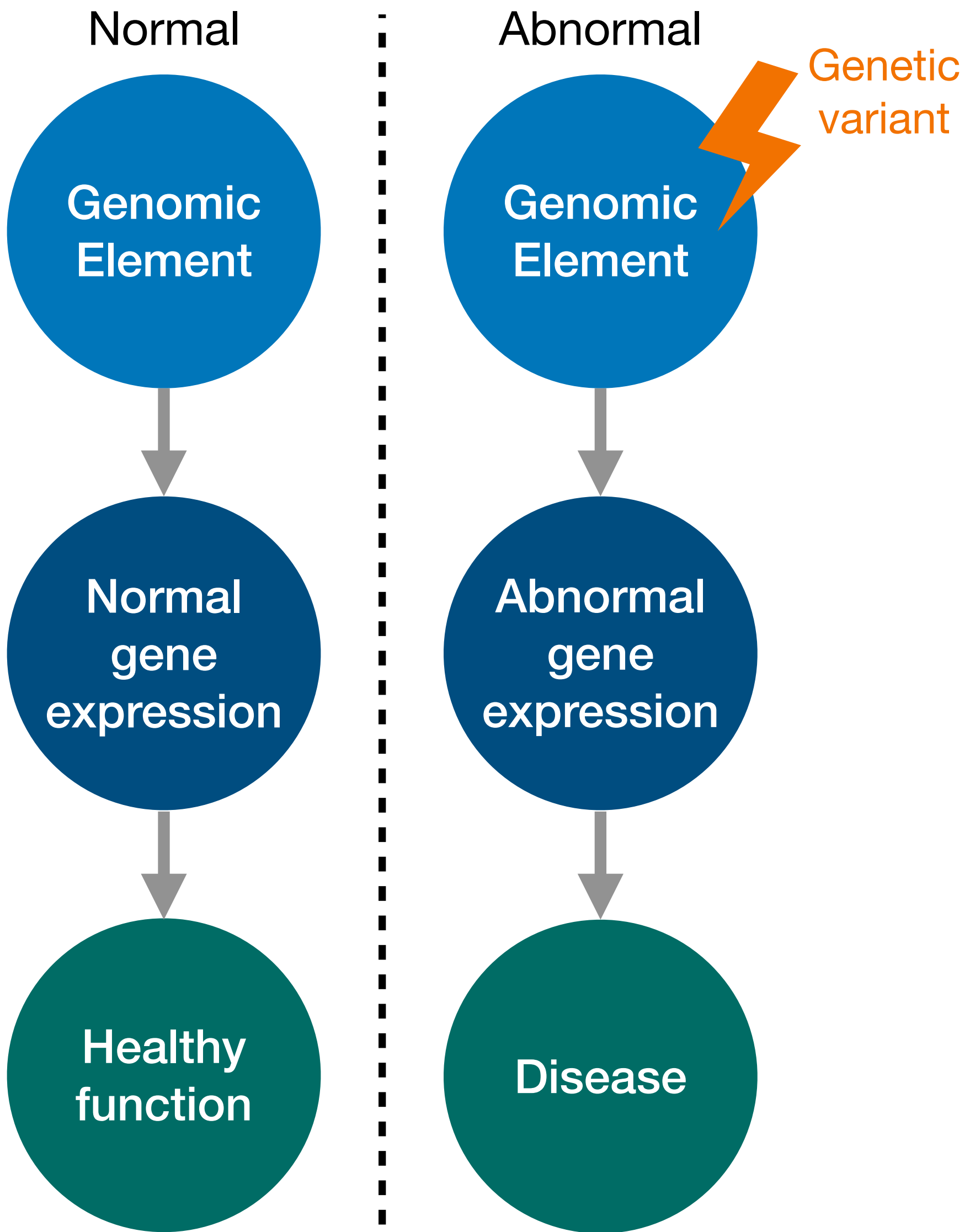
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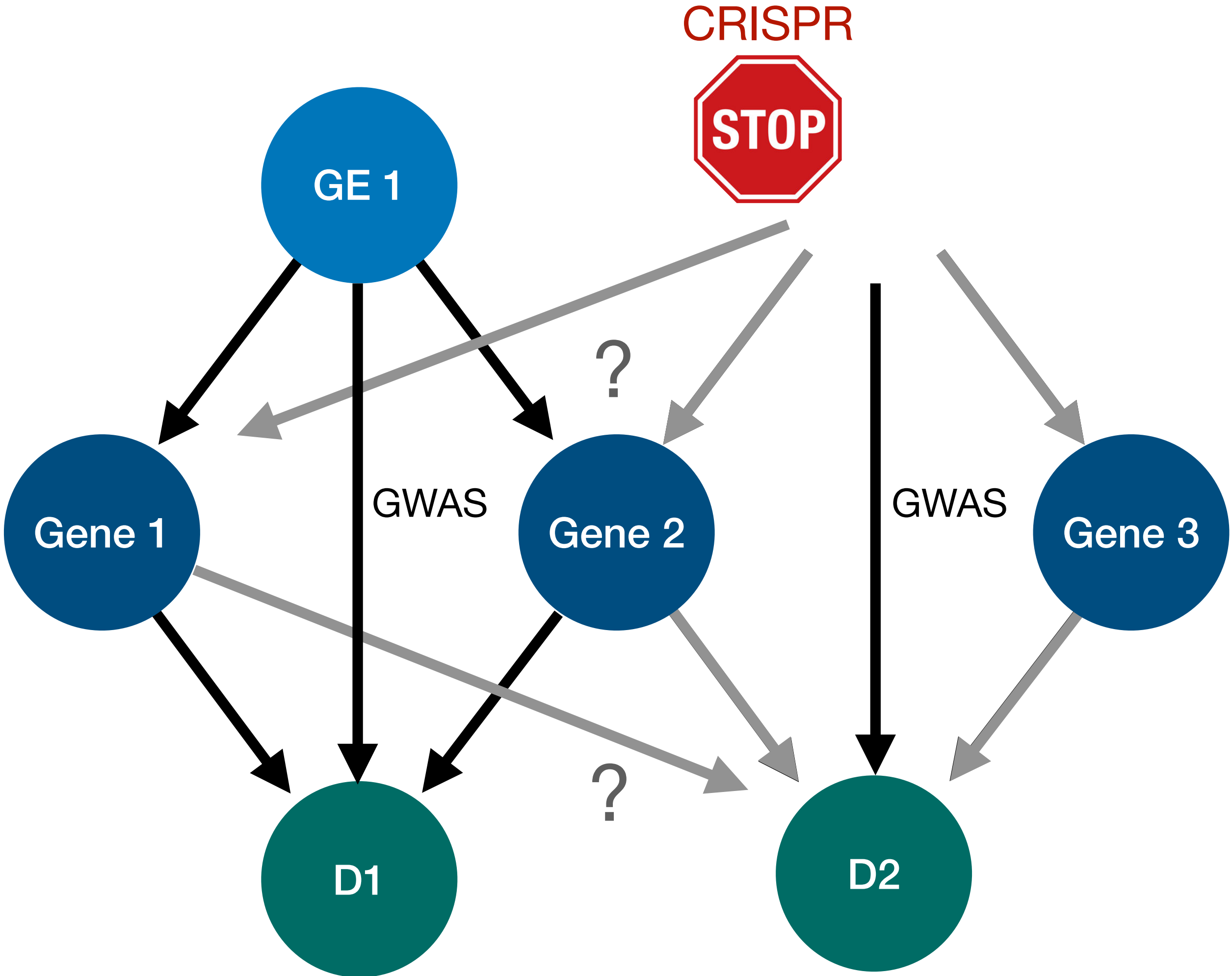
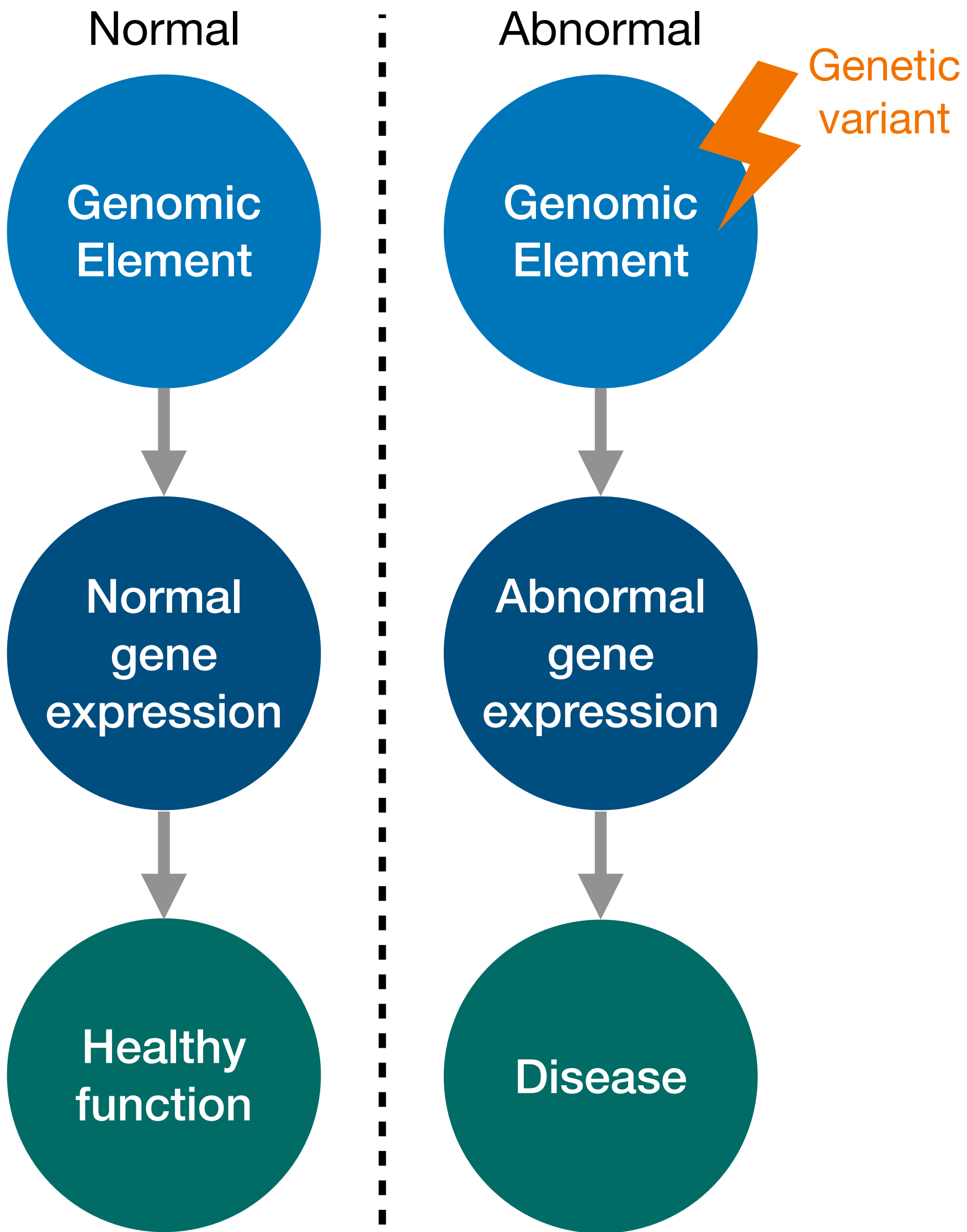
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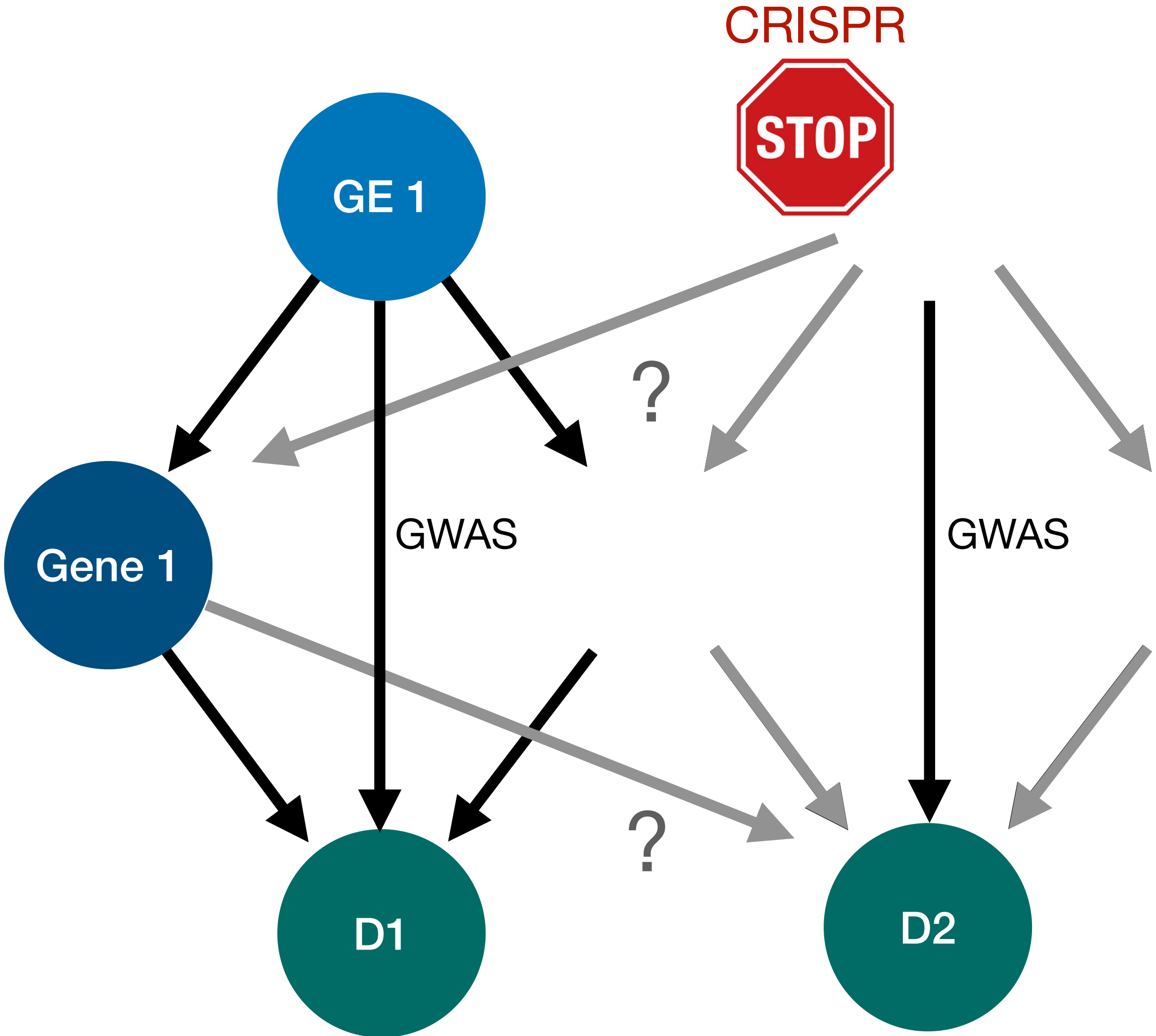
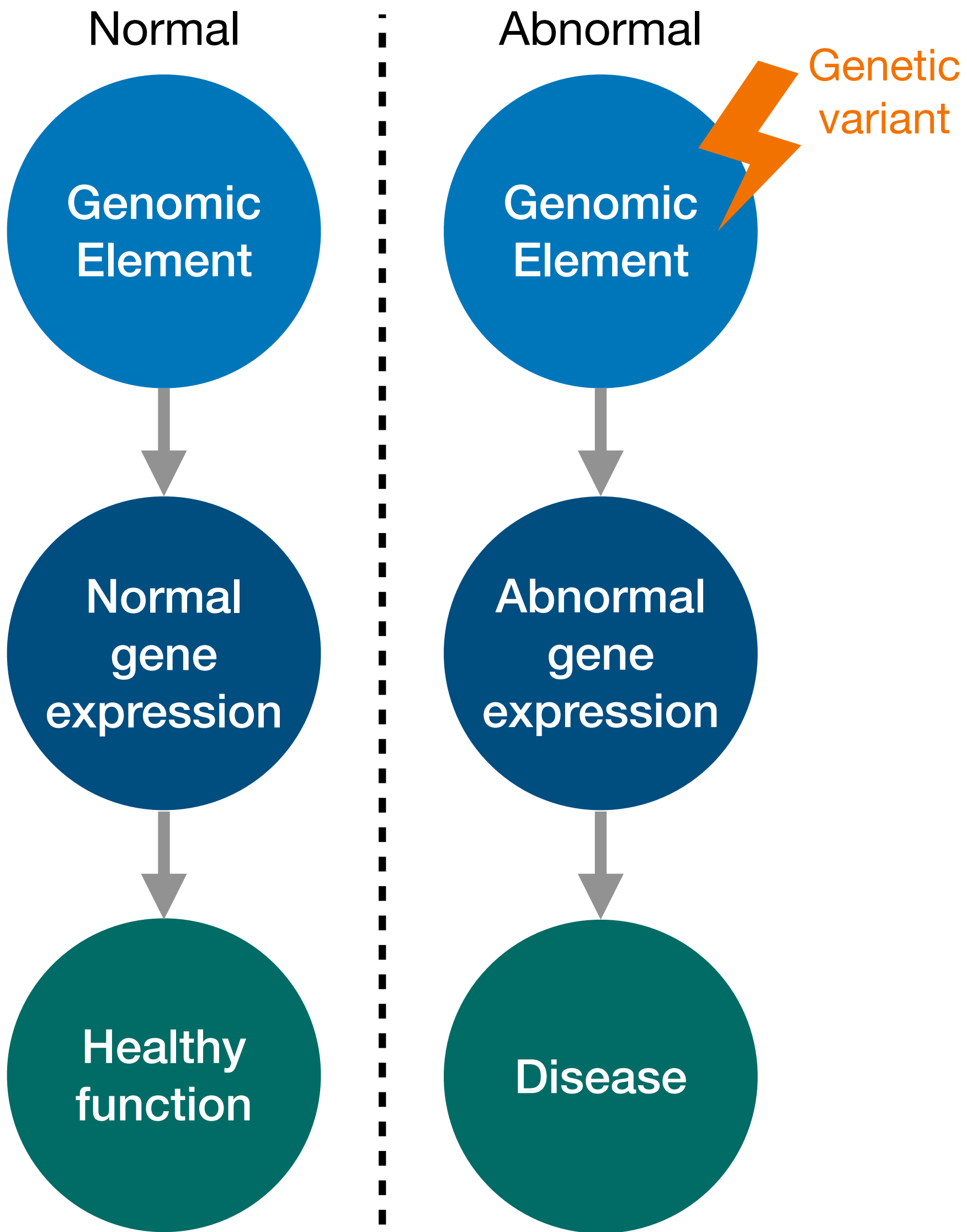
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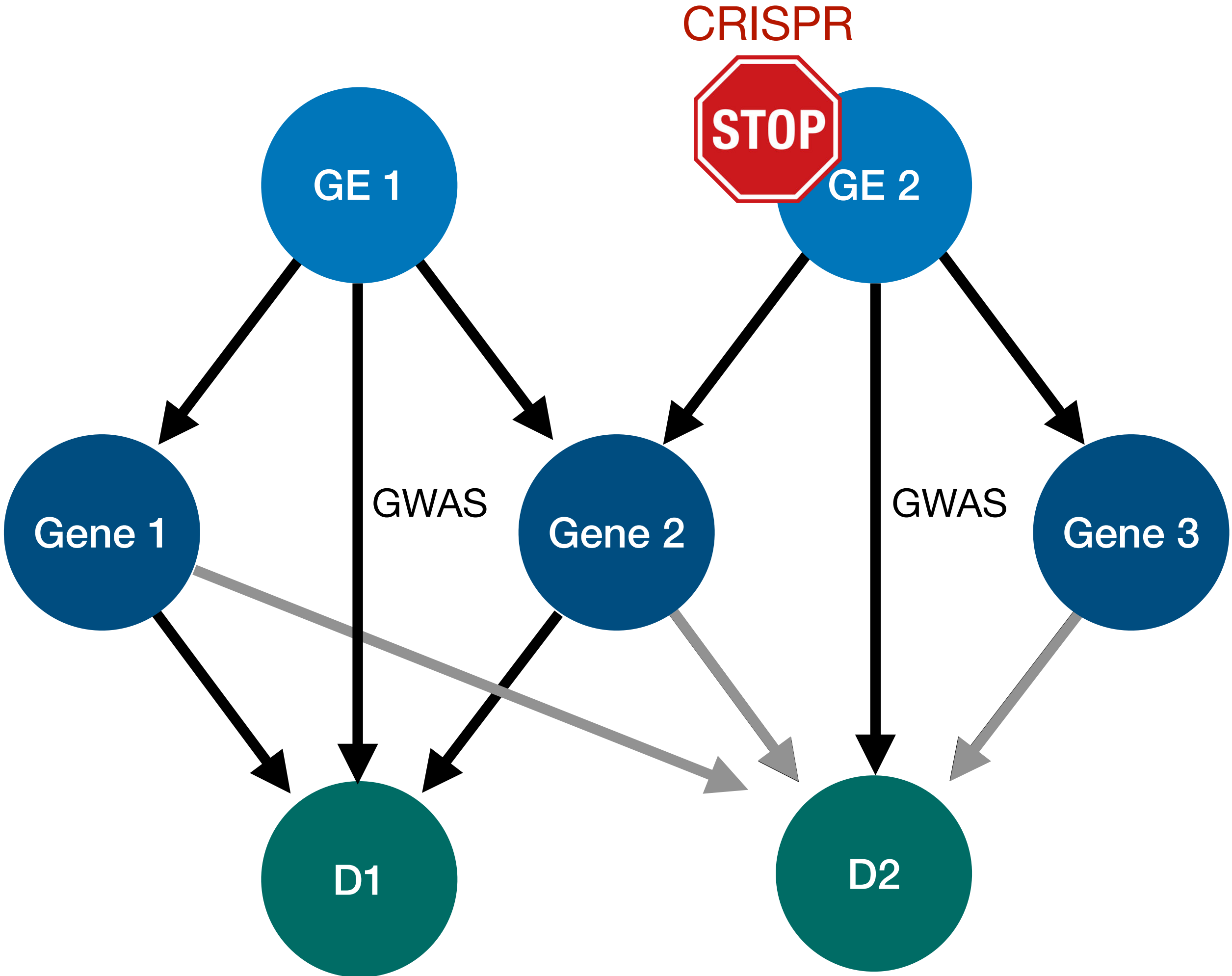
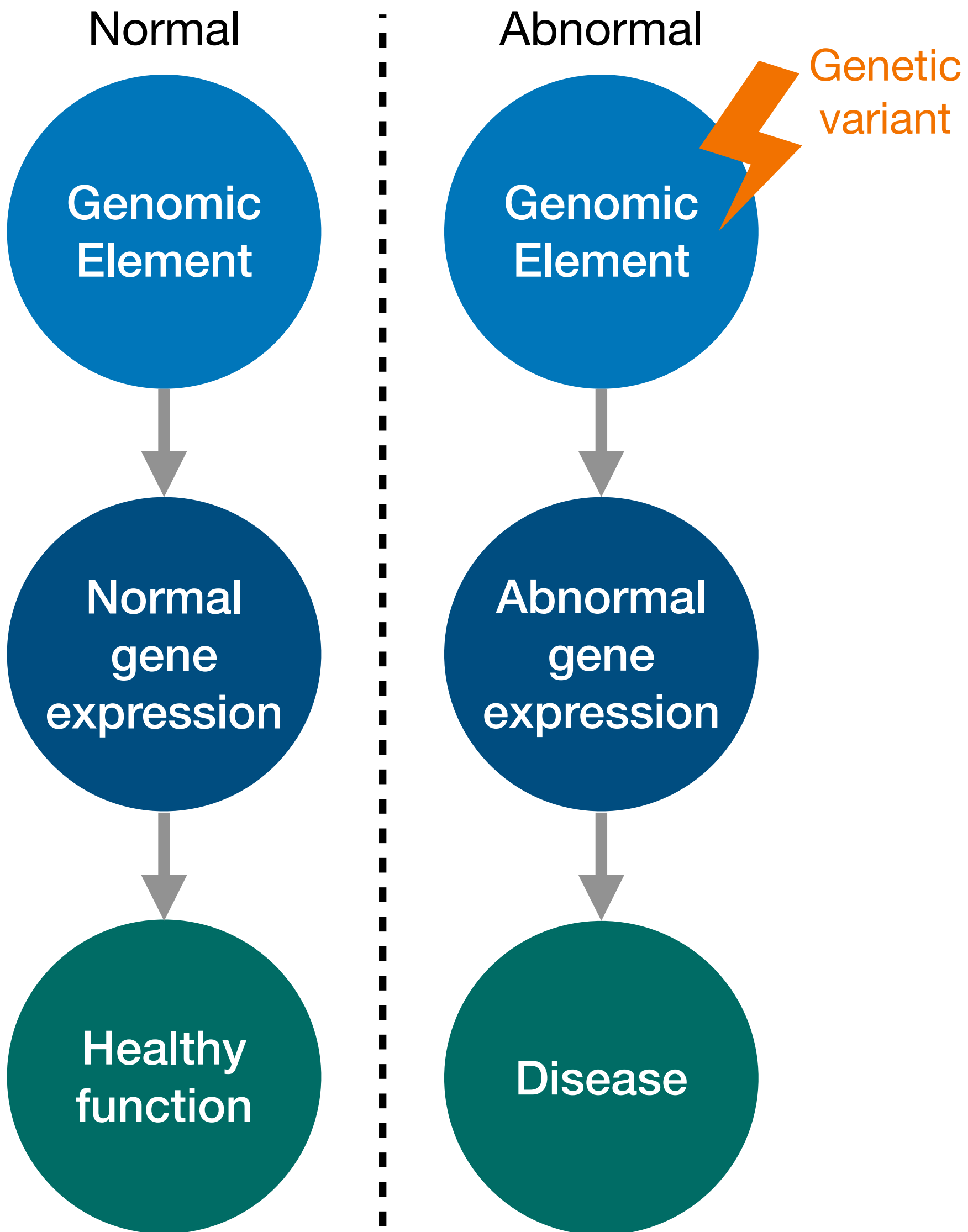
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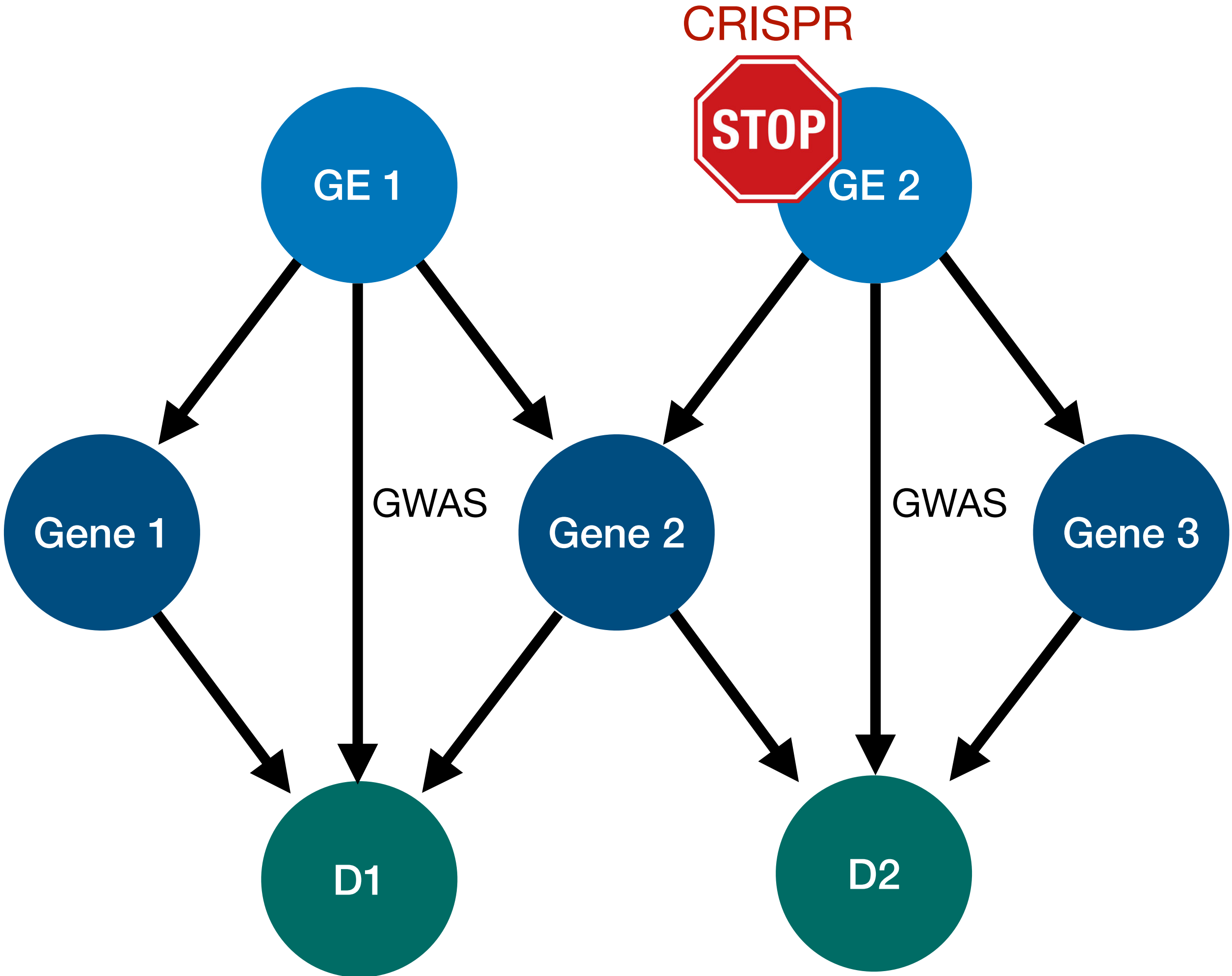
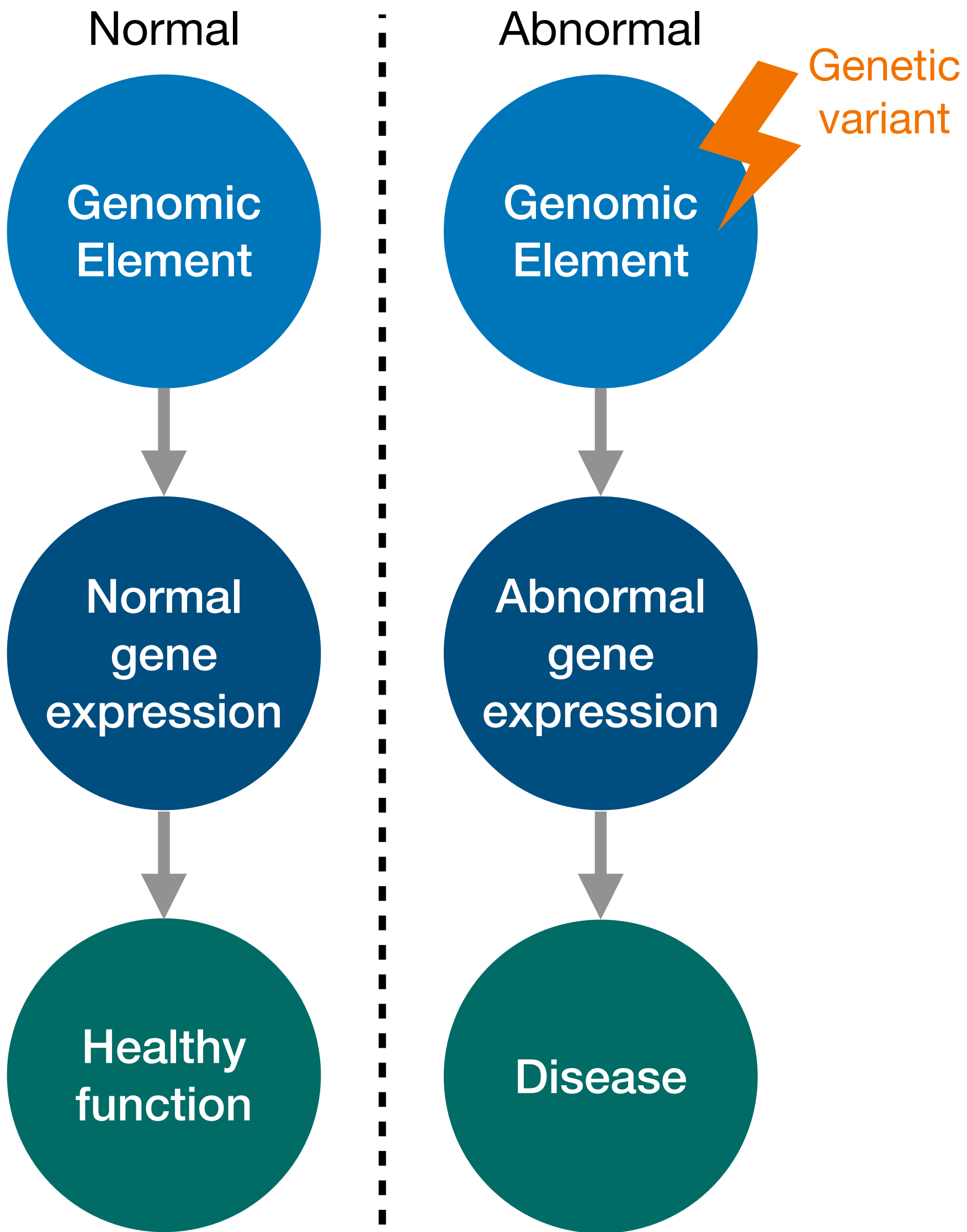
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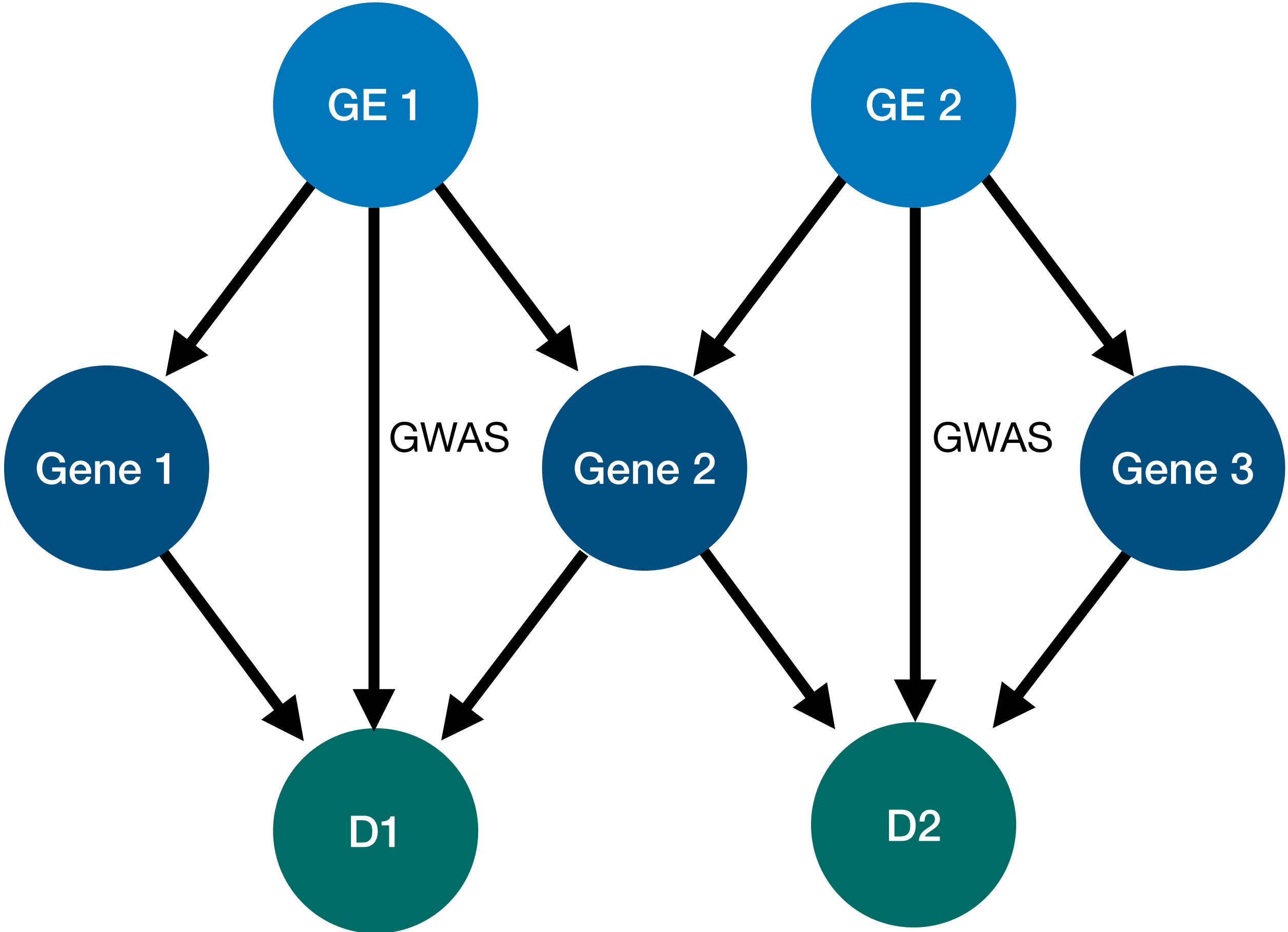
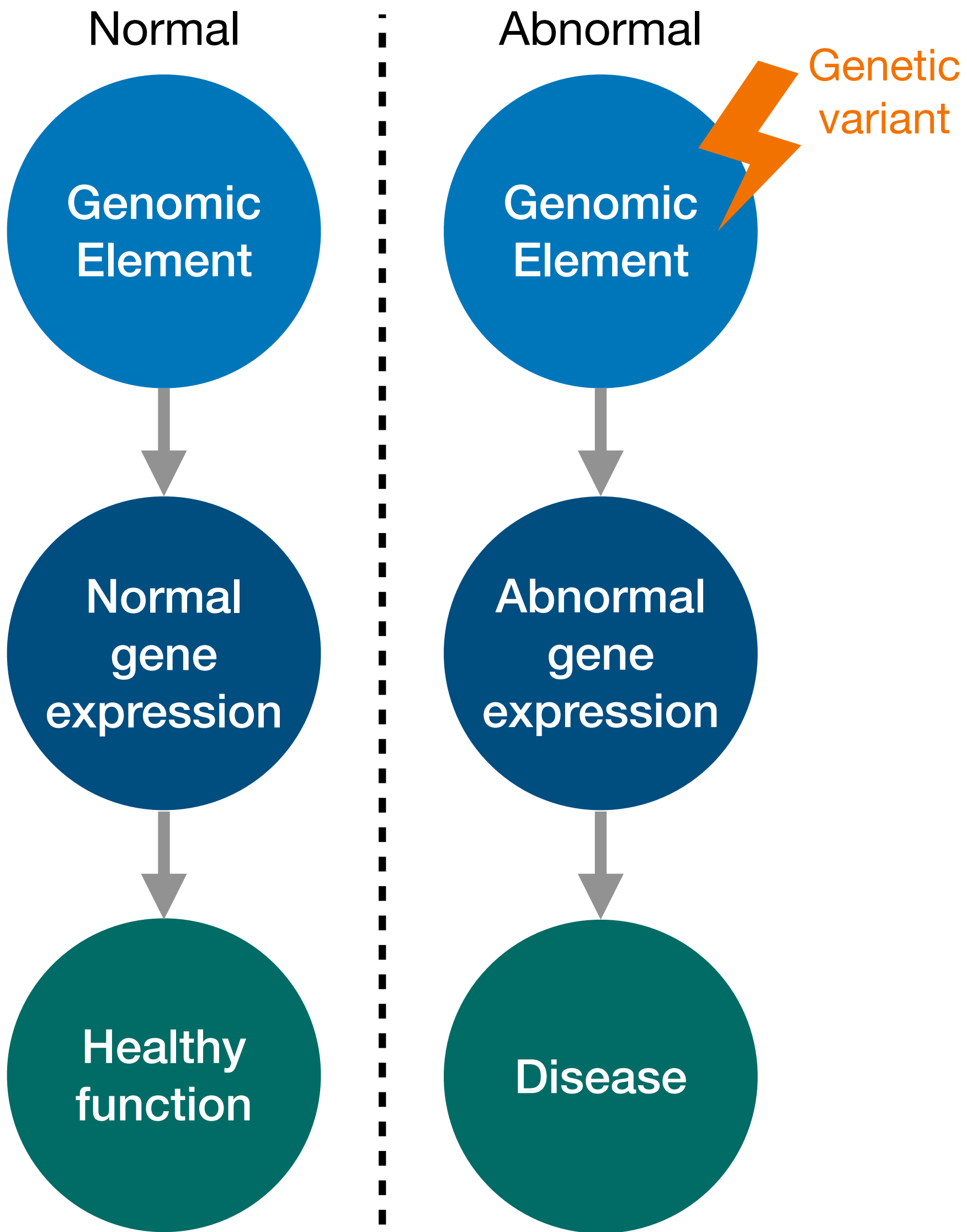
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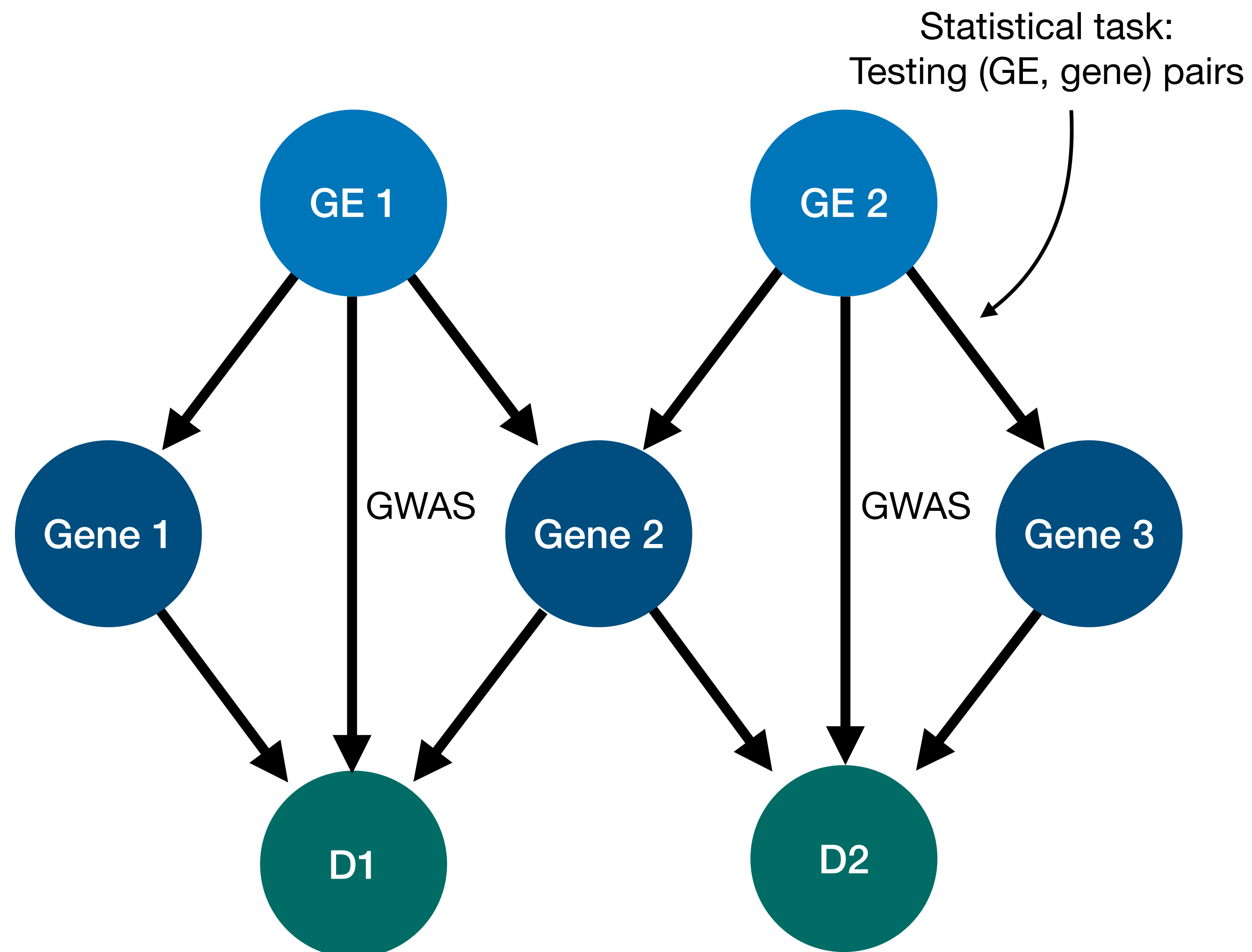
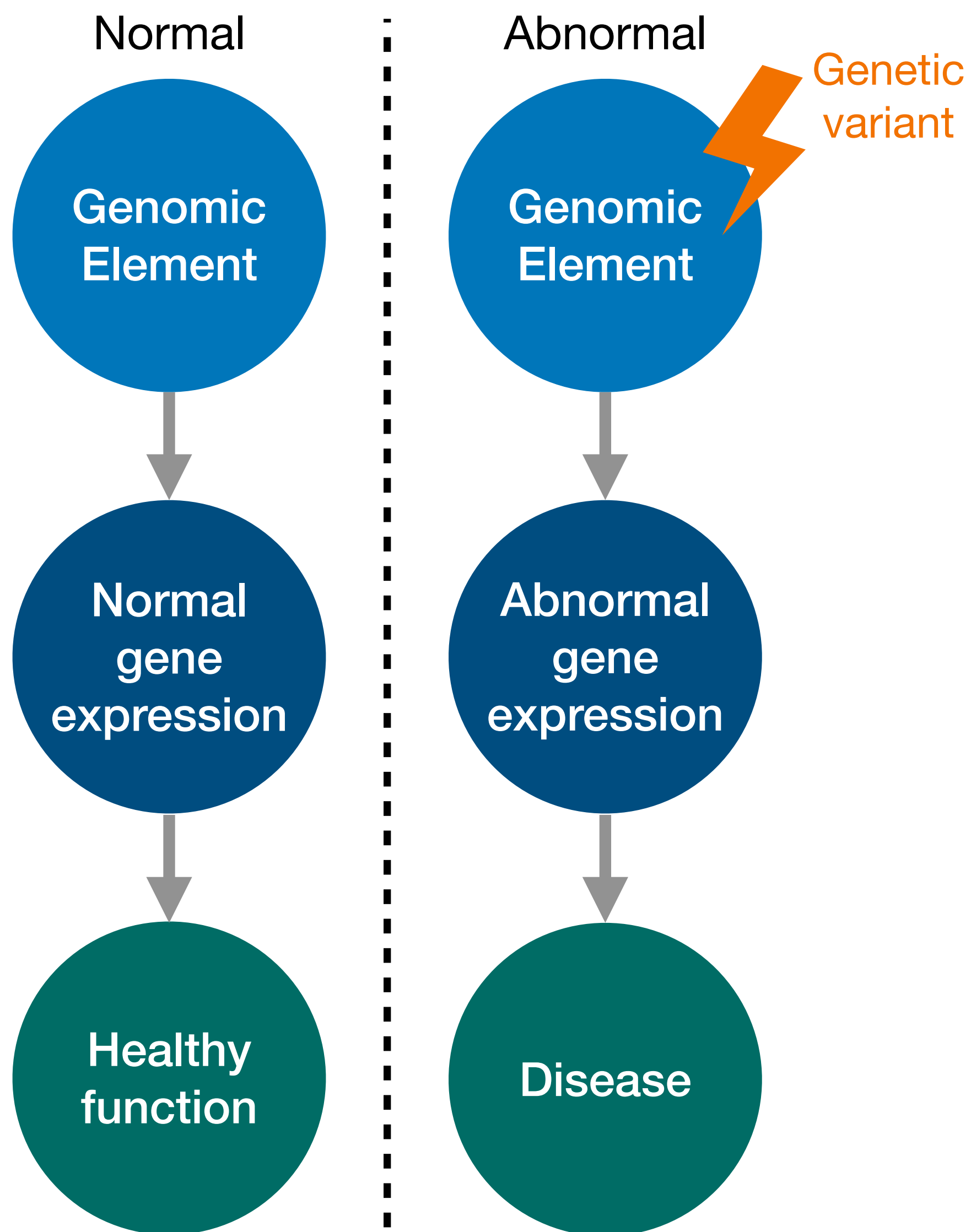
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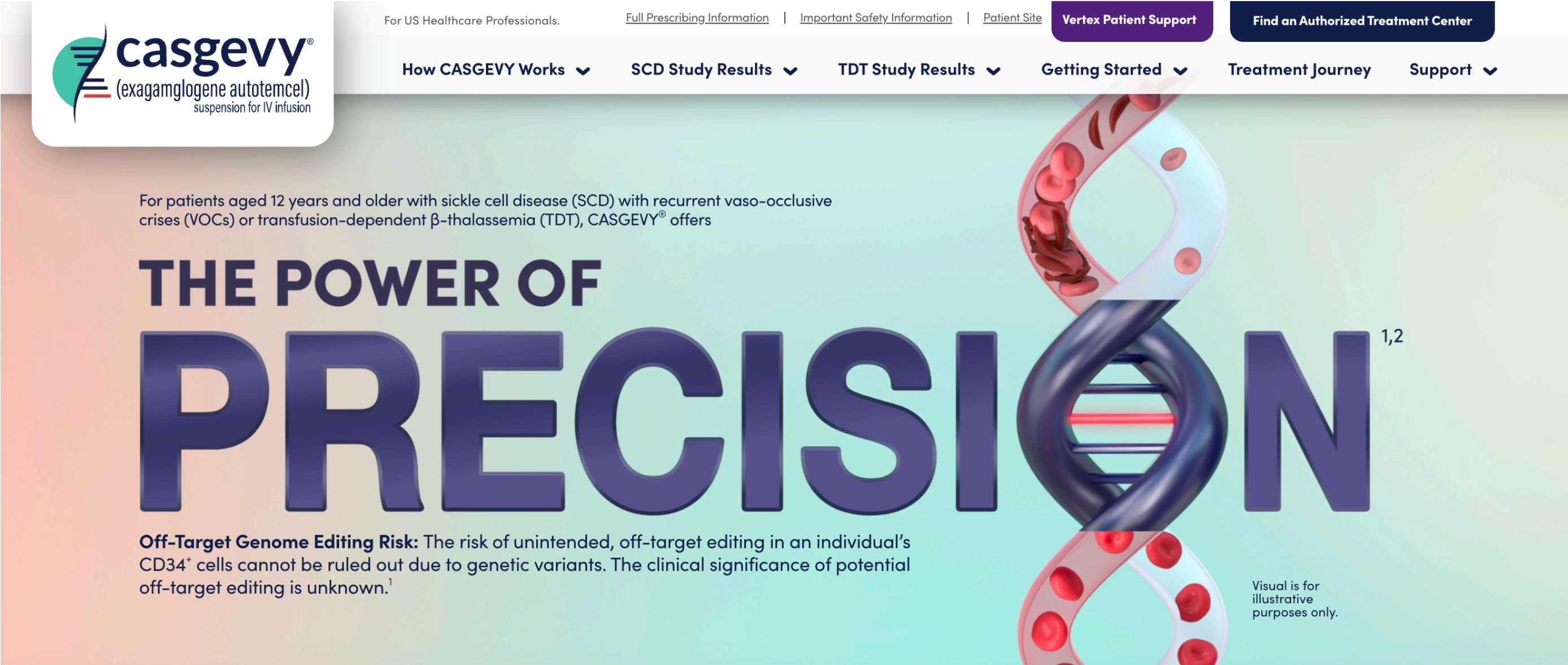
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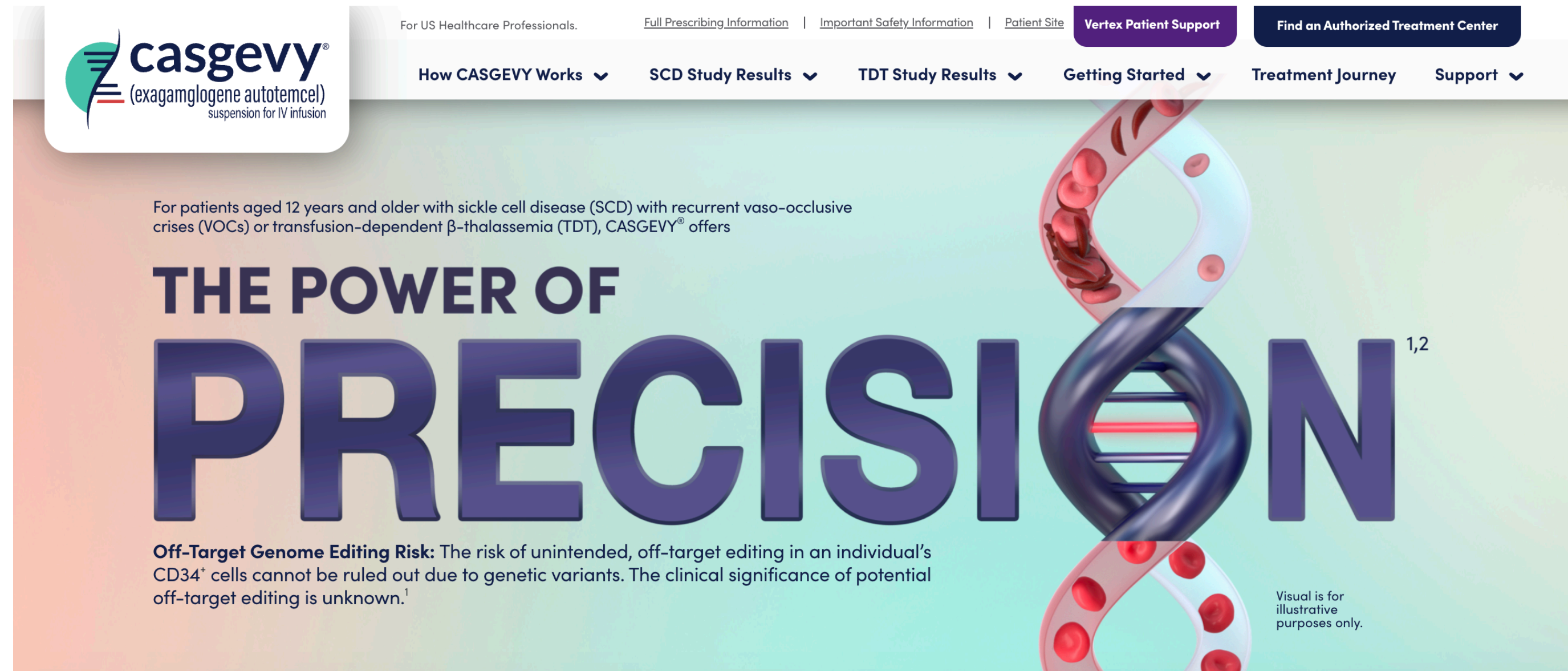
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Clinical promise for monogenic disease

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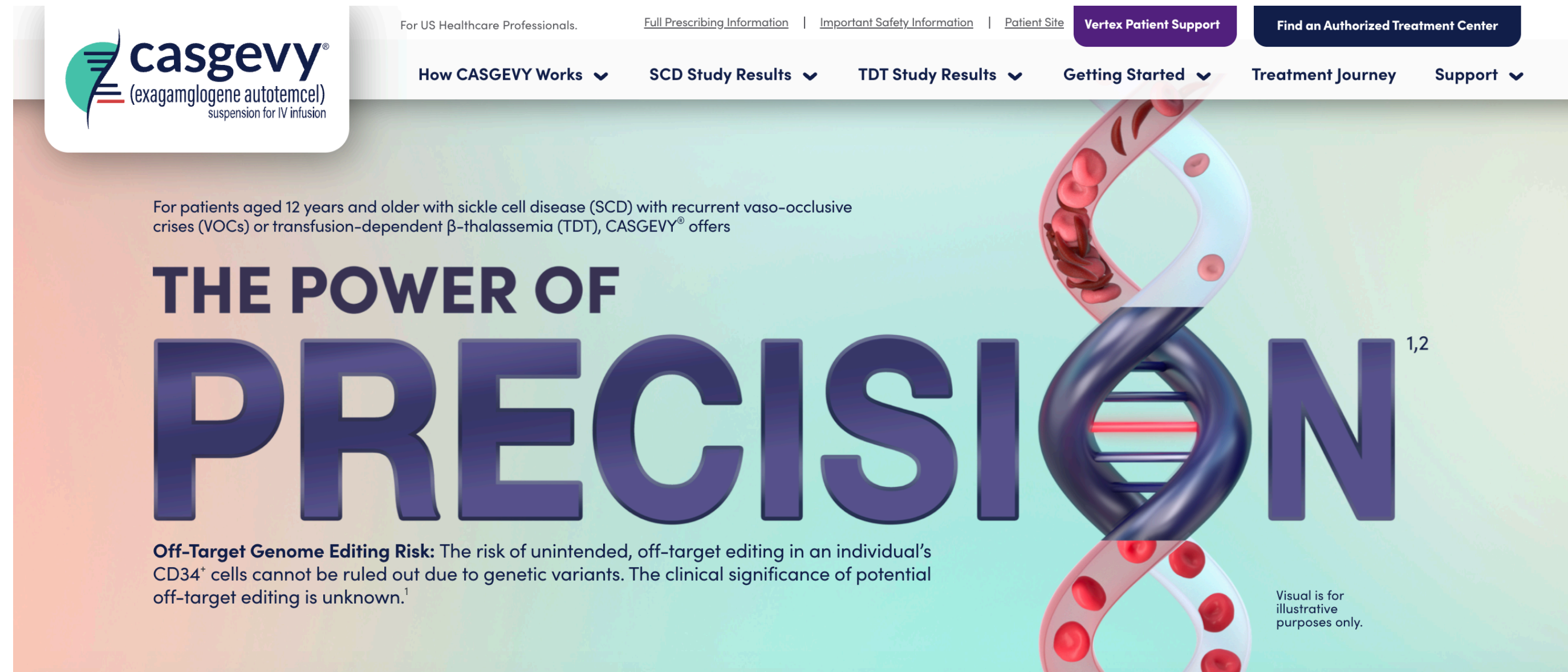


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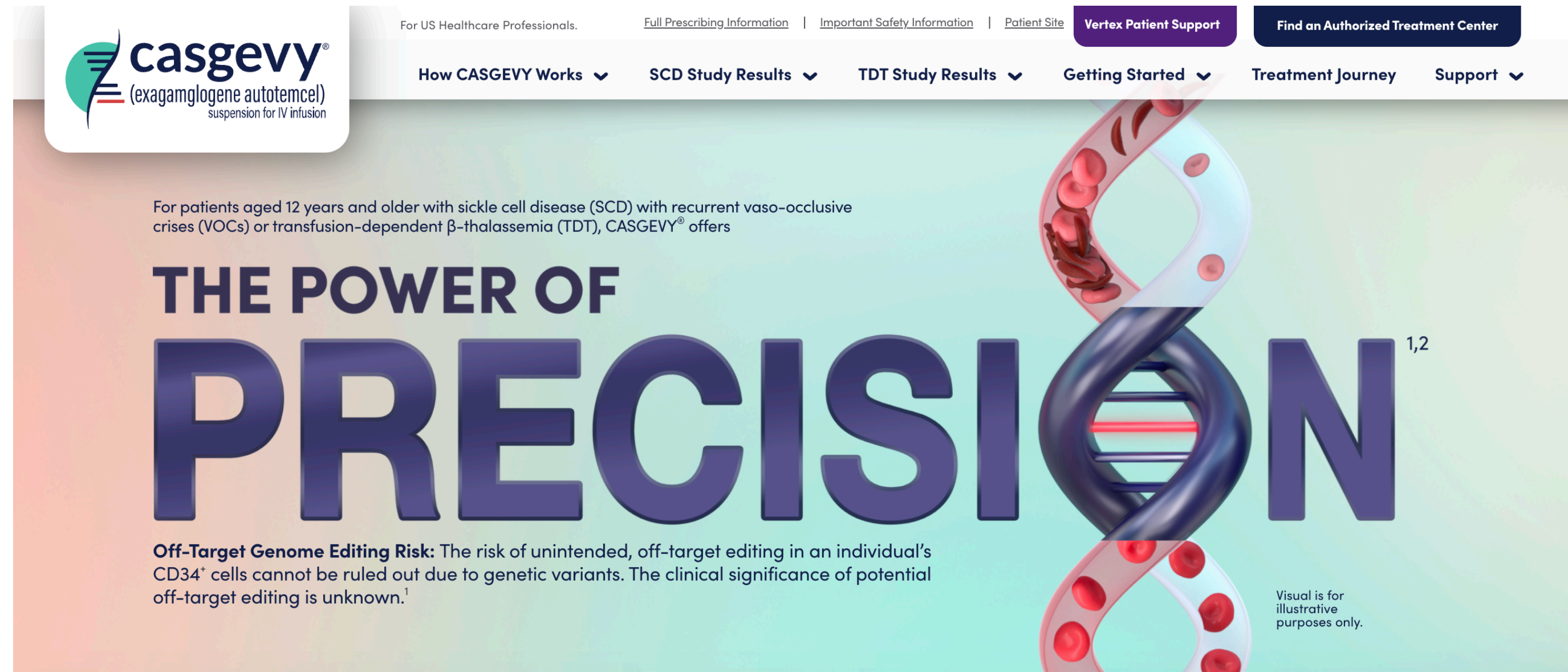
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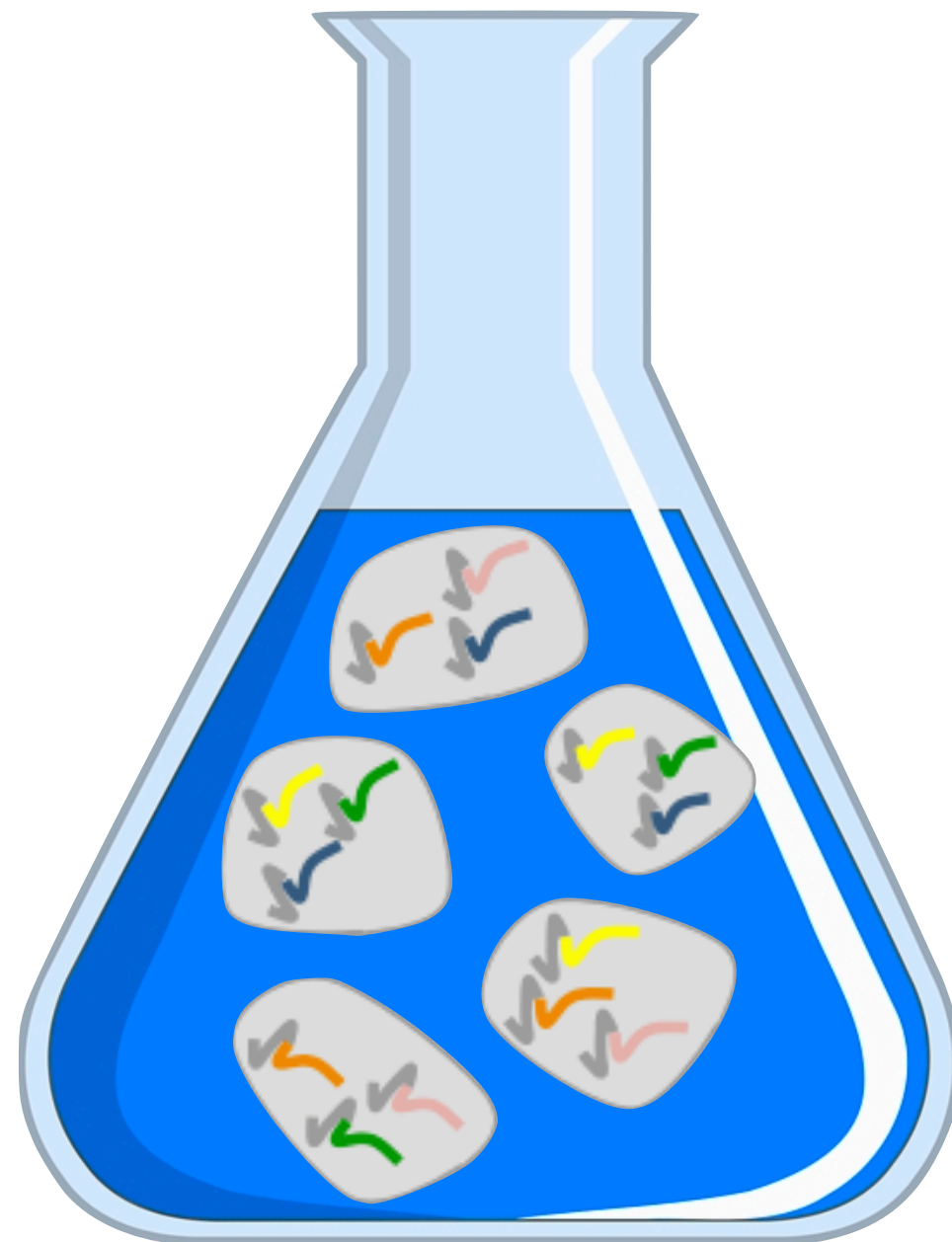
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¹Bauer et al (2013)

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Infect cells with CRISPR gRNAs
targeting genomic elements



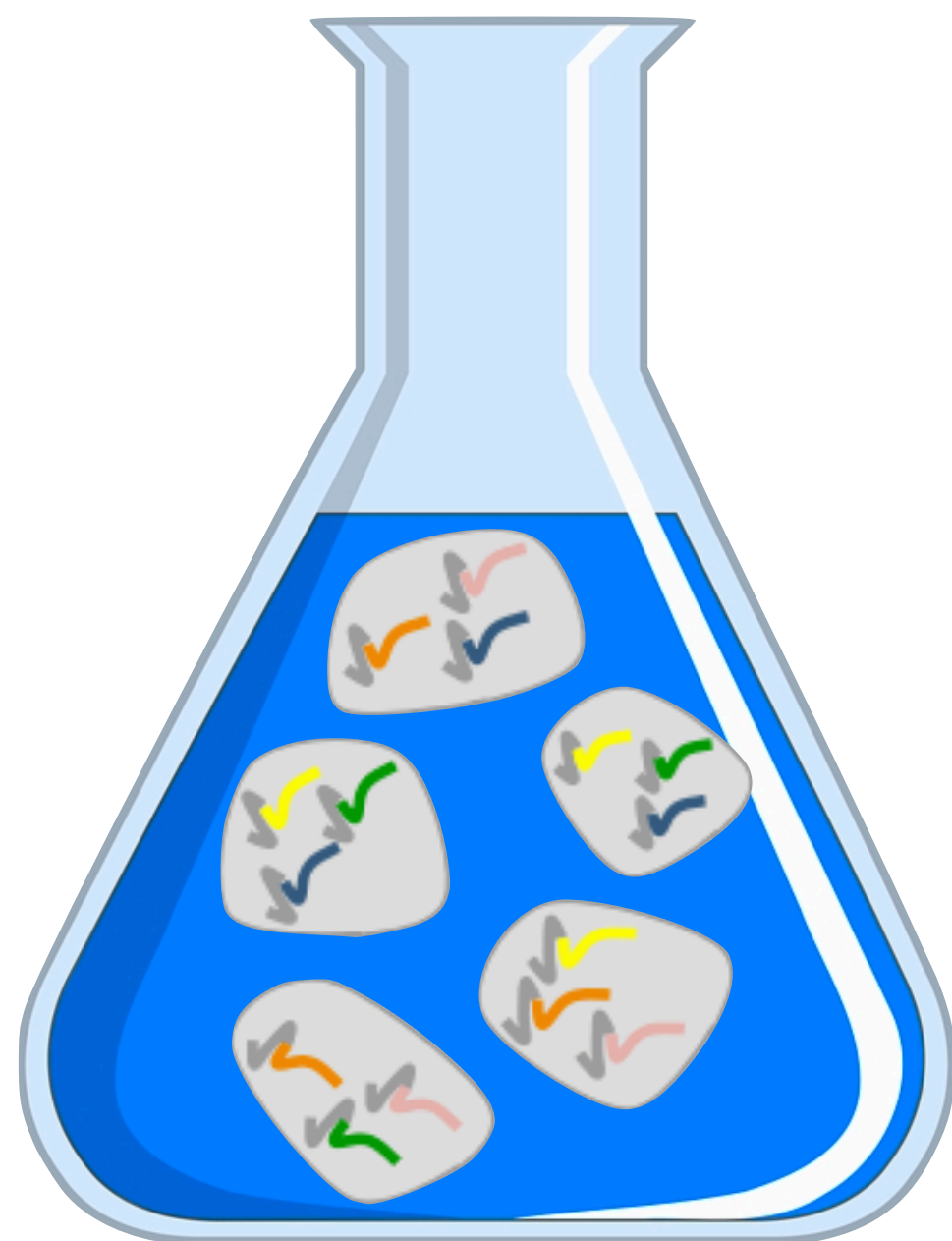
Single-cell
RNA sequencing



For each cell, measure:
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2. Gene expression

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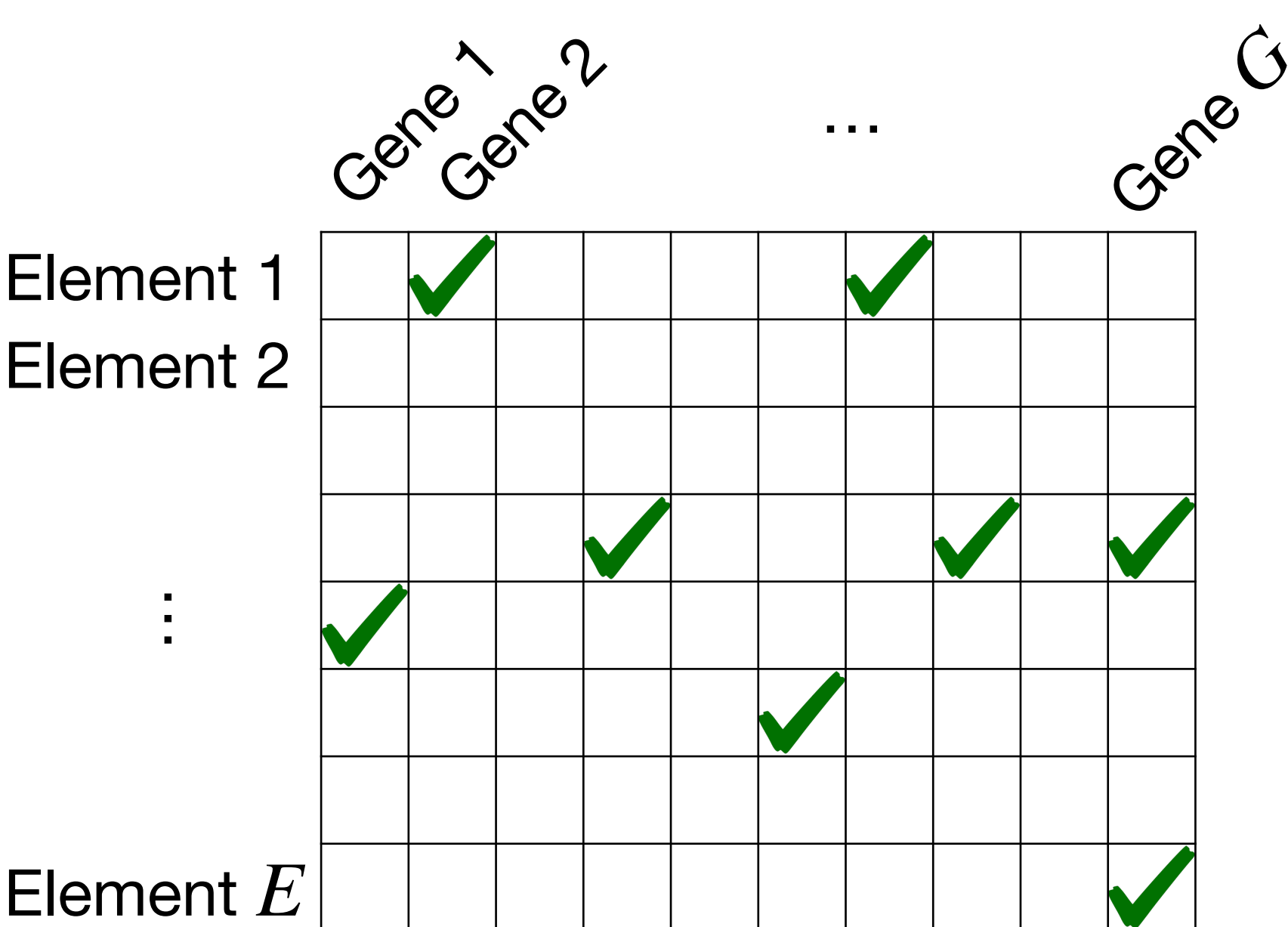


Single-cell RNA sequencing



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Test element-gene pairs for regulatory relationships.



Bottlenecks: scale, cost and sparsity

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Genome-wide gene-targeting:¹

¹Replogle et al (2022)

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Cell perturbed	Gene Expression					
	0	1	2	3	4	8
FALSE	201027	5594	324	32	5	1
TRUE	332	9	0	0	0	0

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Analytical challenge: classical CLT-based test (Wald/Score) fails to control Type-I error.

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Solutions: PerturbPlan and spaCRT

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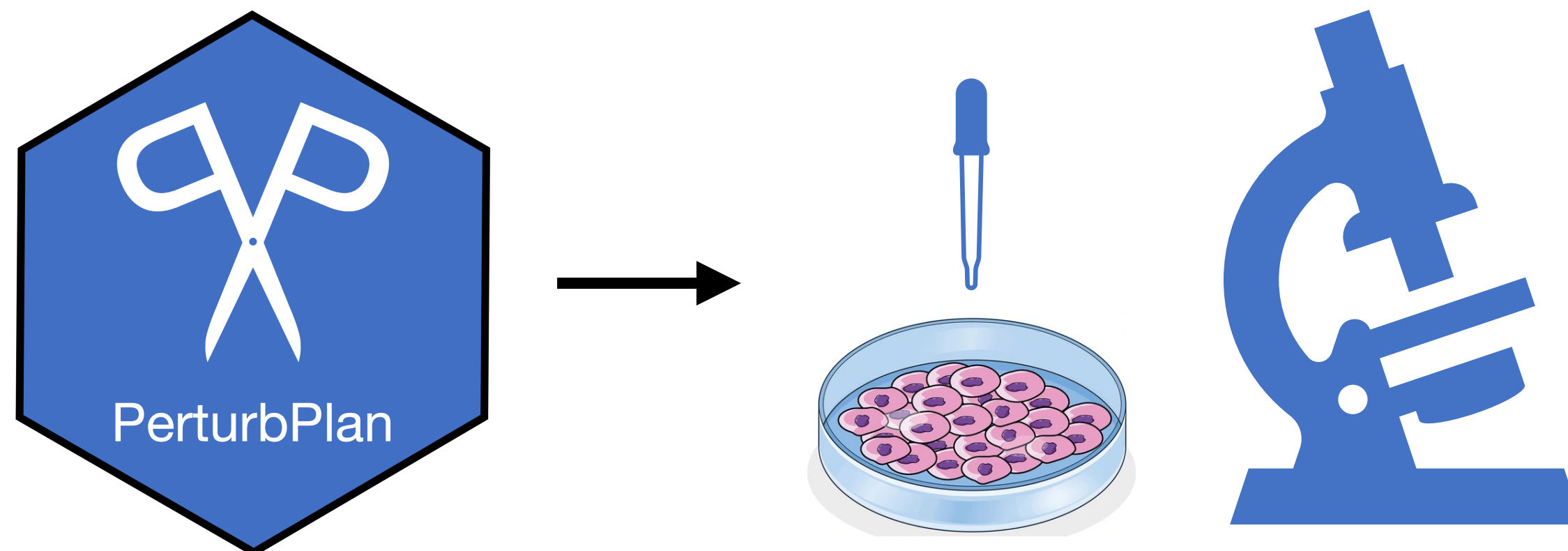


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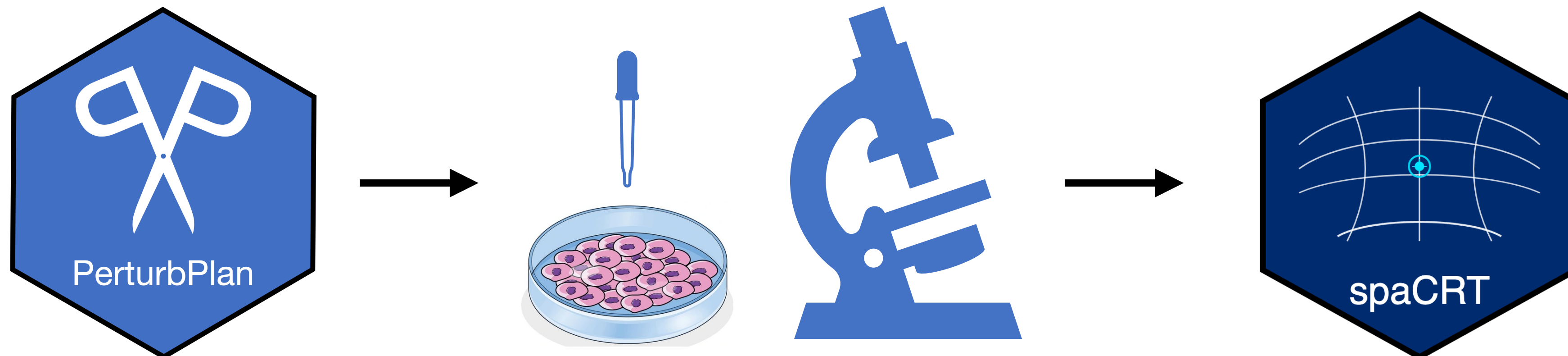
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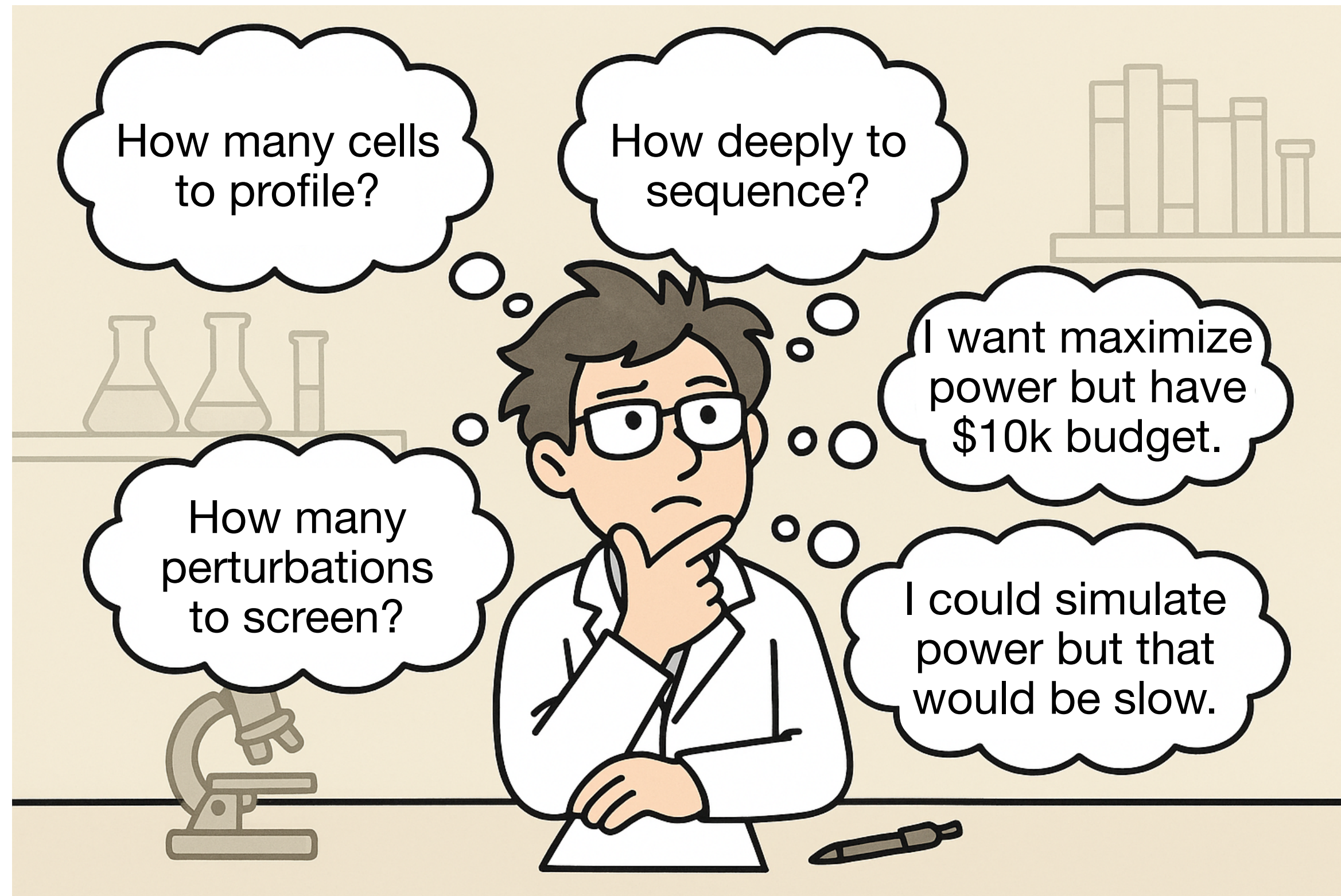
Post-experimental analysis: **spaCRT** supports computationally efficient differential expression analysis, with strong statistical validity guarantee.



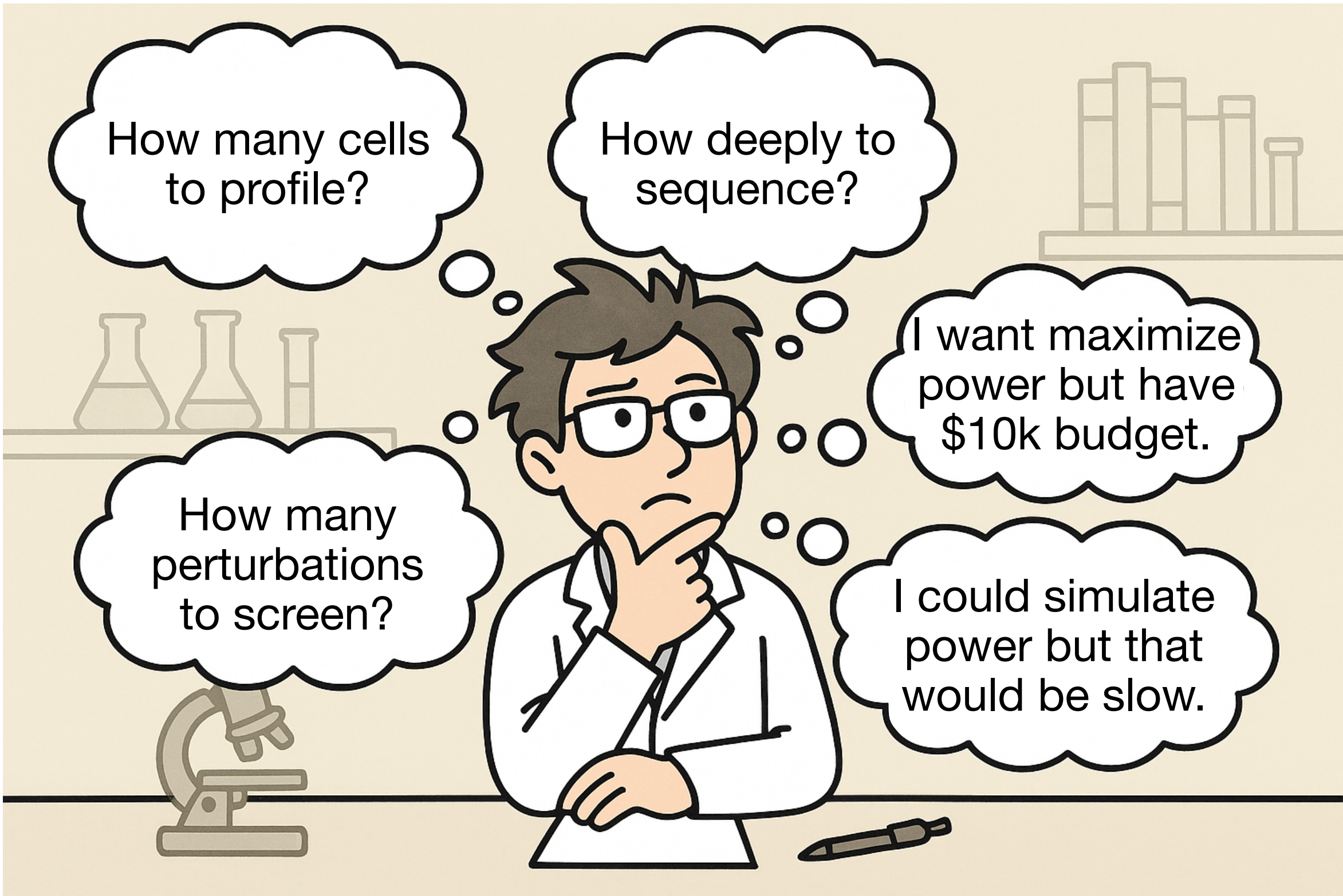
Part I: PerturbPlan

Designing a Perturb-seq experiment is hard

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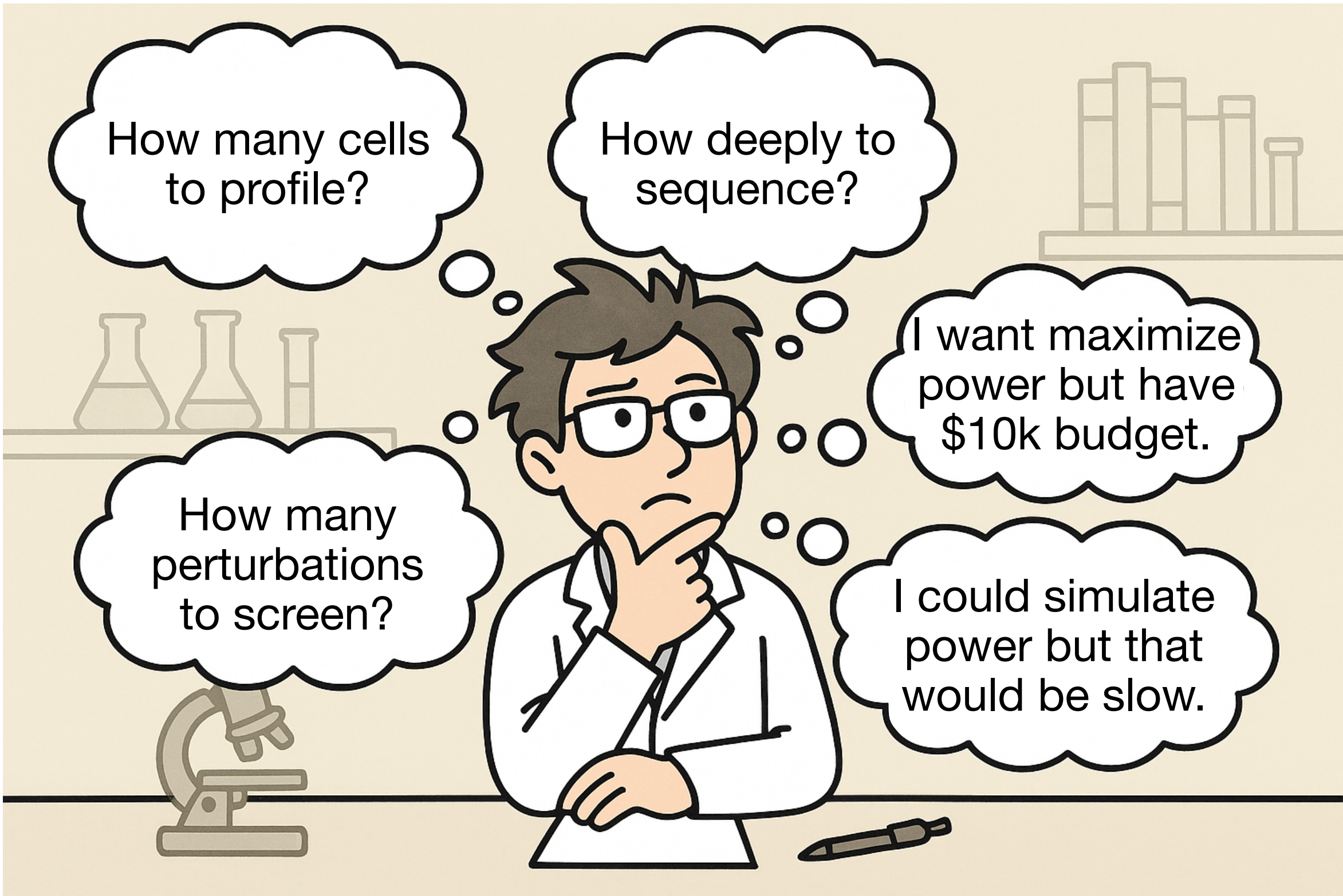


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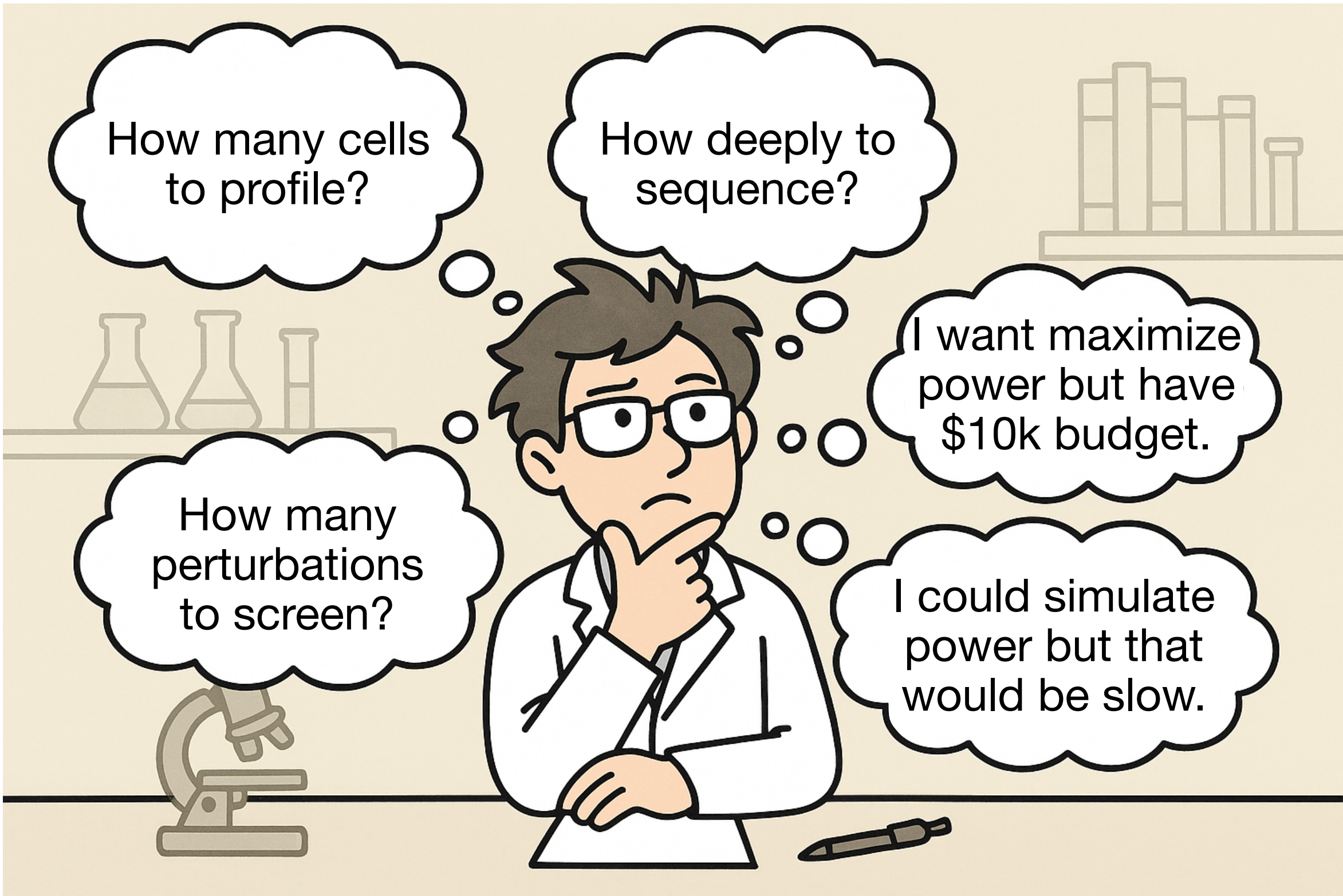
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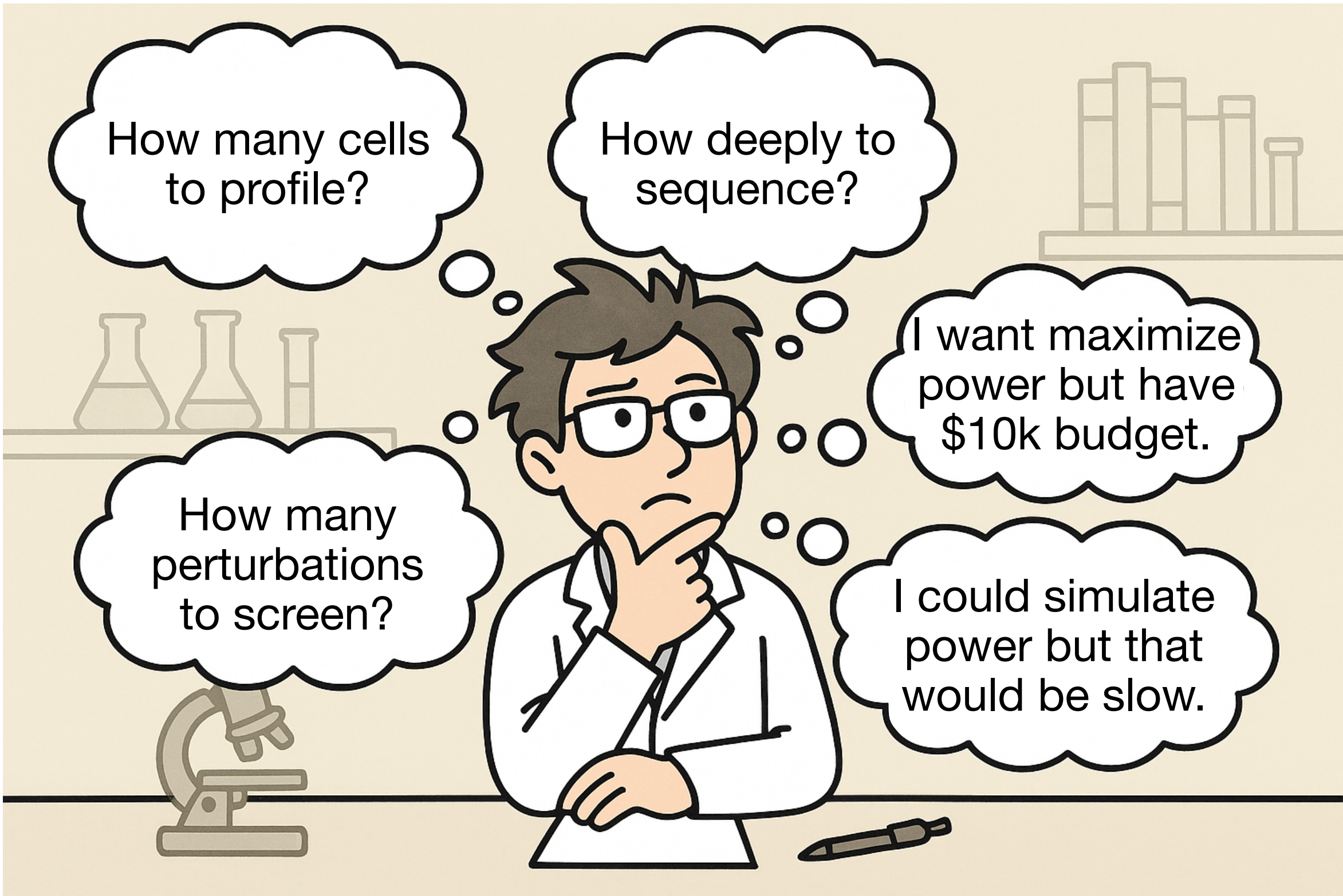
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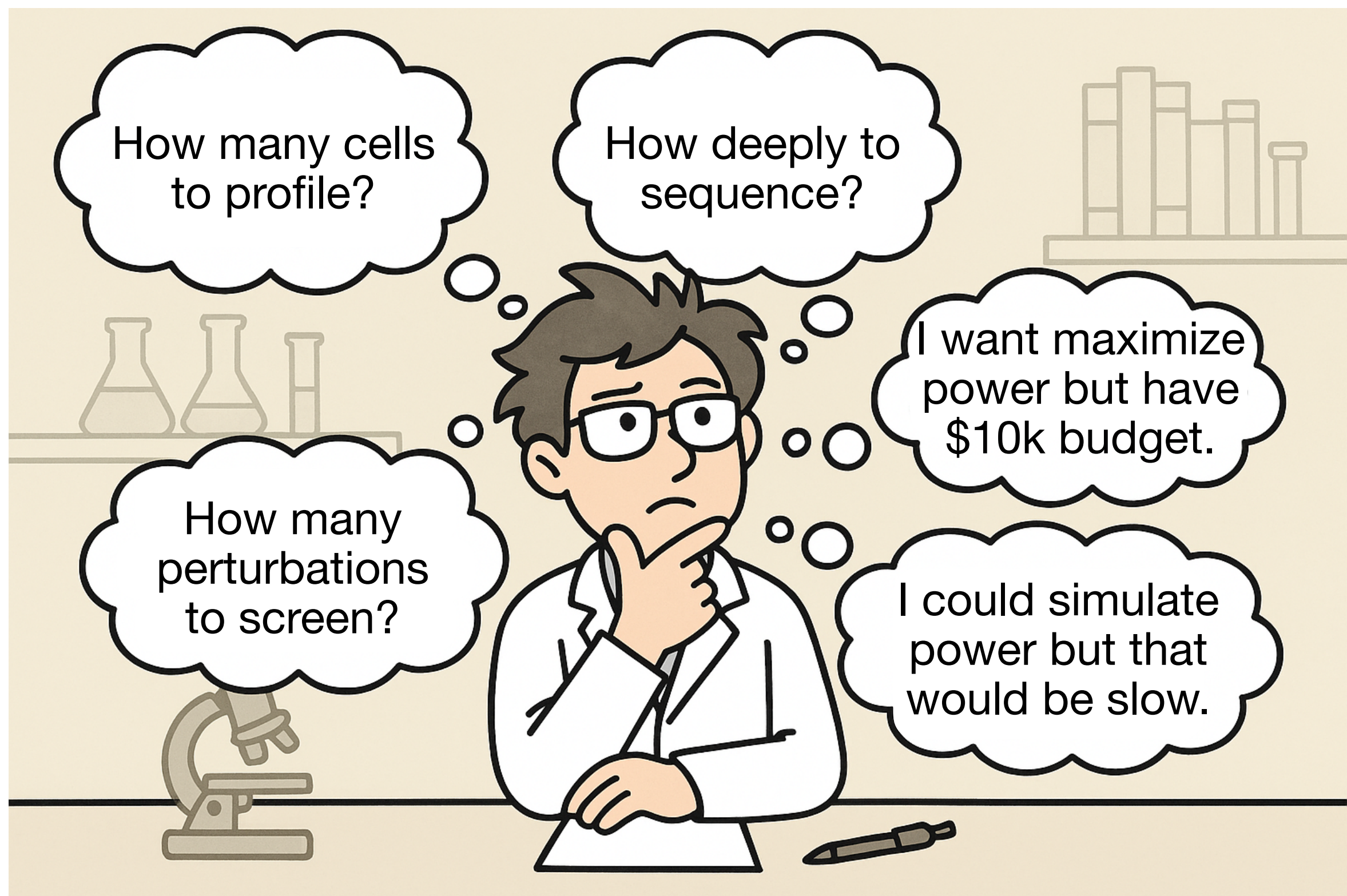
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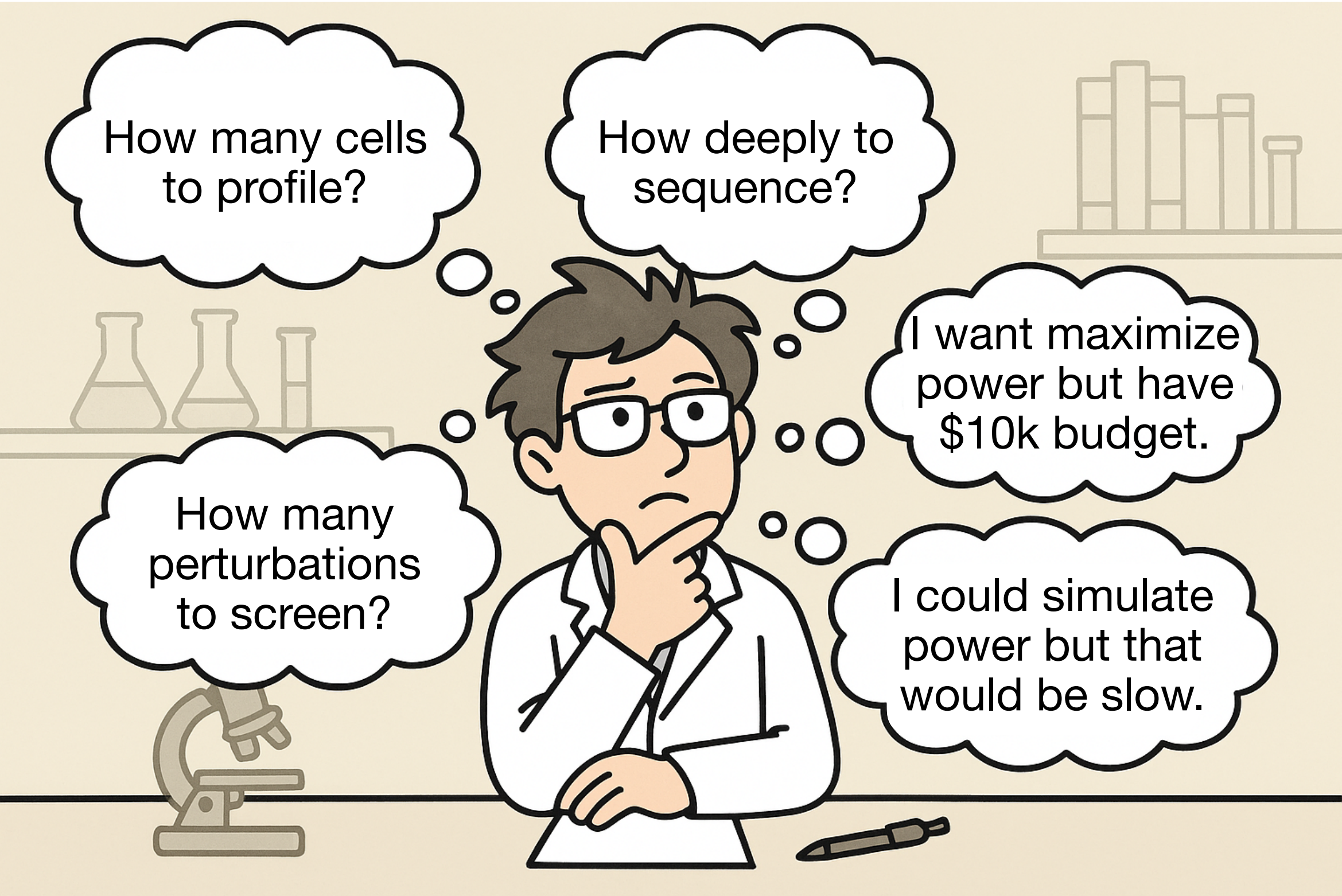
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Technical challenge

Compute power quickly and accurately.

A satisfactory solution is not yet available

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[Ray 2025]

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- Not directly applicable to Perturb-seq

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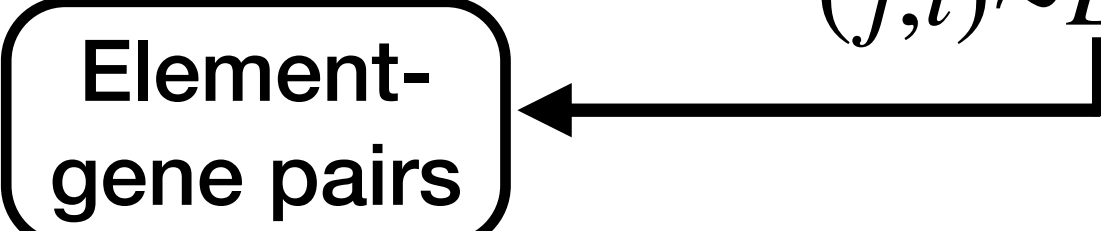
Element-gene pairs

The diagram illustrates the relationship between the expectation formula and the element-gene pairs. A rounded rectangle labeled "Element-gene pairs" is connected by an arrow to the term (j,l) in the expectation formula $\mathbb{E}_{(j,l) \sim E}[\text{power of detecting element } l \text{ on gene } j]$. The arrow points from the rectangle to the (j,l) term, indicating that the expectation is taken over the distribution of element-gene pairs.

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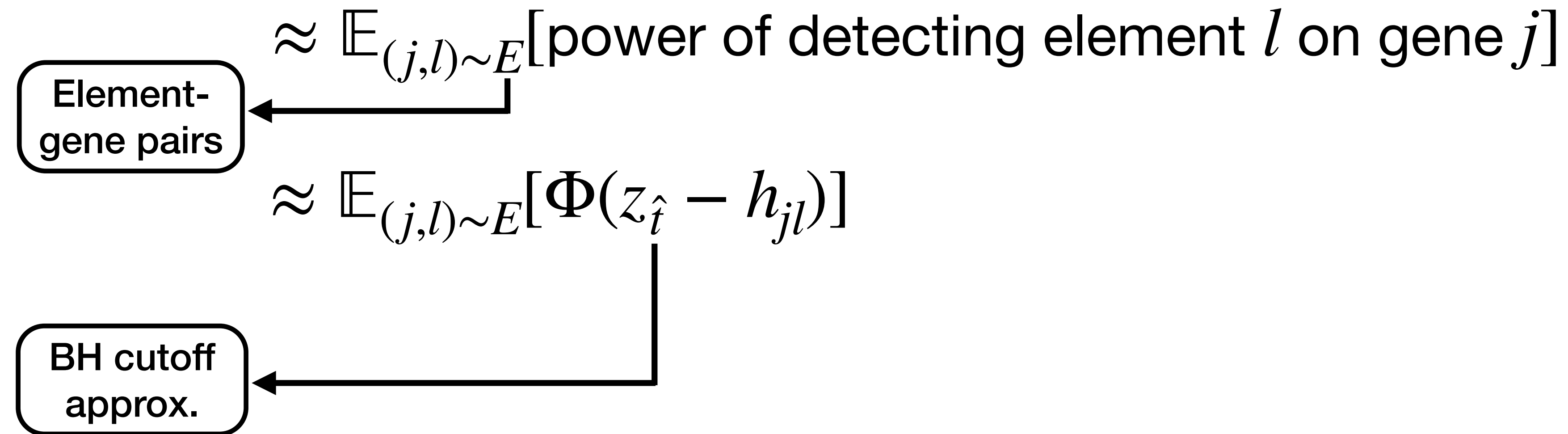
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$$\approx \mathbb{E}_{(j,l) \sim E}[\Phi(z_{\hat{t}} - h_{jl})]$$


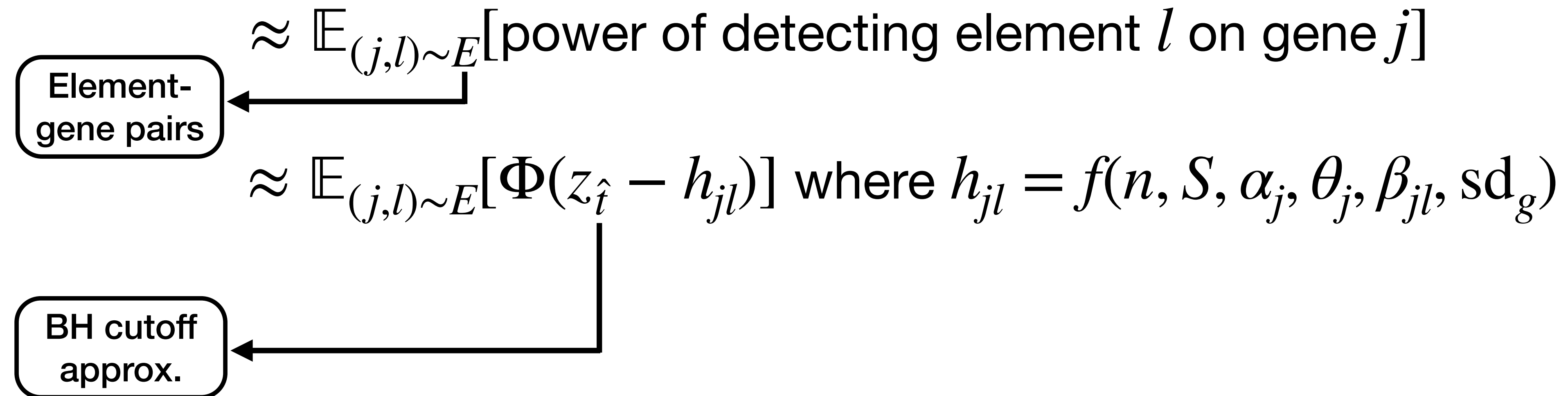
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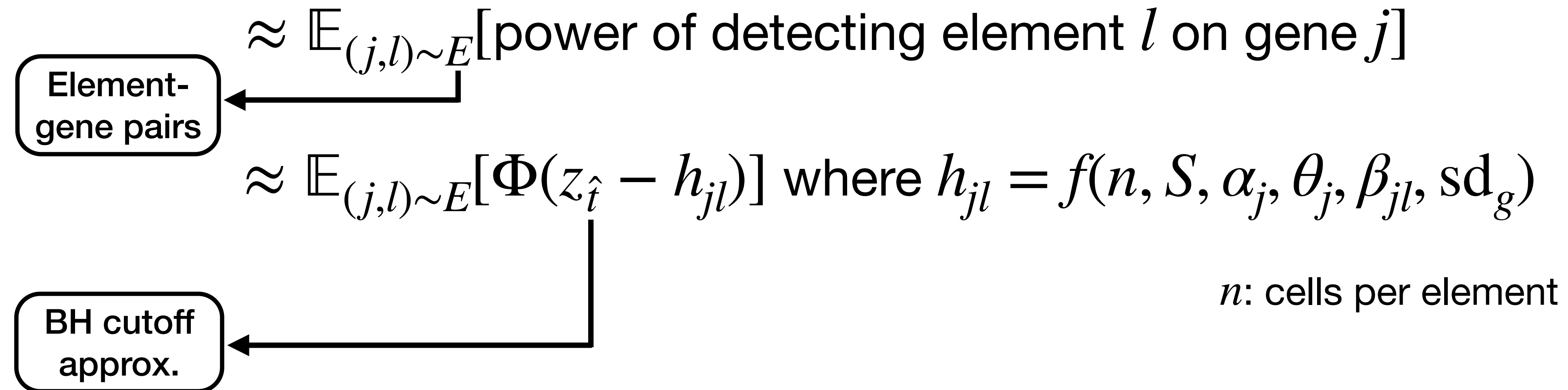
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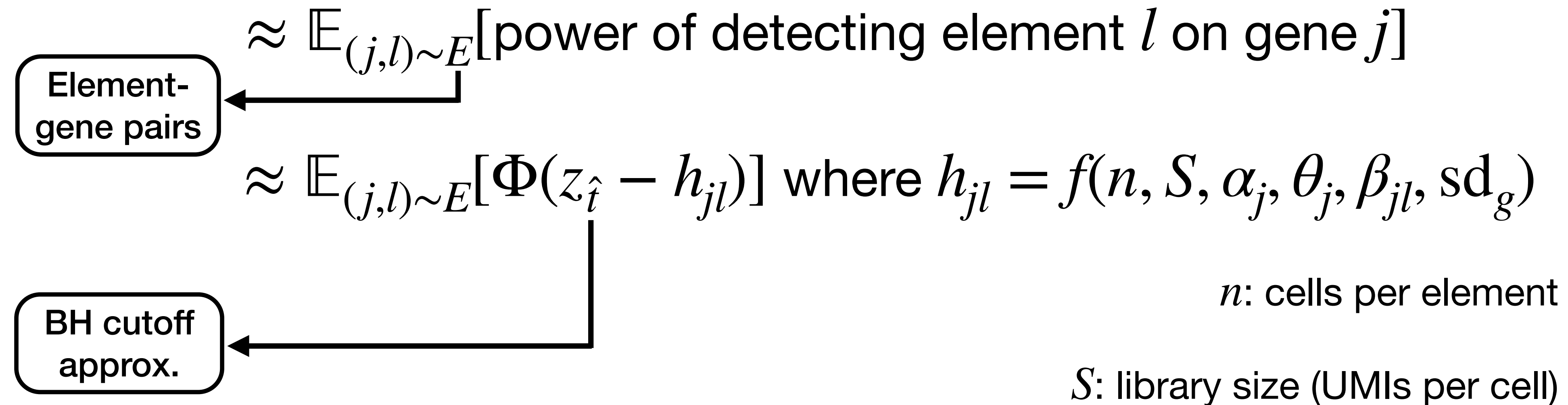
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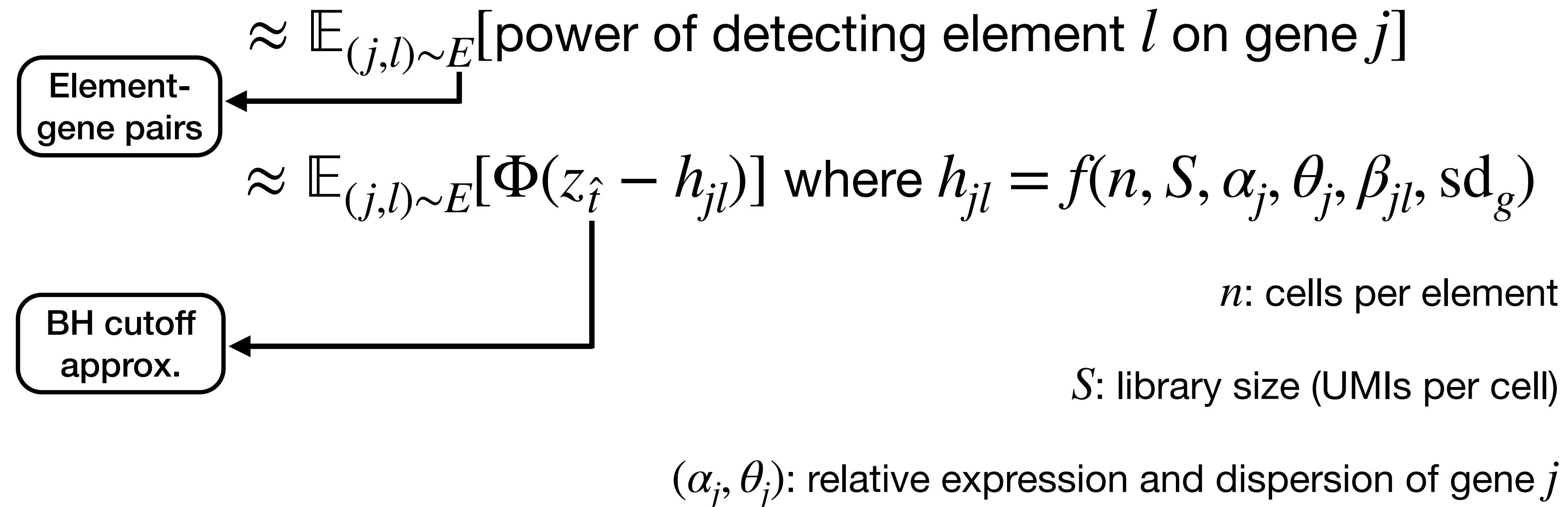
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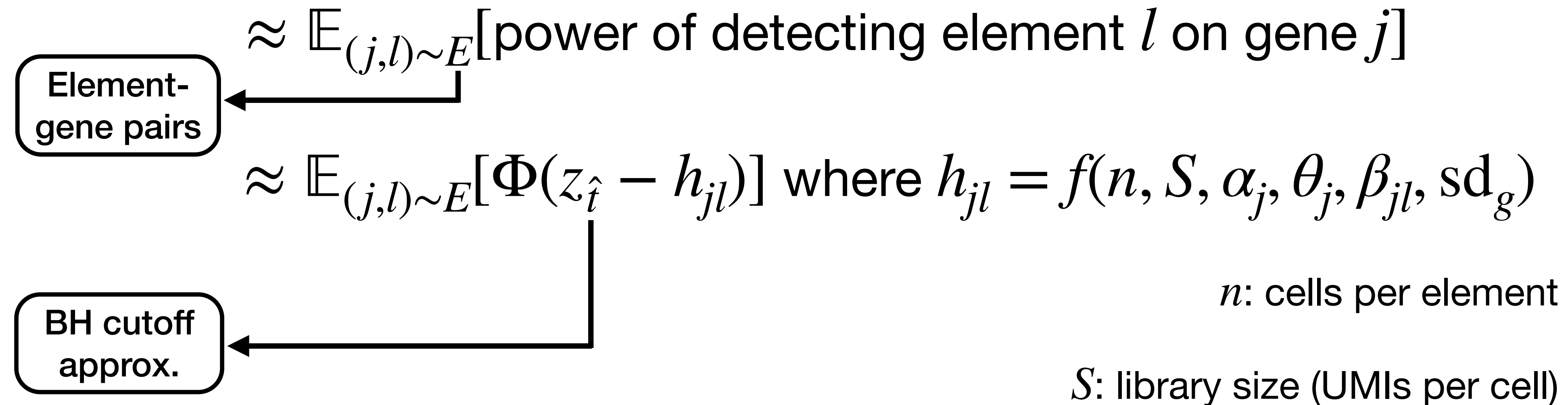
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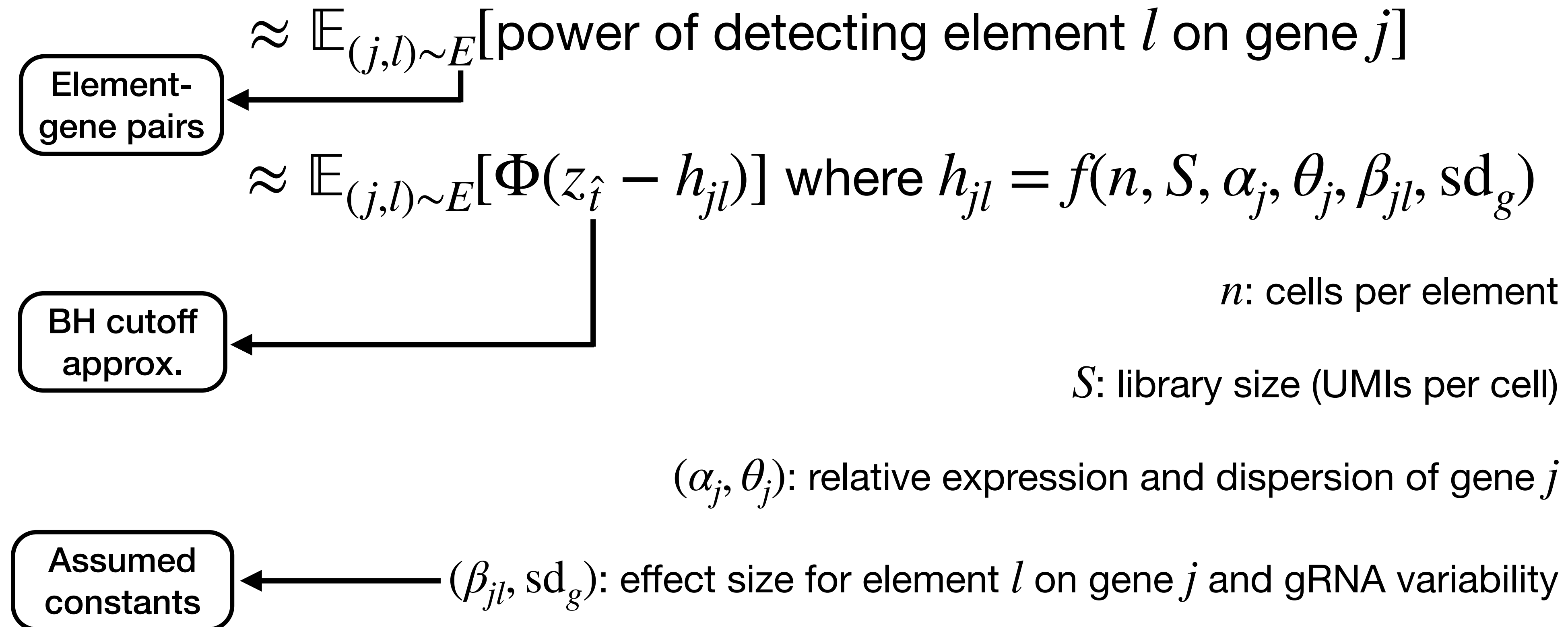


(α_j, θ_j) : relative expression and dispersion of gene j

$(\beta_{jl}, \text{sd}_g)$: effect size for element l on gene j and gRNA variability

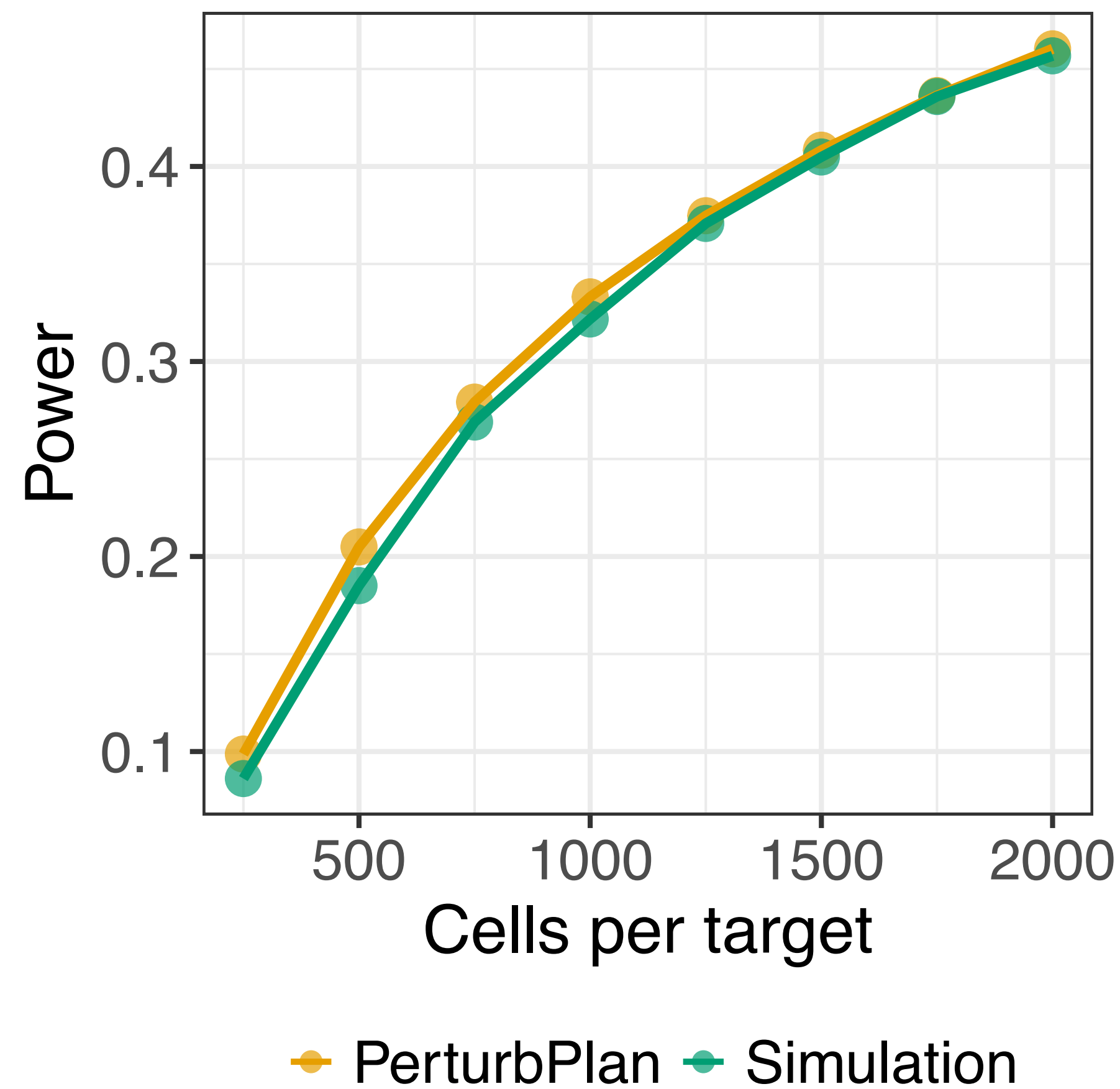
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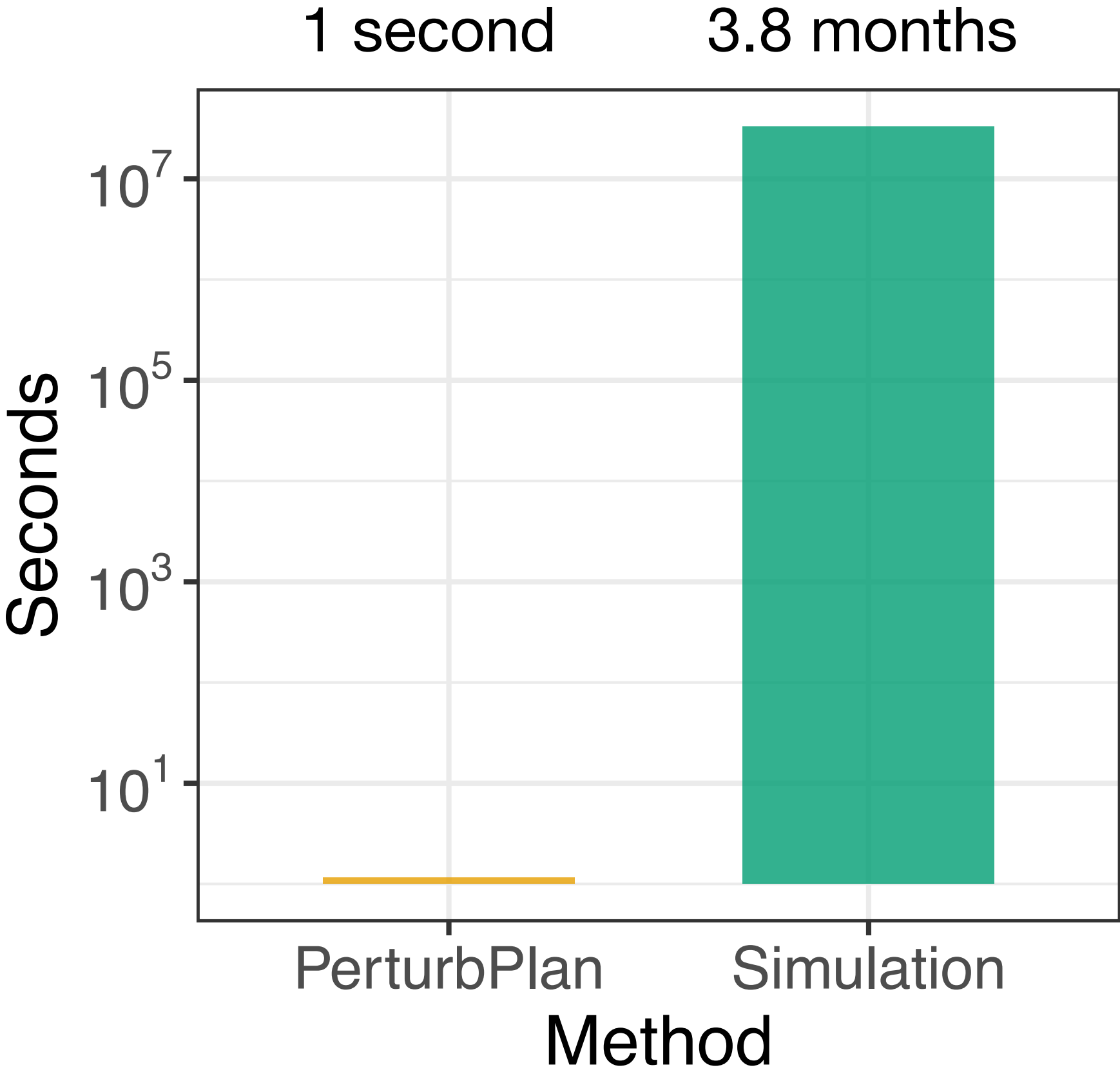
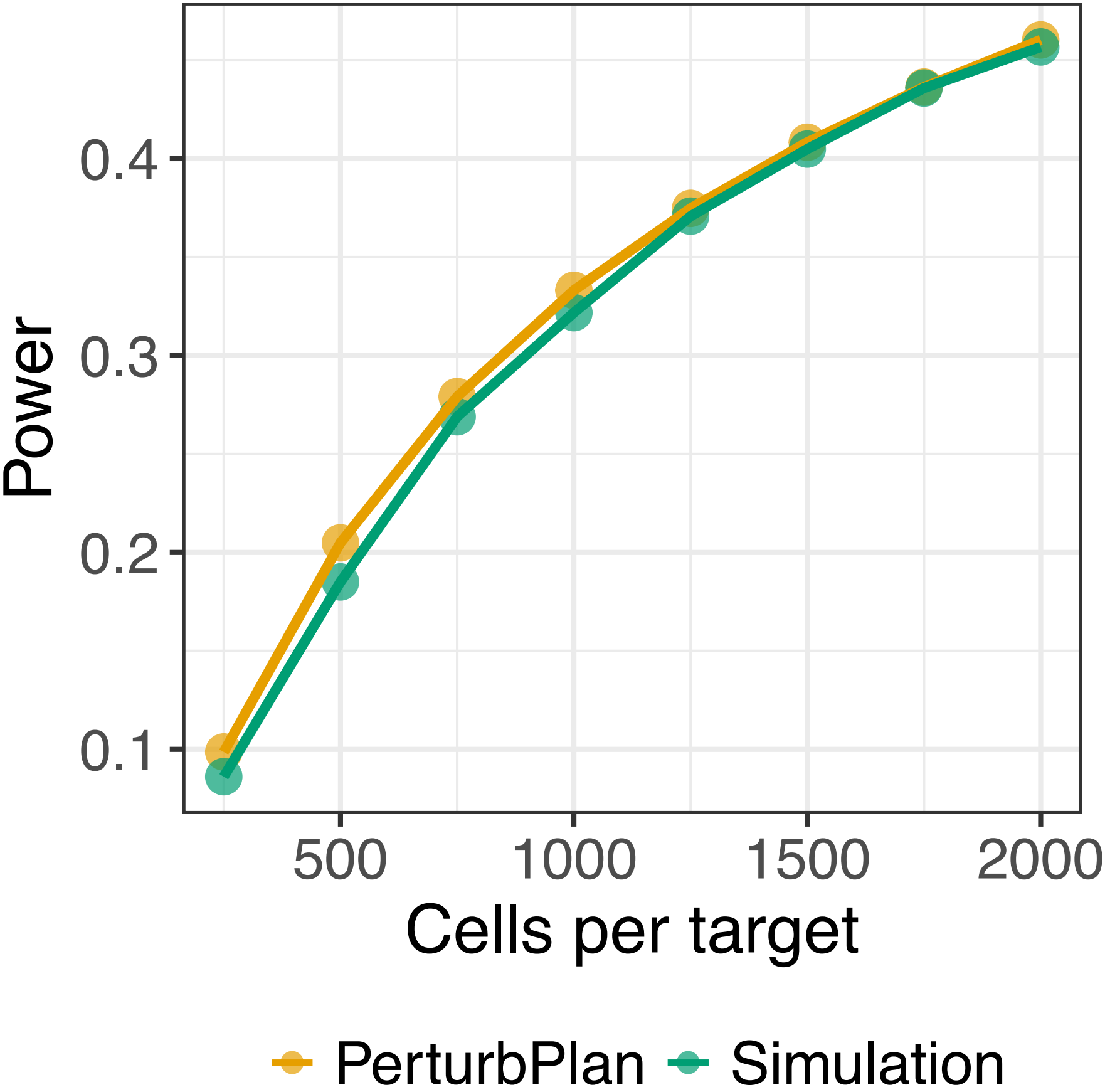


Formula matches simulations

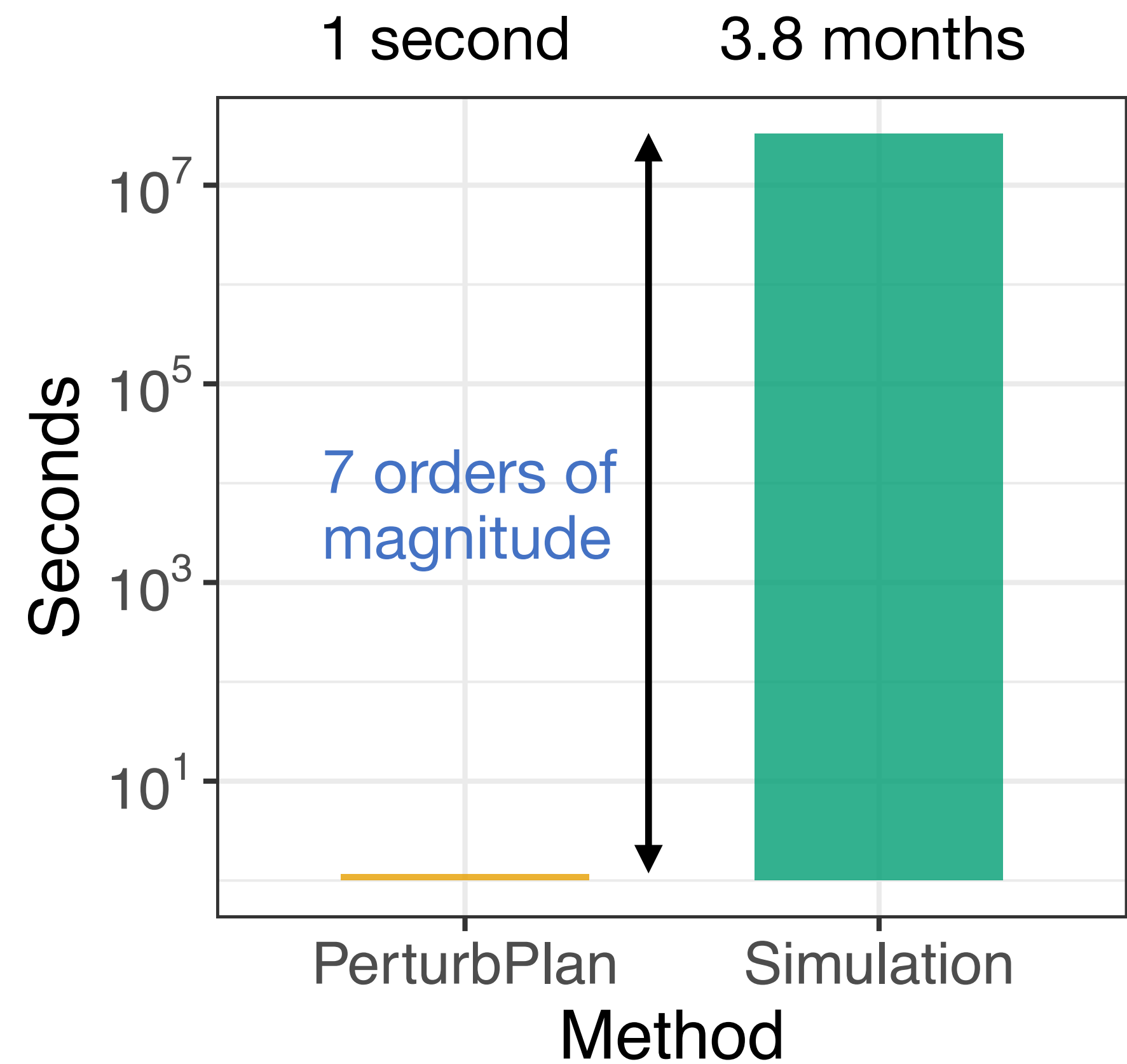
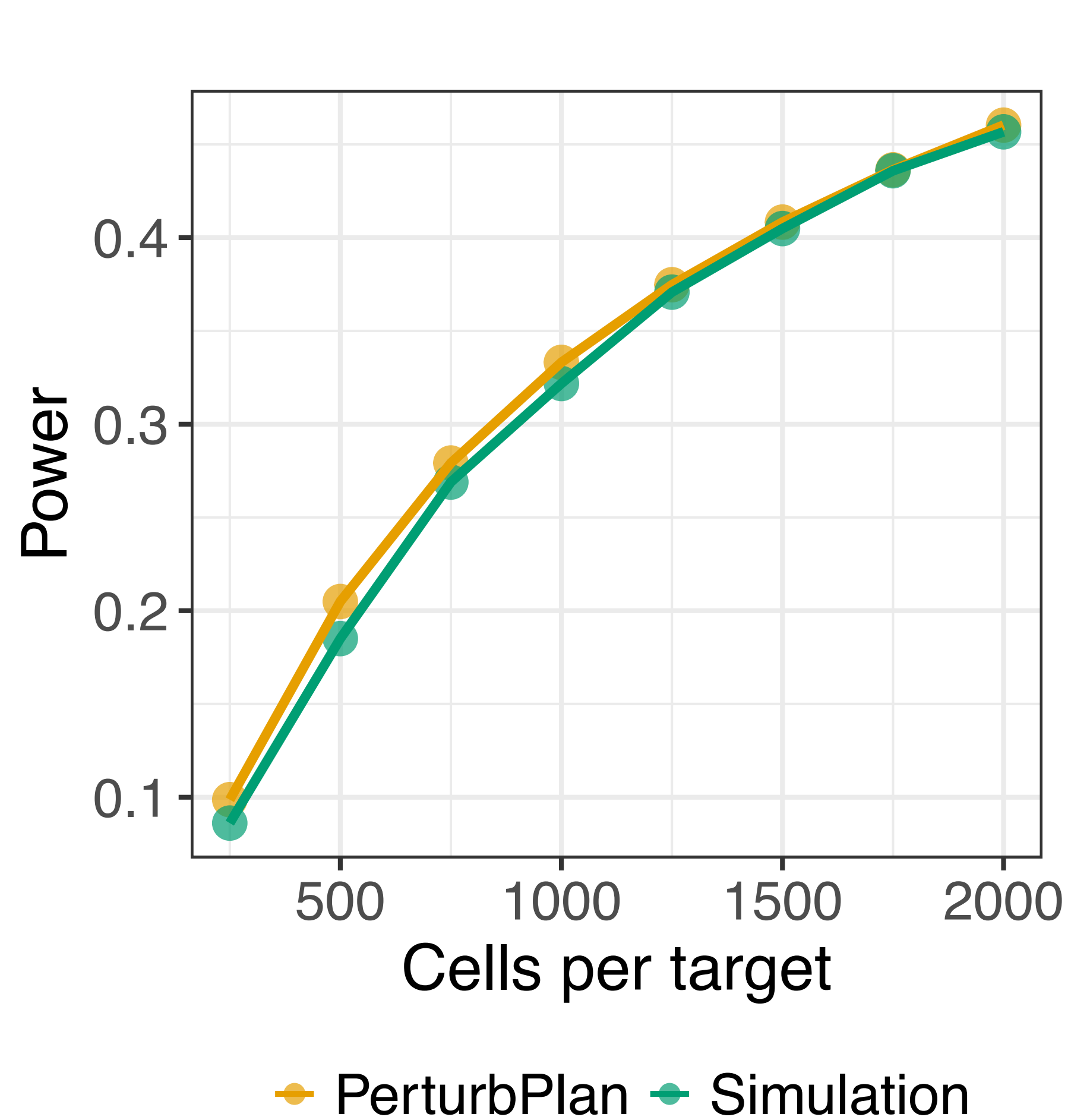
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Formula matches simulations but is much faster



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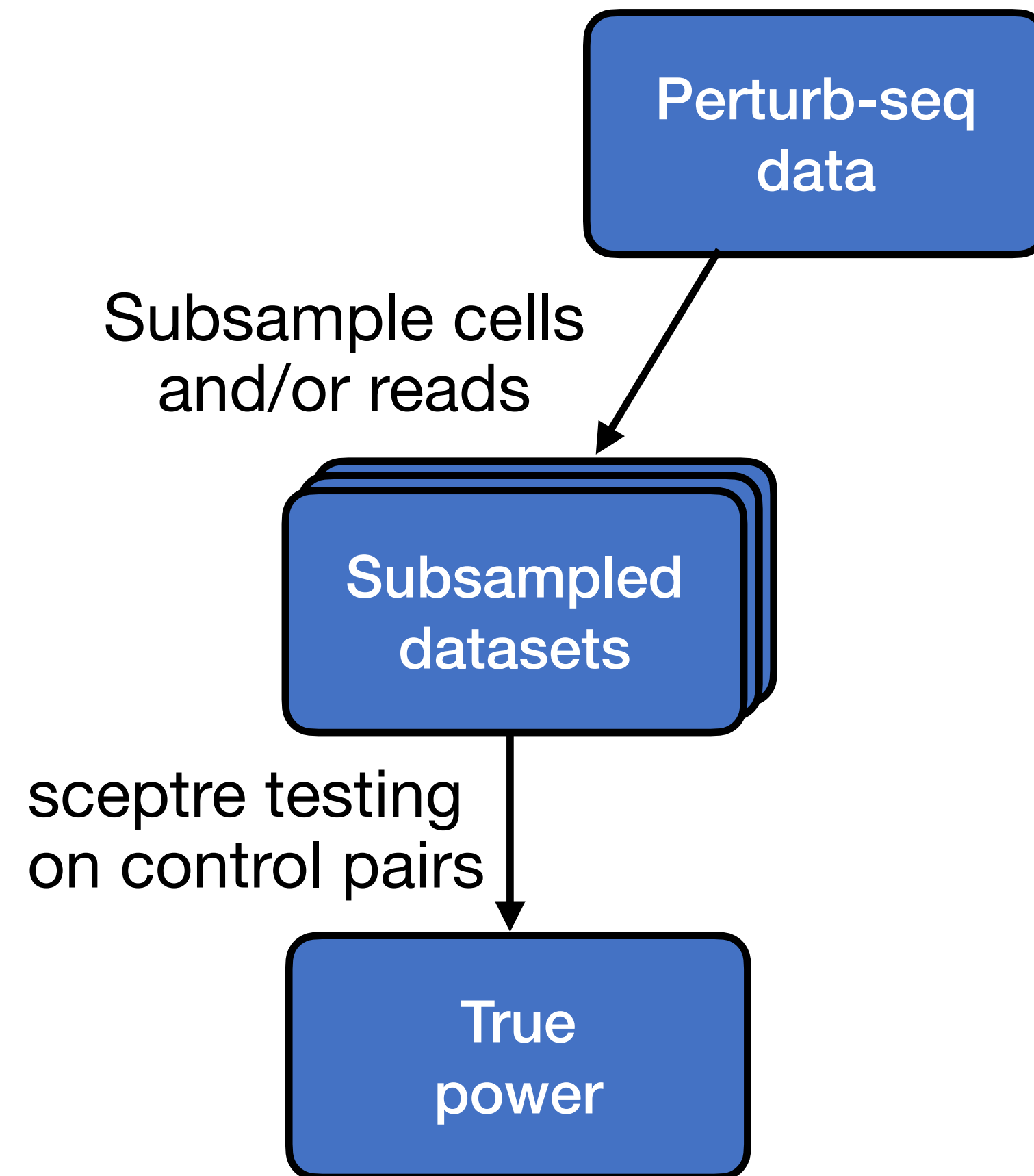
PerturbPlan tracks downsampling on real data

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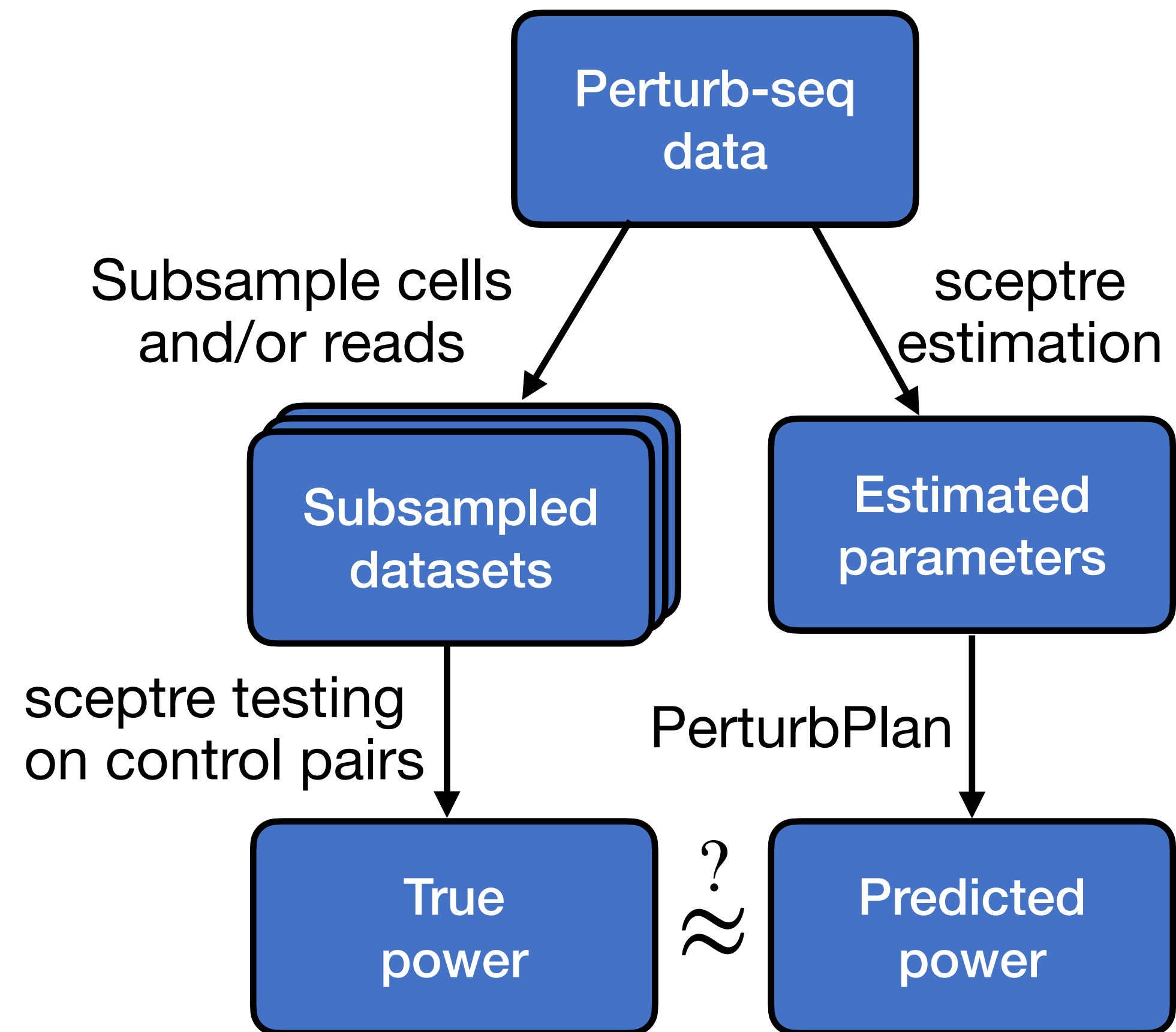


Perturb-seq
data

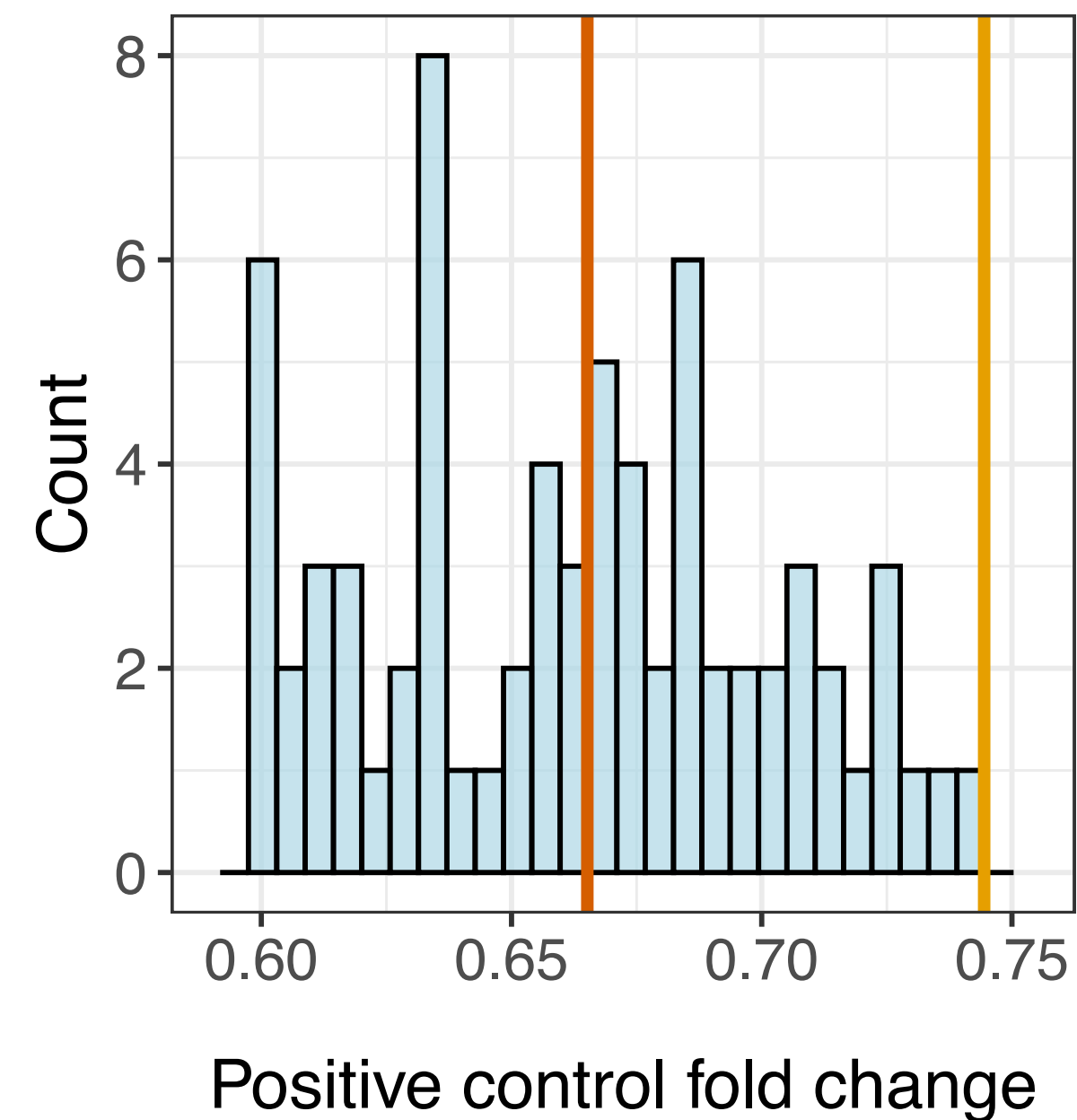
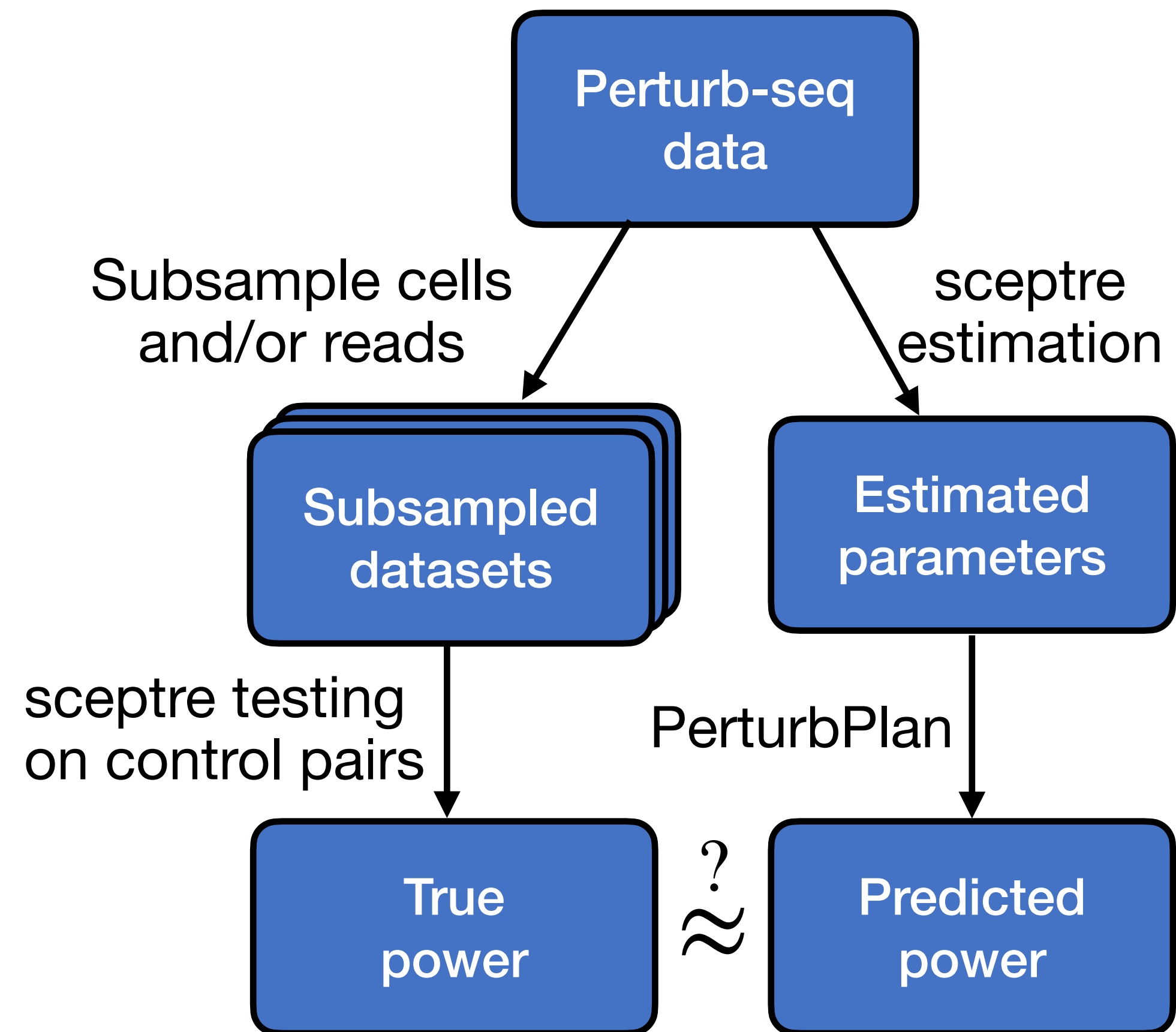
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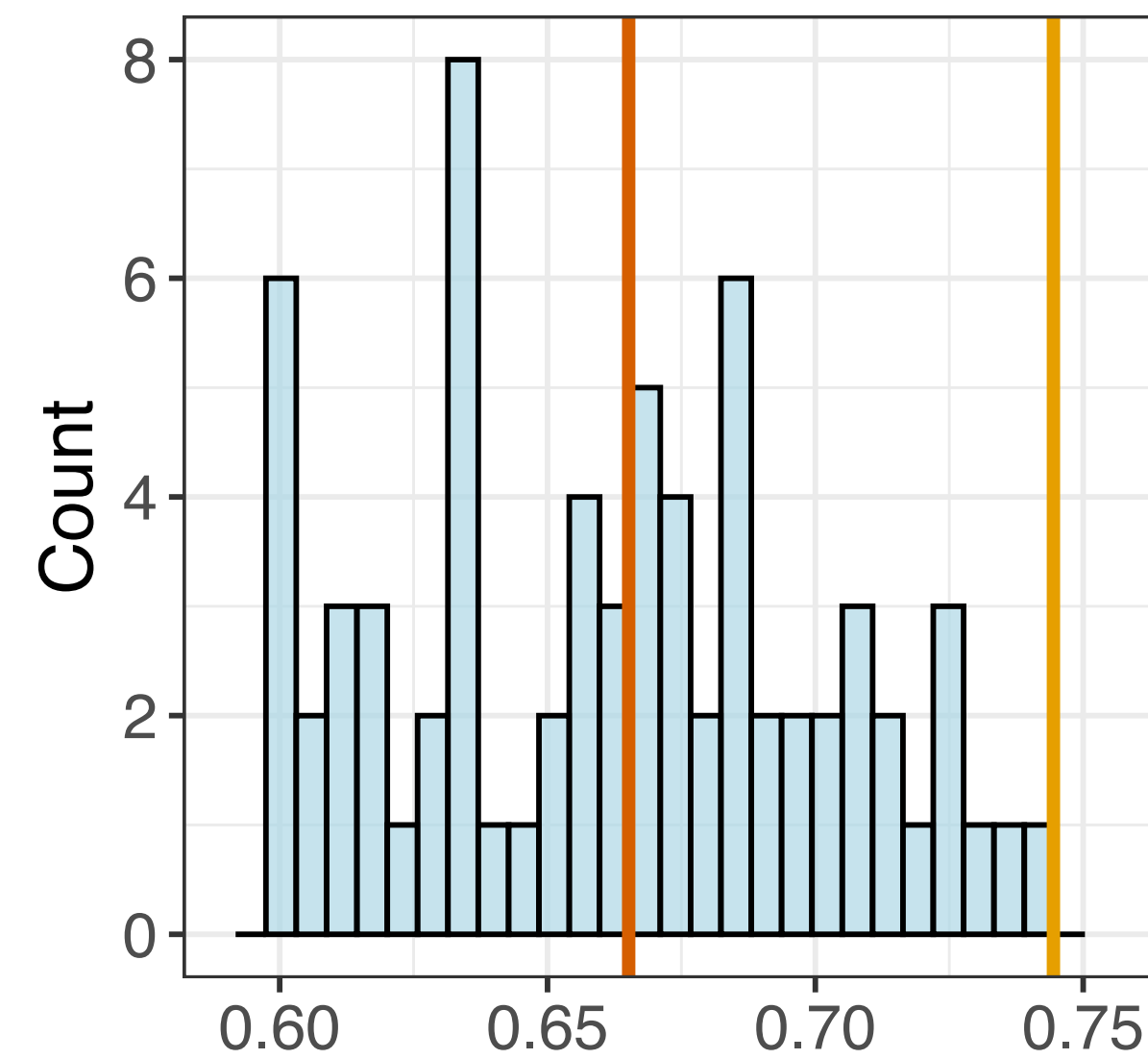
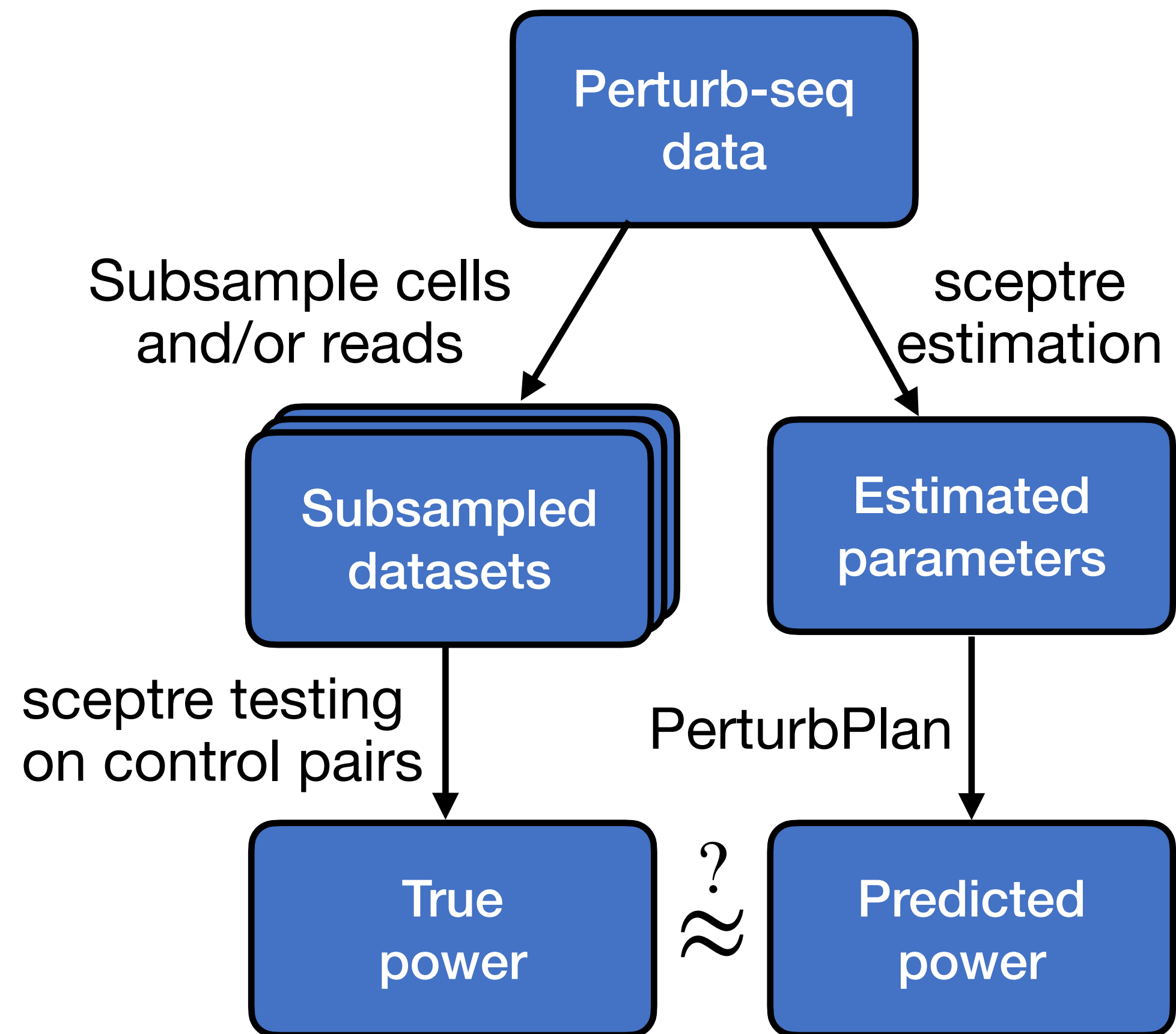
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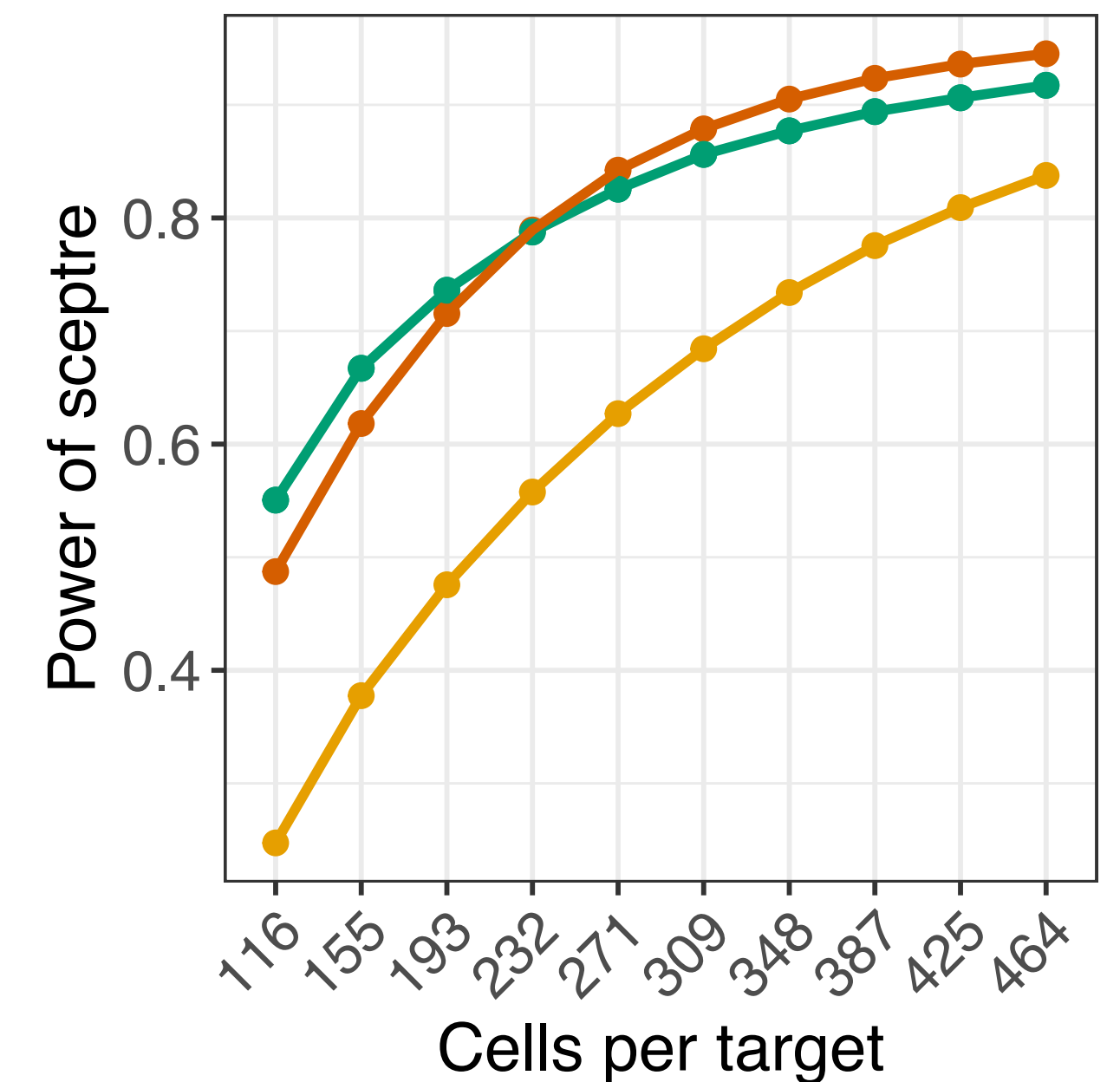
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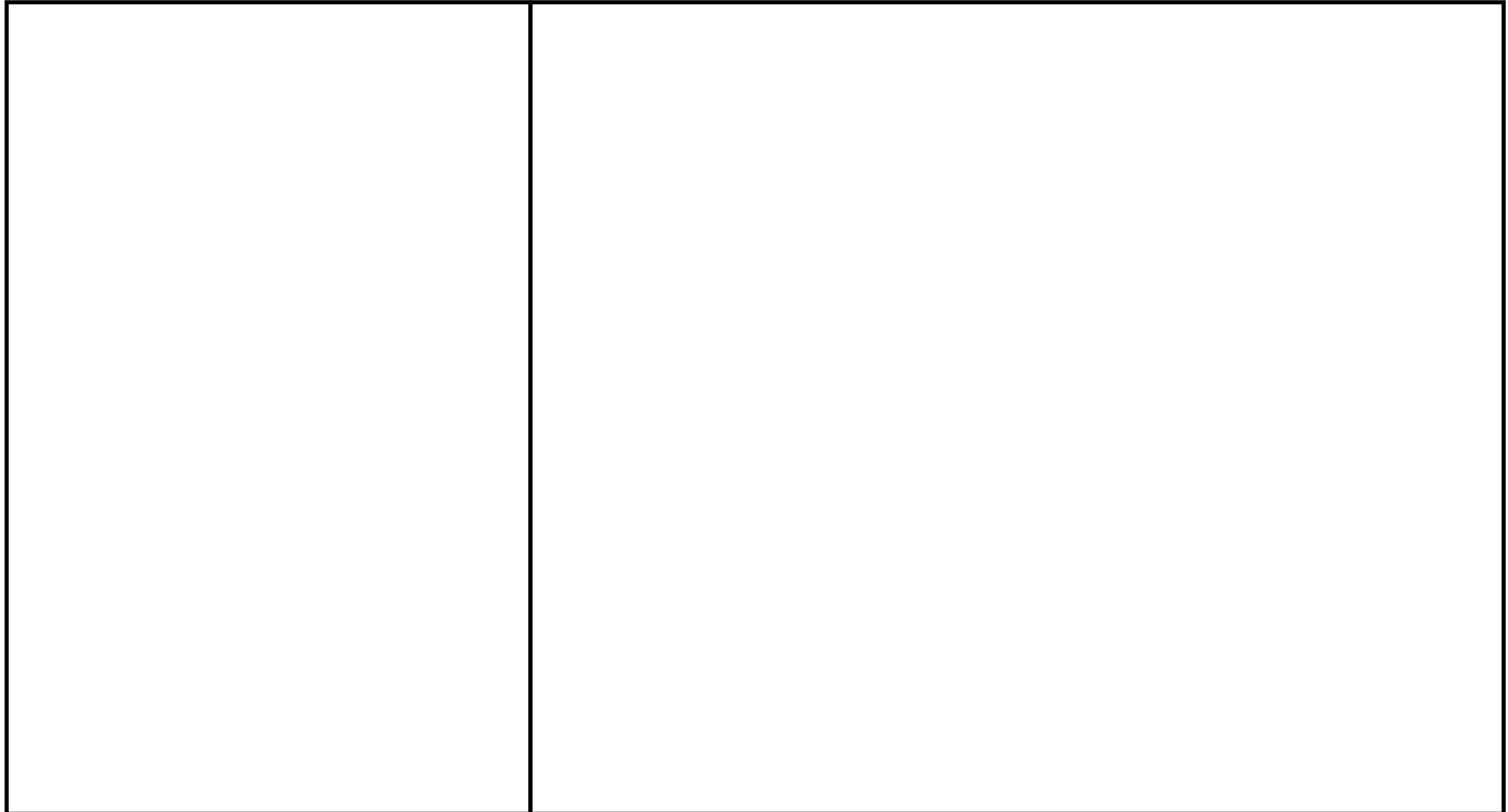
Positive control fold change



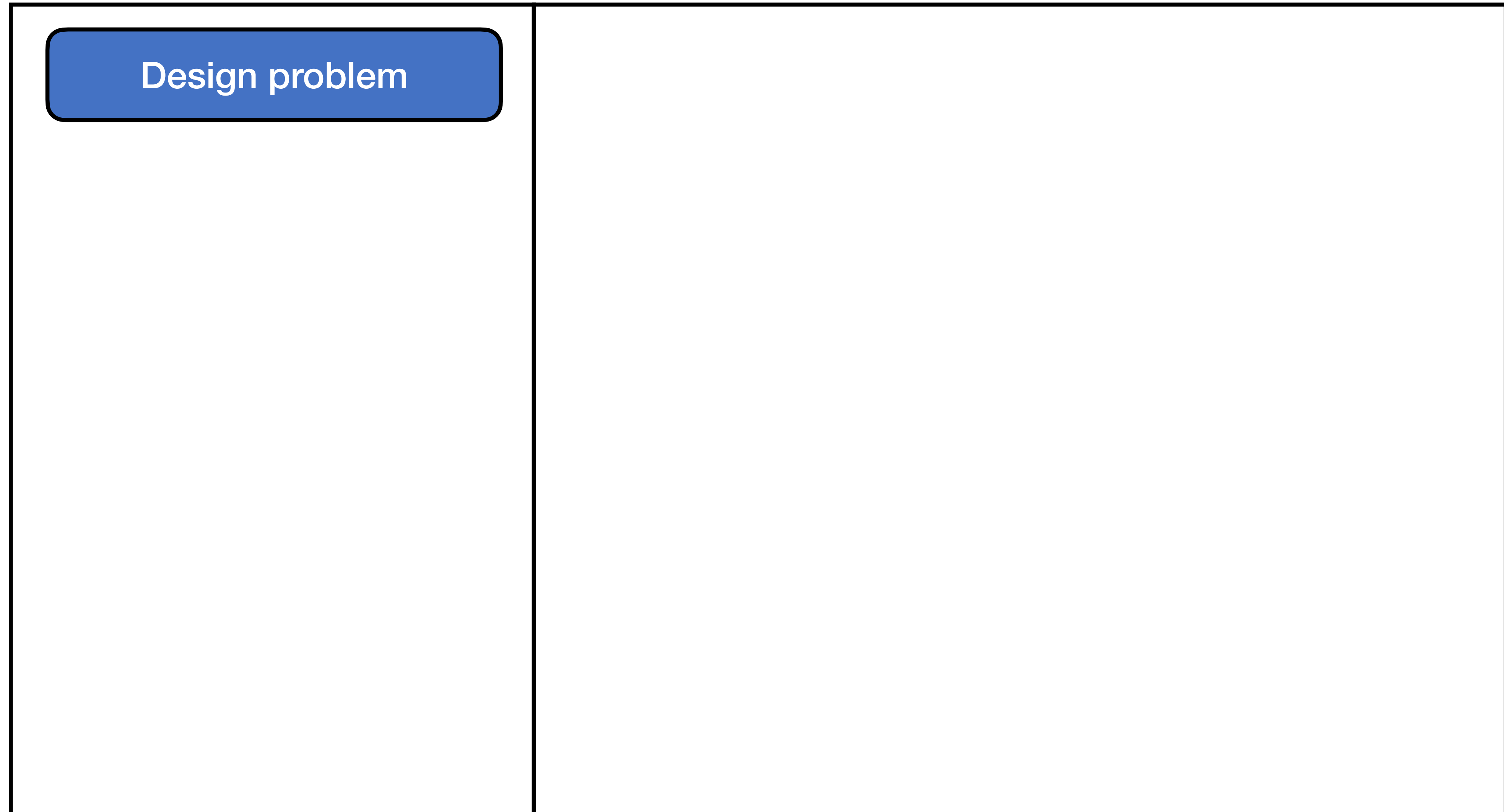
Downsampling PerturbPlan (max FC) PerturbPlan (median FC)

PerturbPlan web app facilitates interactive design

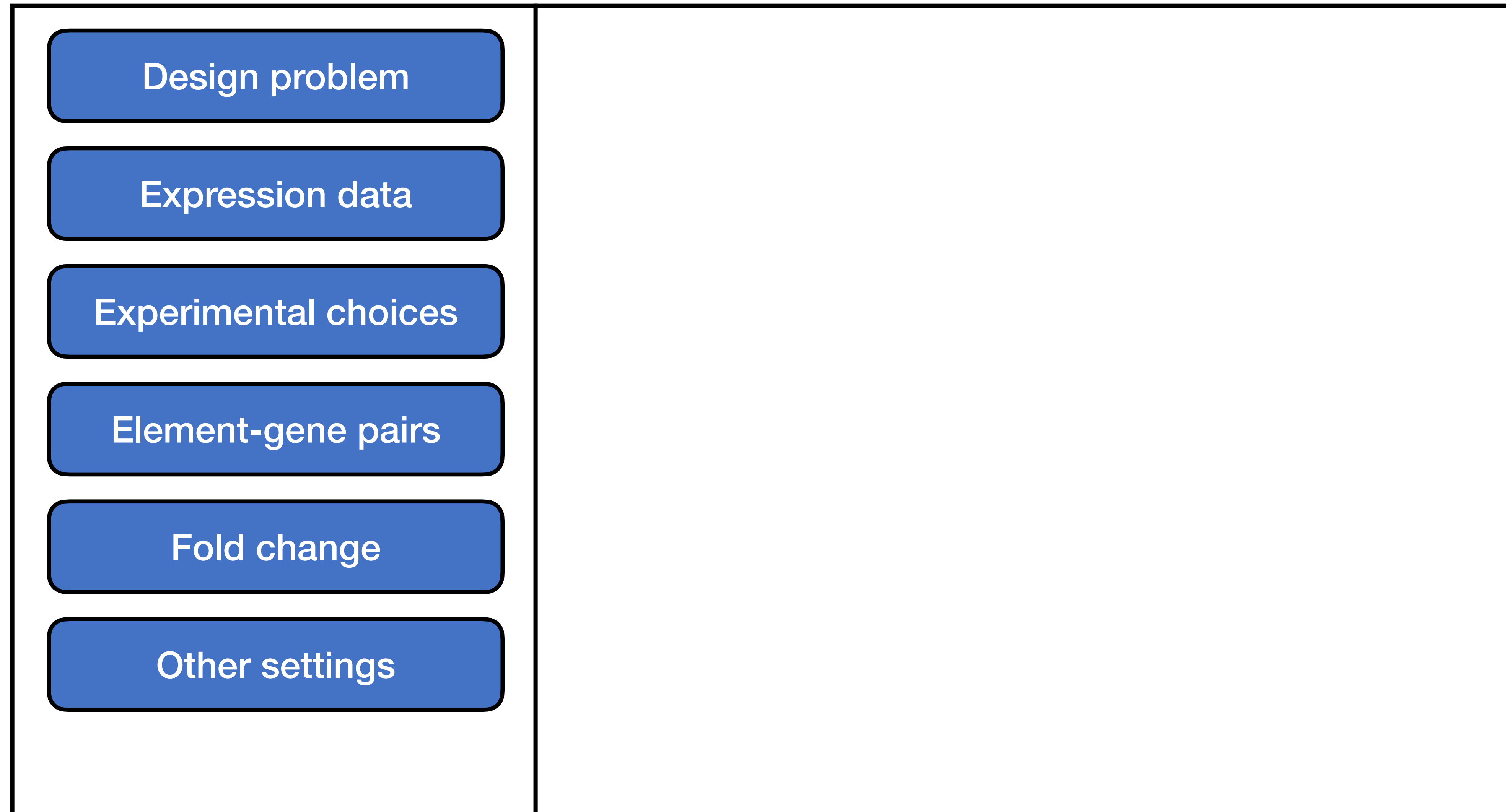
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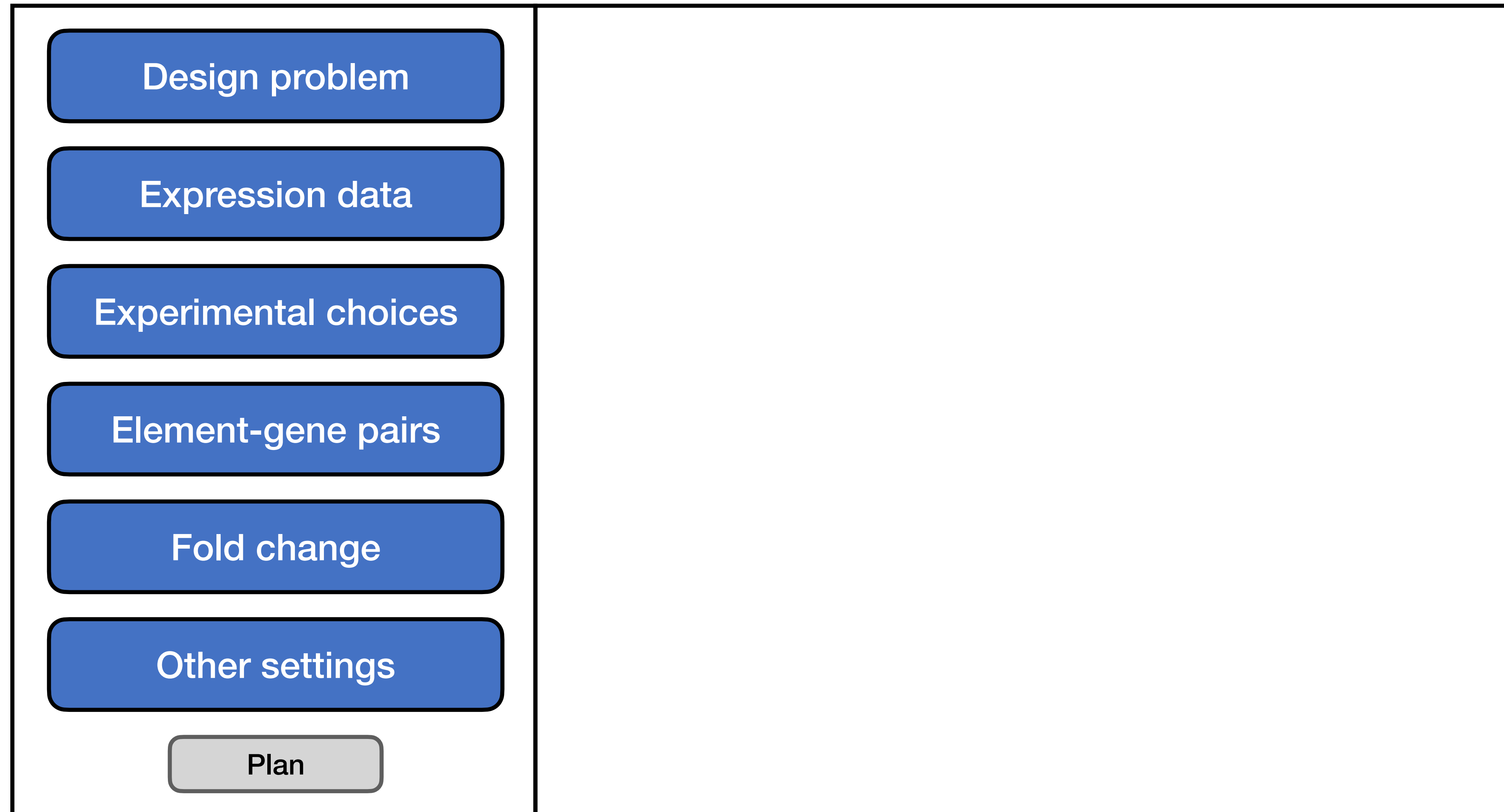
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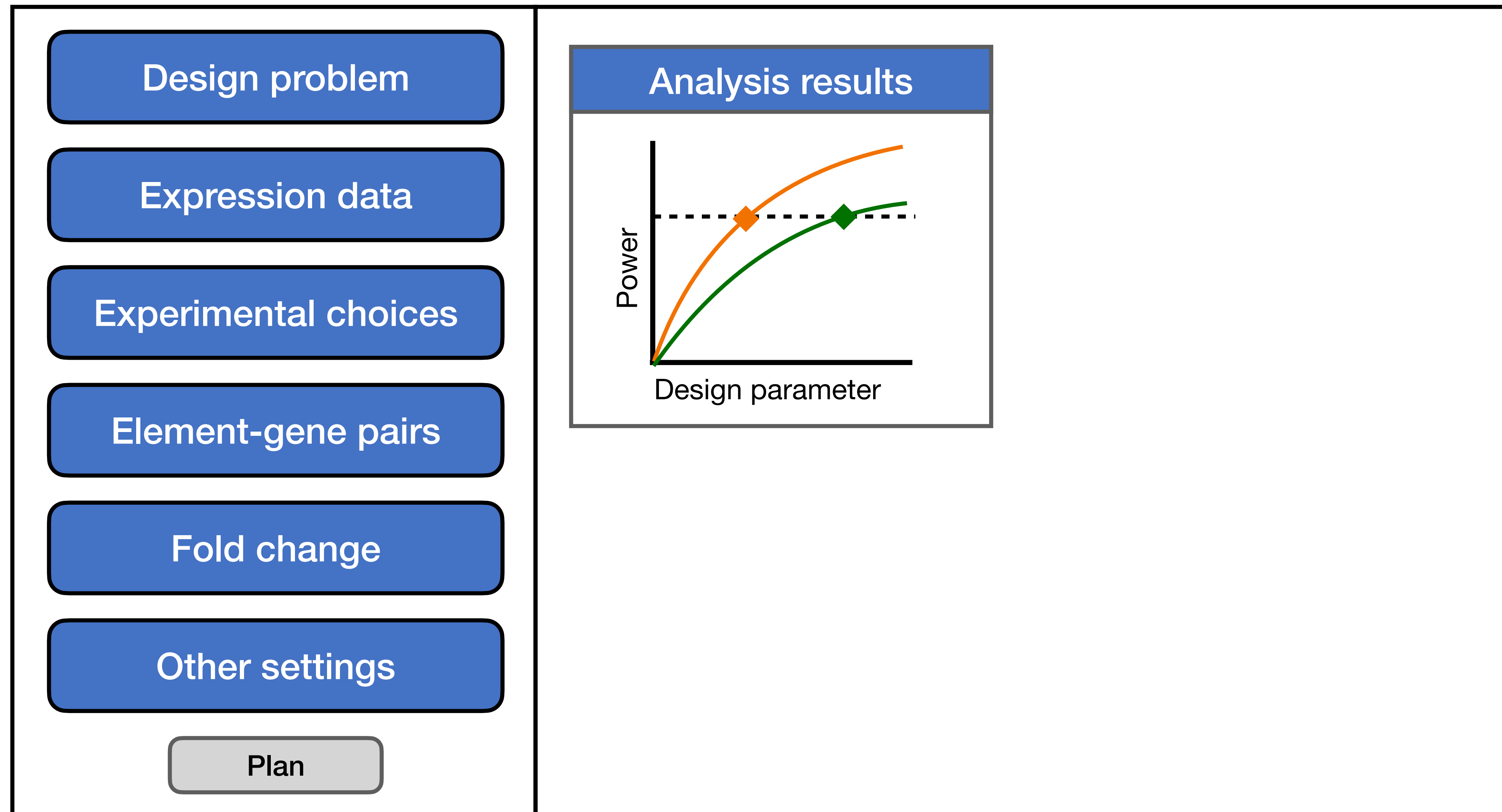
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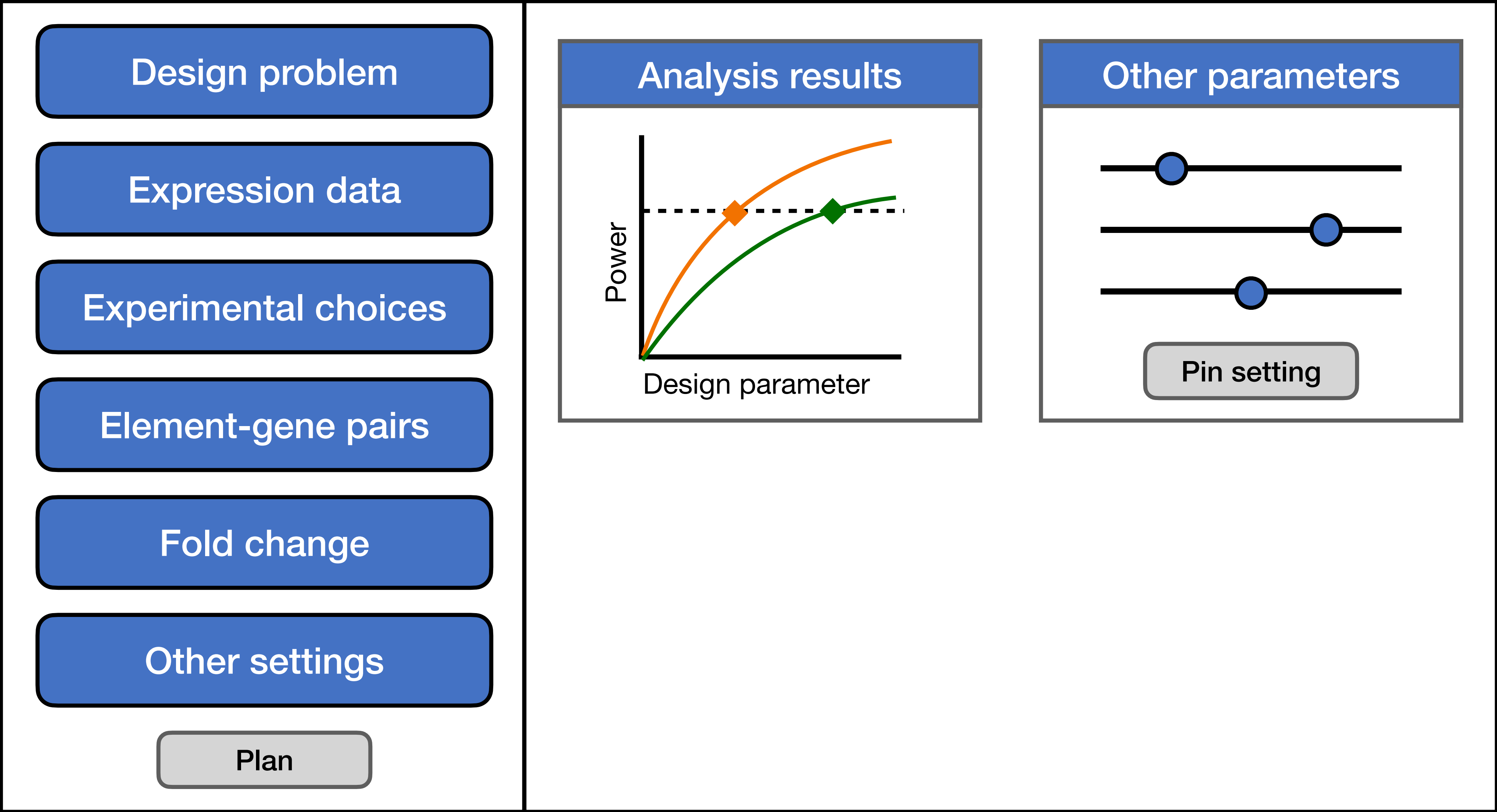
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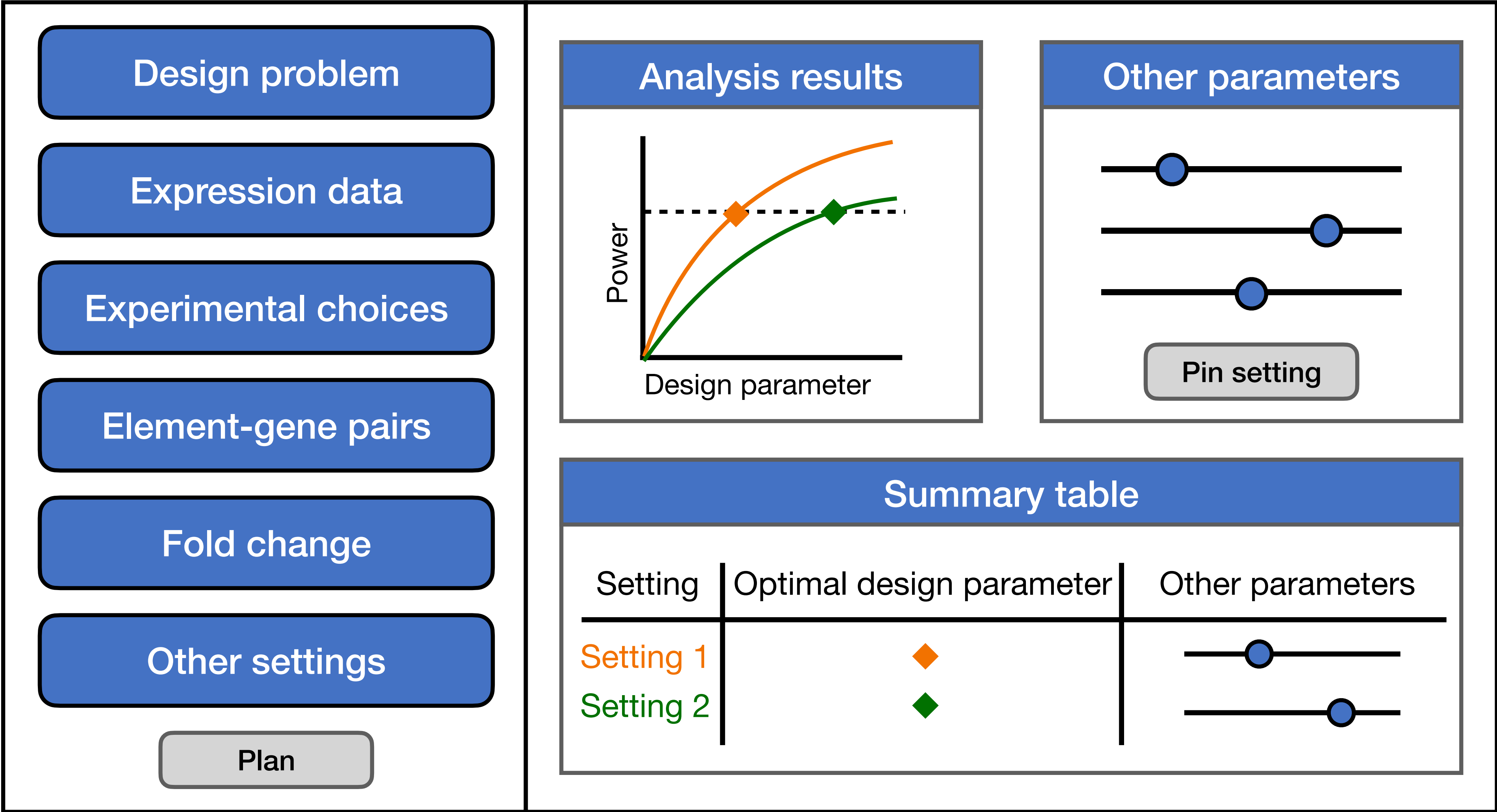
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PerturbPlan addresses 11 design problems

Minimize

- Cells per target
- Reads per cell
- Expression threshold
- Fold change
- ✓ Total cost

Example: Find the minimum cost for which power is at least 80%, varying cells per target and reads per cell.

Subject to constraints on

- ✓ Power
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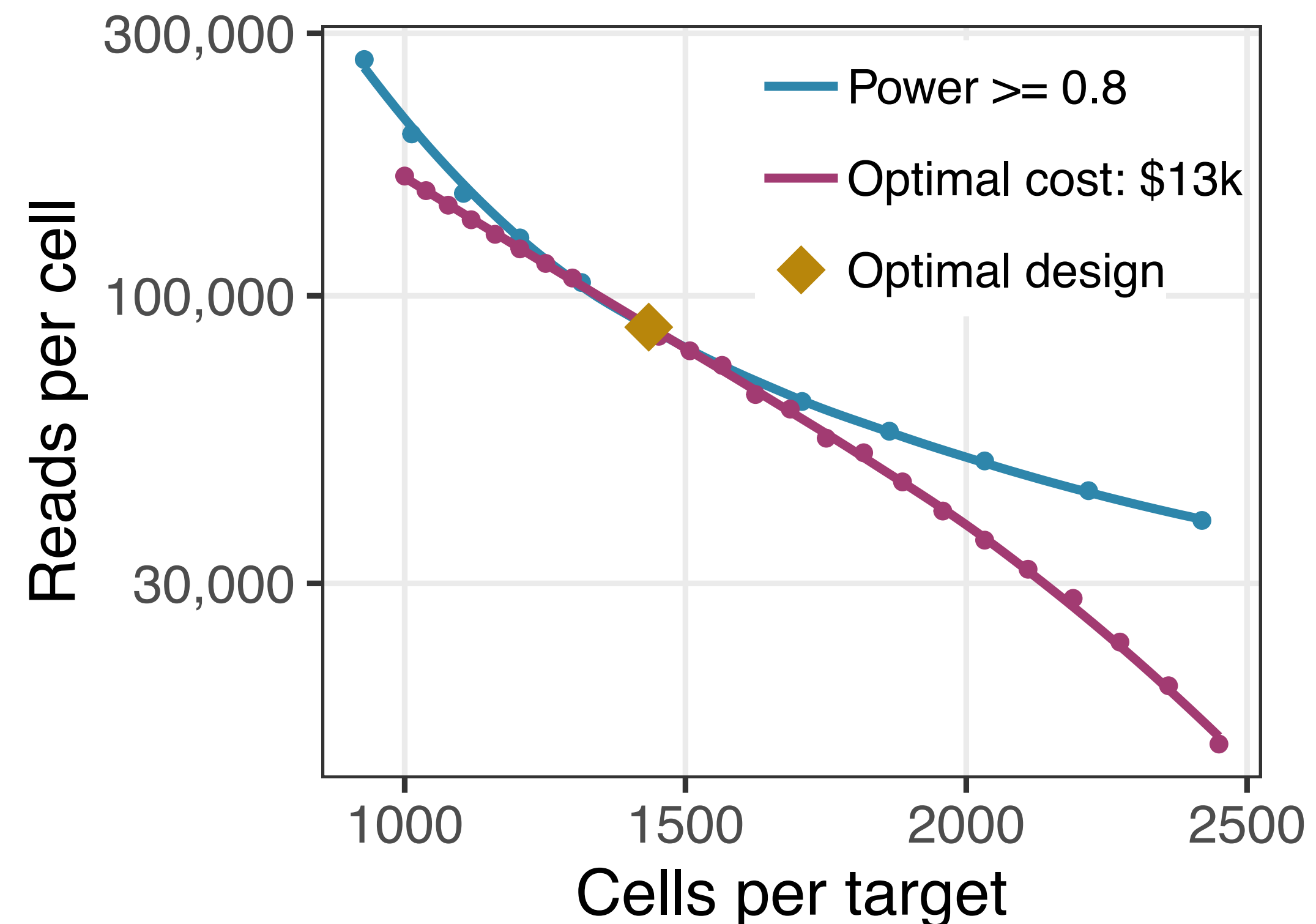
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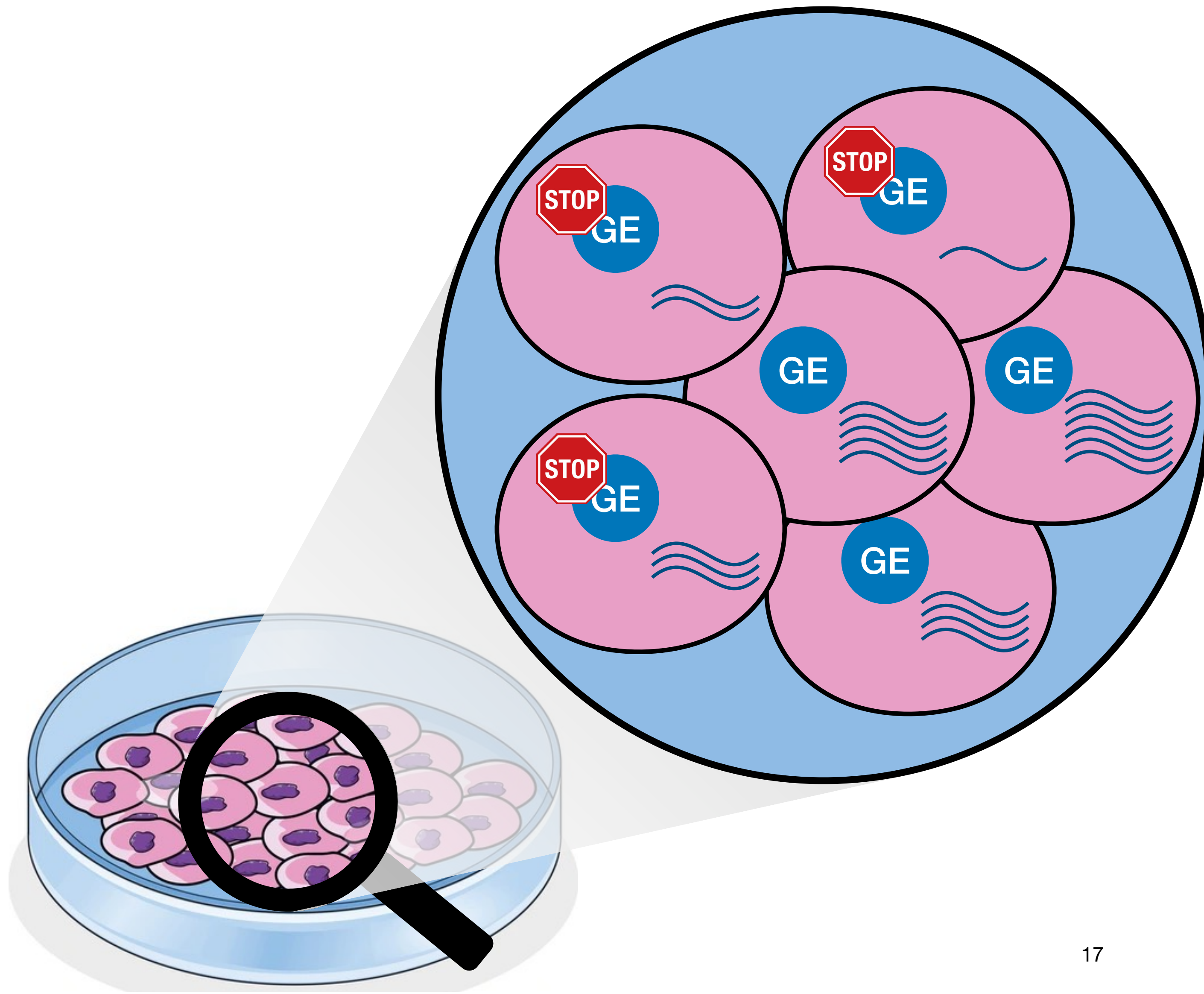
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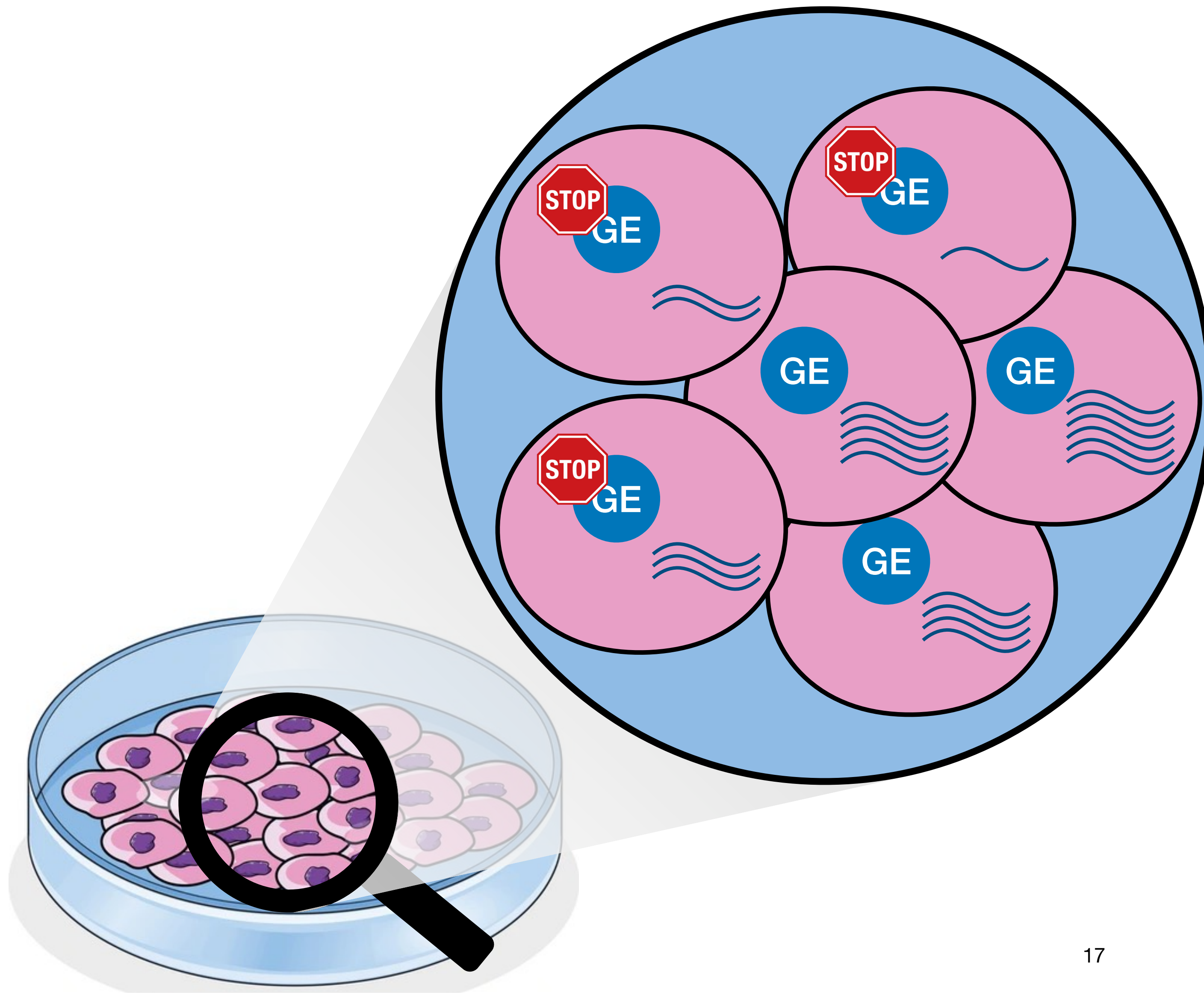
Part II: spaCRT

Perturbation-gene association testing



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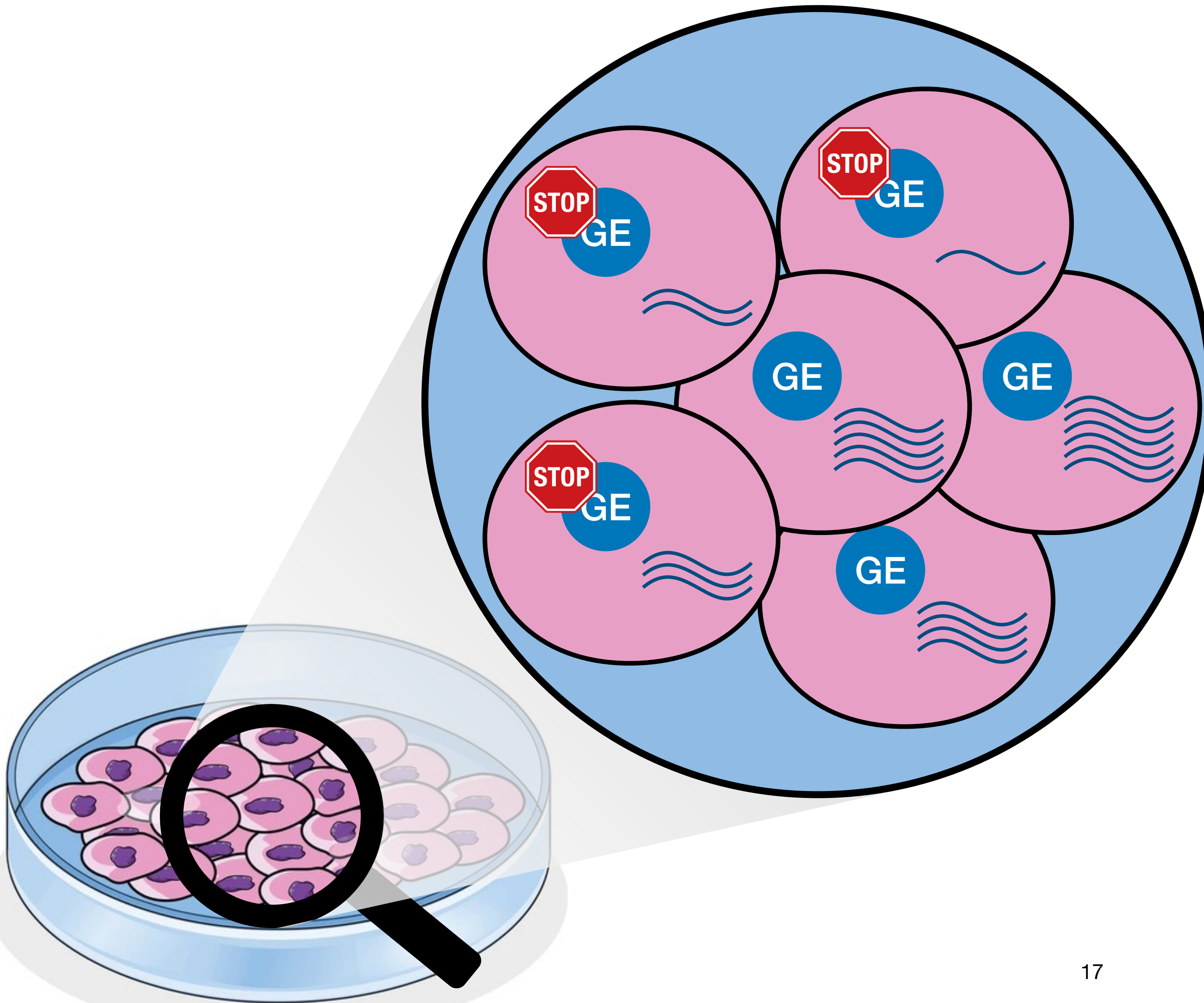
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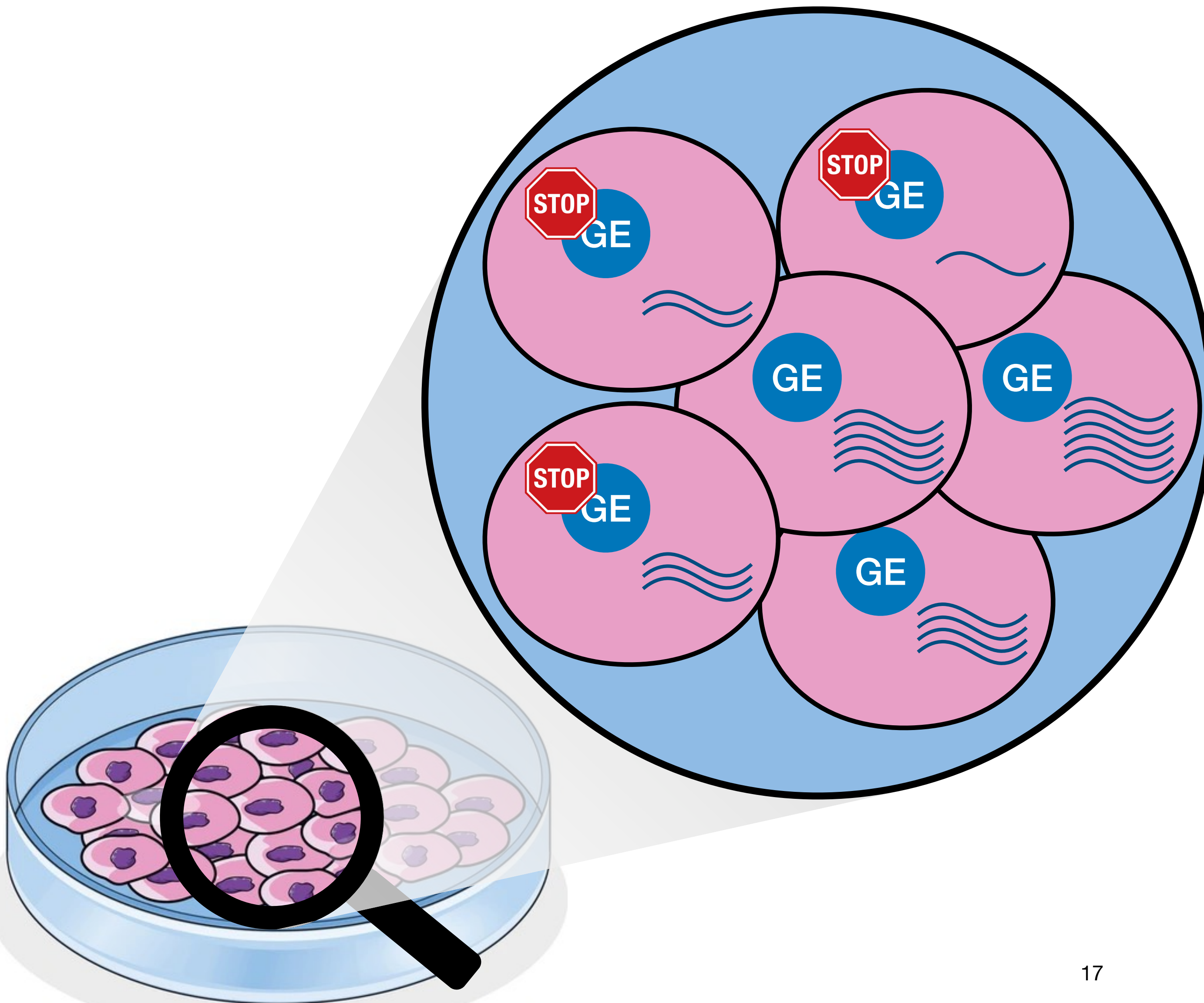


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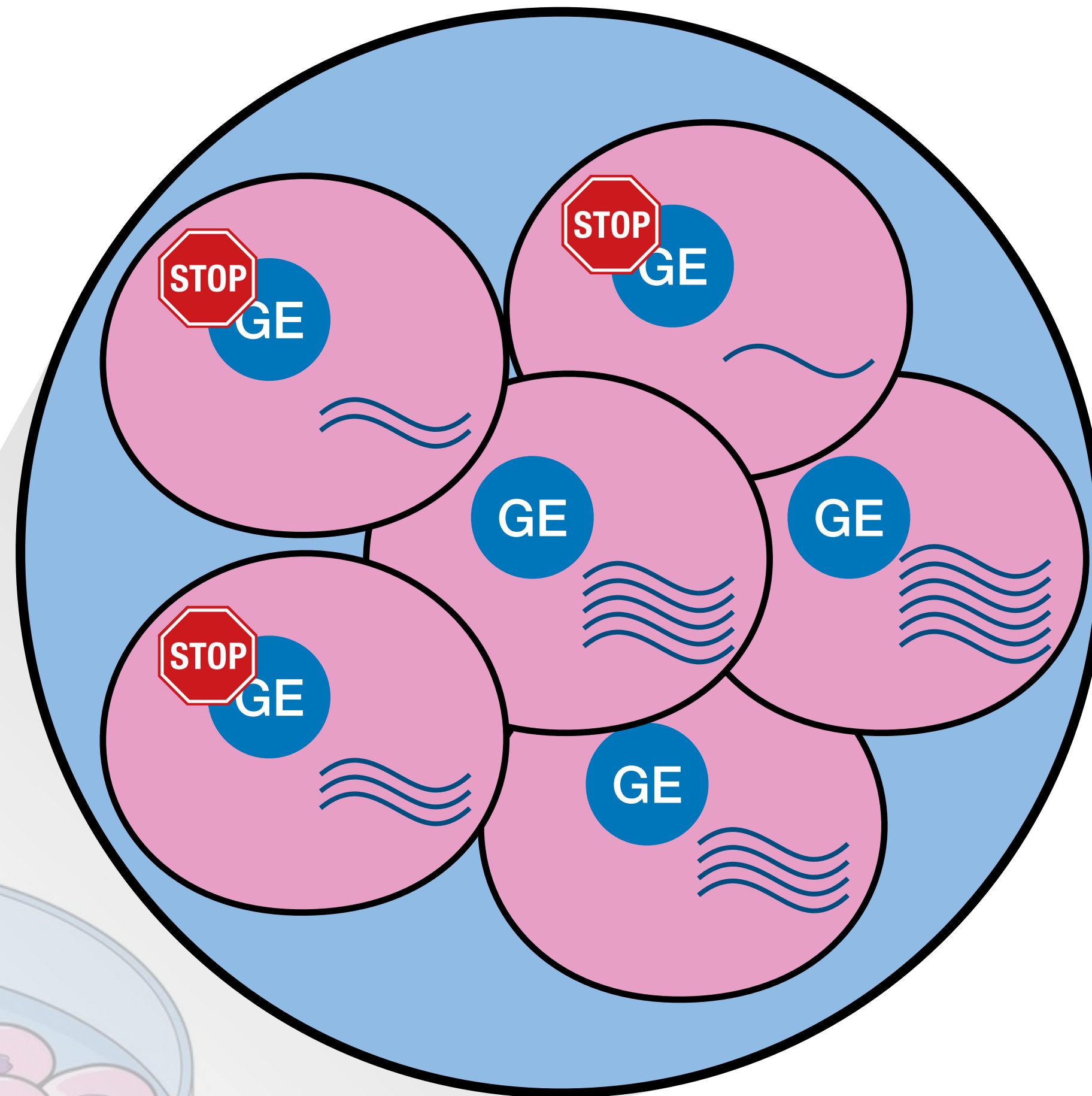


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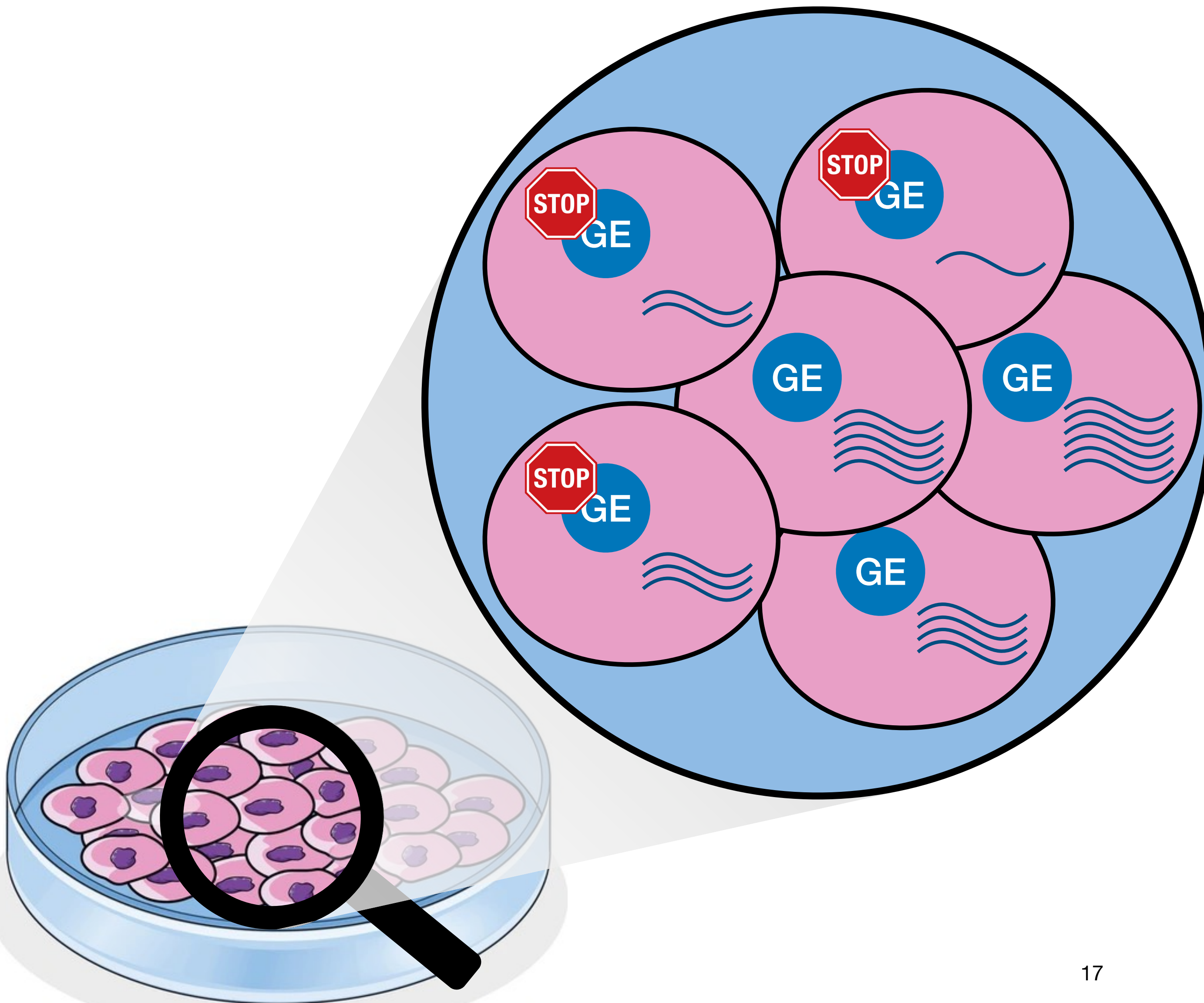


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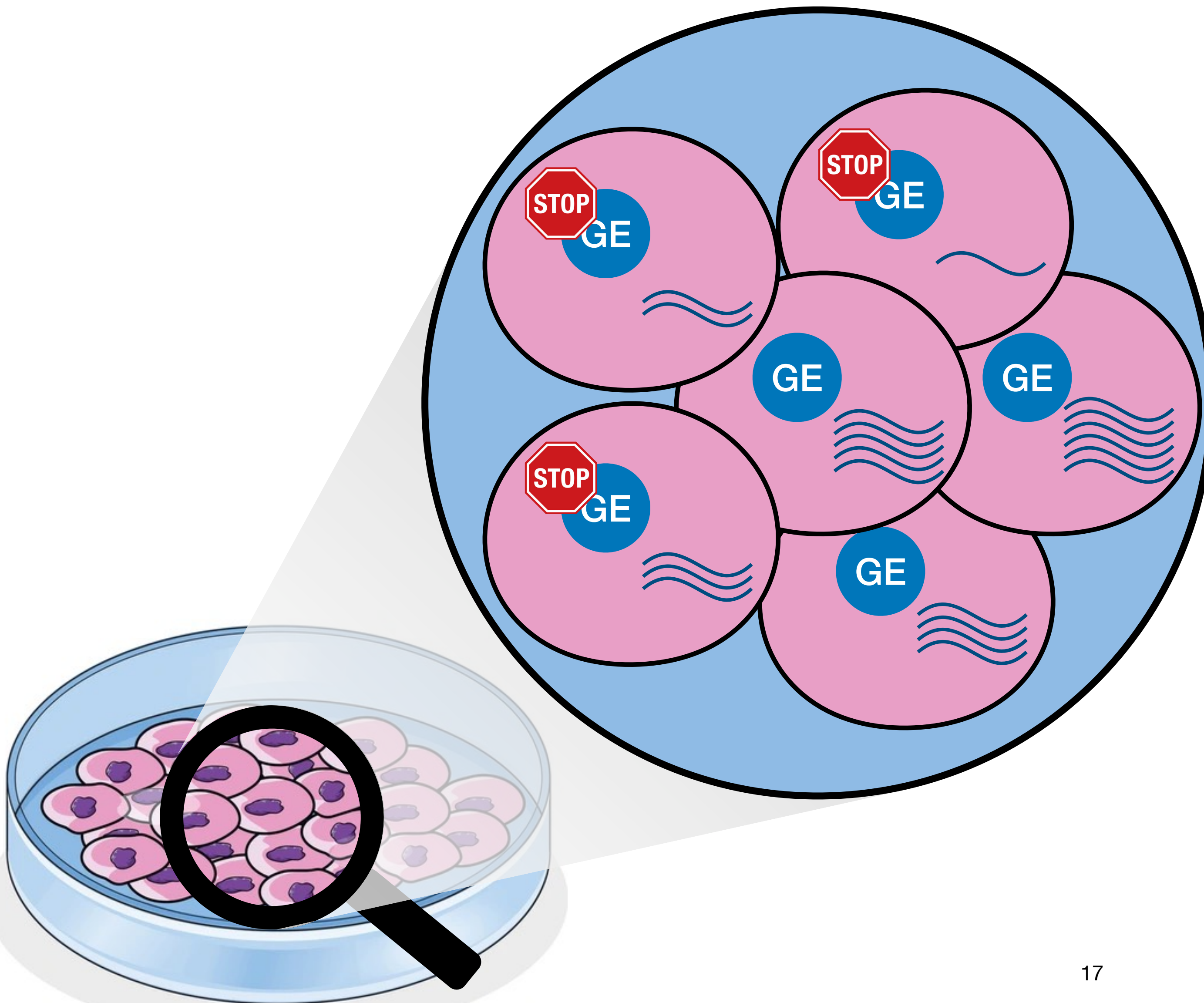
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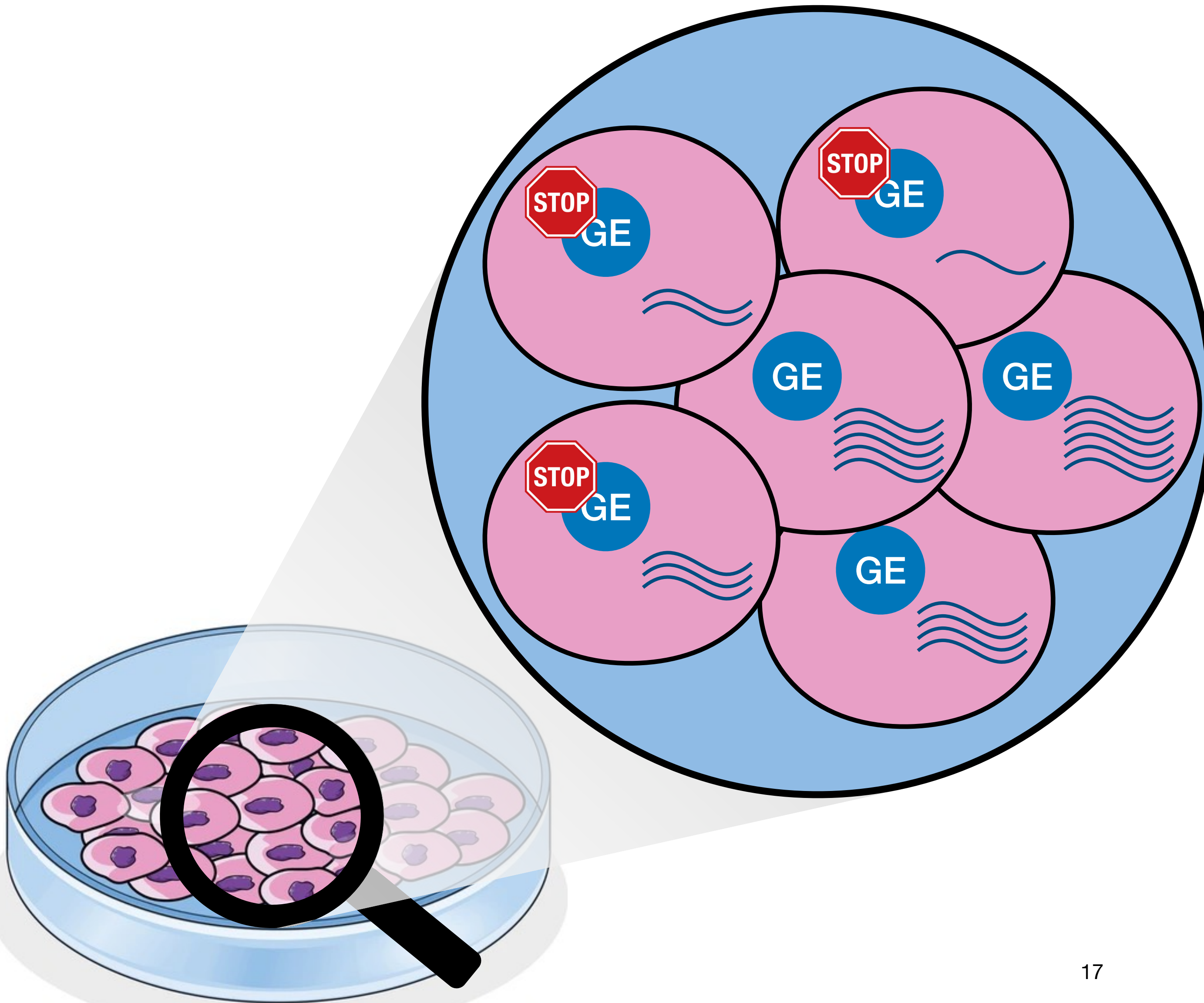
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$$H_0 : X_i \perp\!\!\!\perp Y_i \mid Z_i.$$



Model for the count-based CRISPR data

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Working model:
$$\begin{cases} X_i | Z_i \stackrel{\text{ind}}{\sim} \text{Ber}(\pi_i); & \text{logit}(\pi_i) = Z_i^T \delta \\ Y_i | X_i, Z_i \stackrel{\text{ind}}{\sim} \text{NegBin}(\mu_i, \theta); & \log \mu_i = X_i \beta + Z_i^T \gamma \end{cases}.$$

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¹Shah and Peters (2020)

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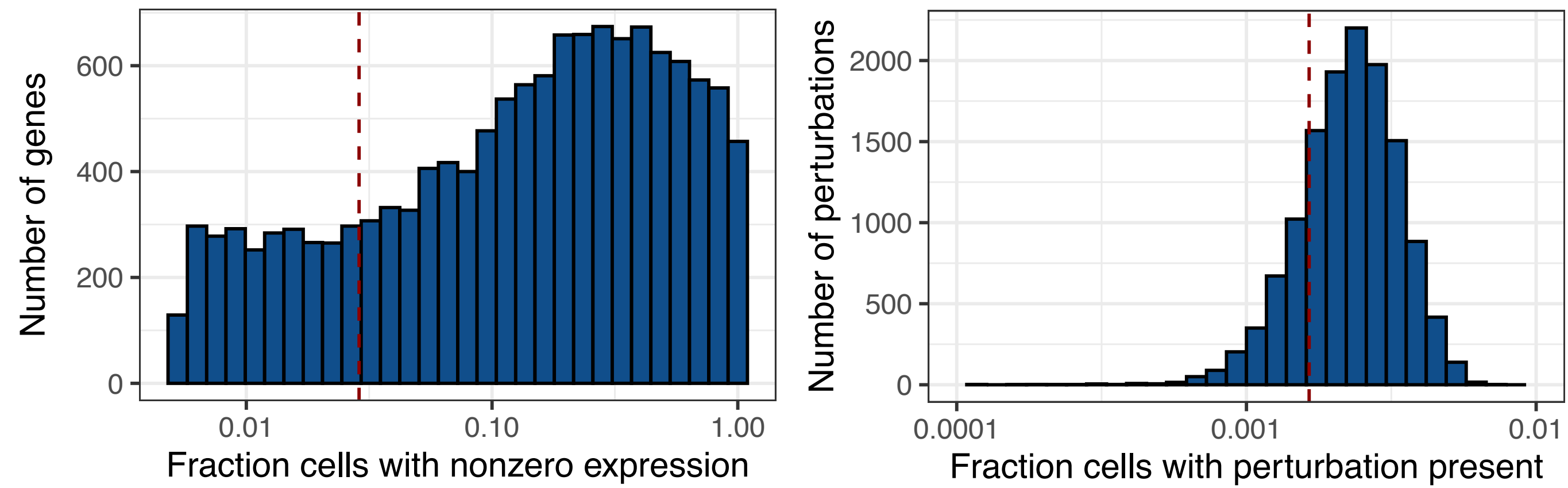
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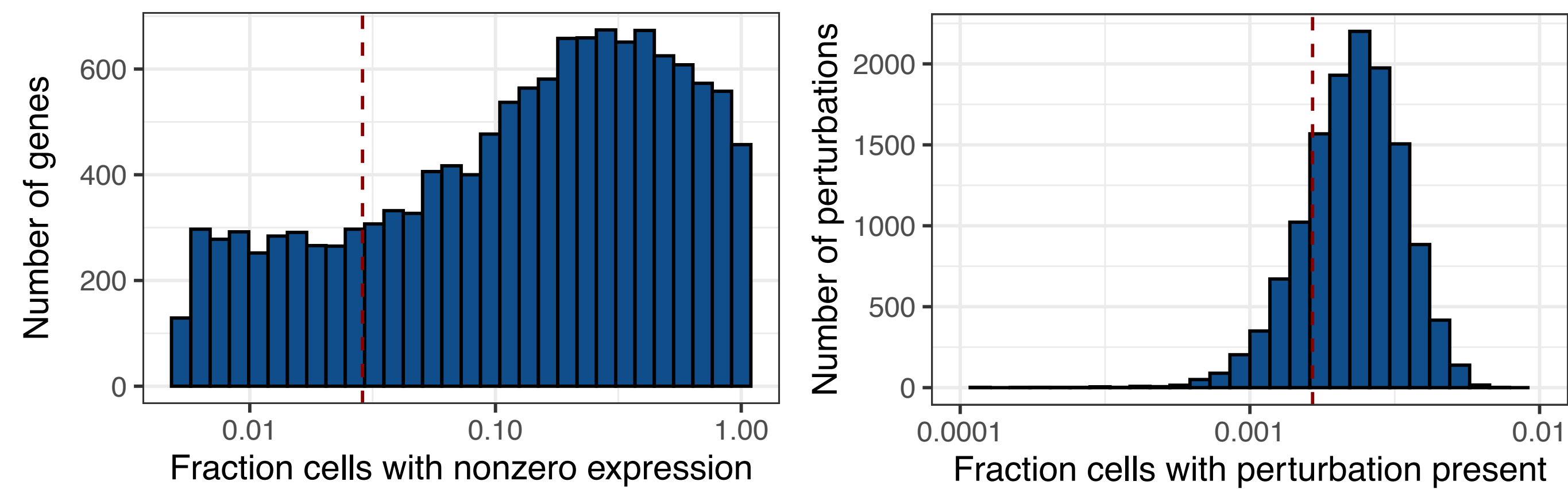
²Liu et al. (2022), Barry et al. (2021)

Challenges: scale and sparsity

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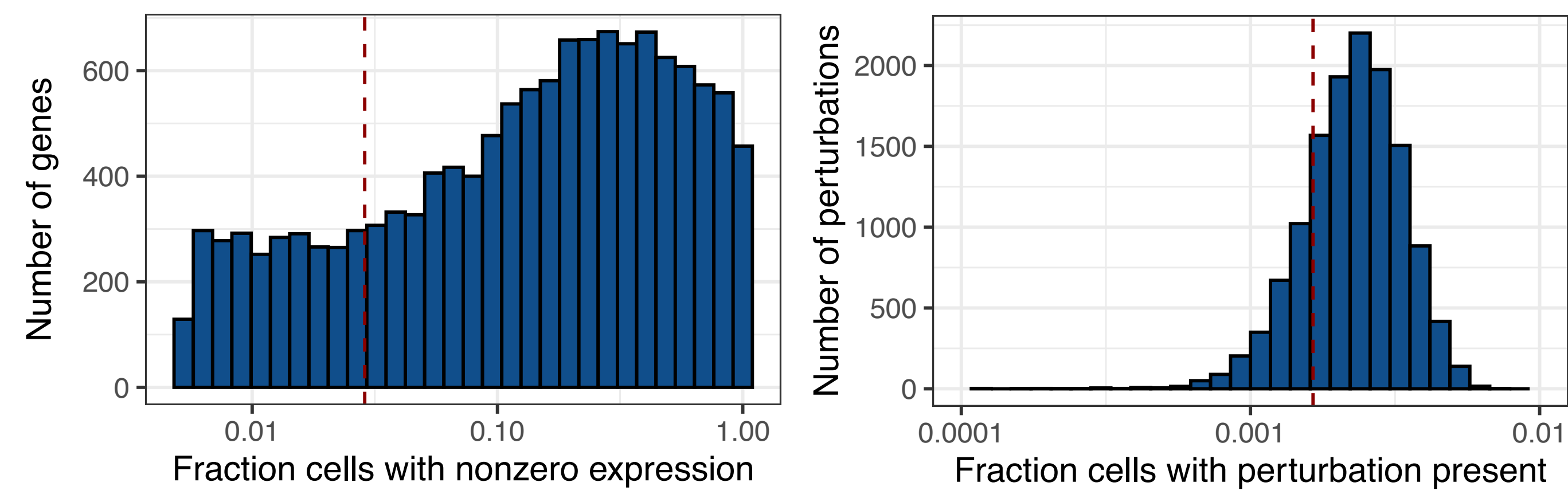


Challenges: scale and sparsity



Gene Expression						
Cell perturbed	0	1	2	3	4	8
FALSE	201027	5594	324	32	5	1
TRUE	332	9	0	0	0	0

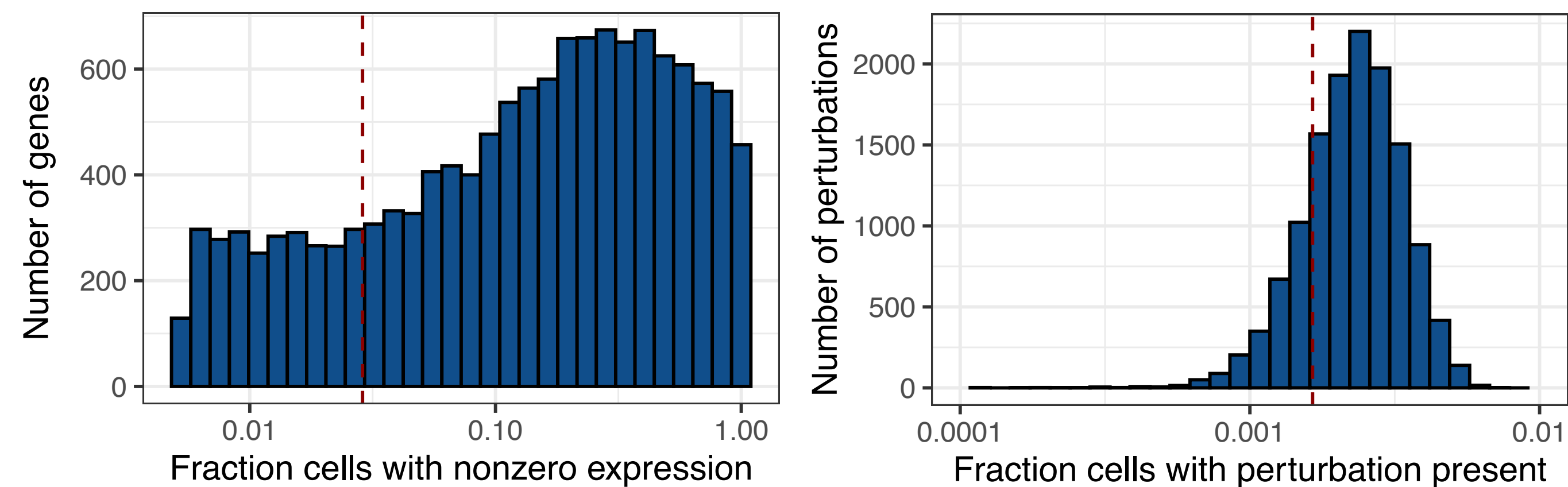
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Statistical challenge: CLT does not kick in.

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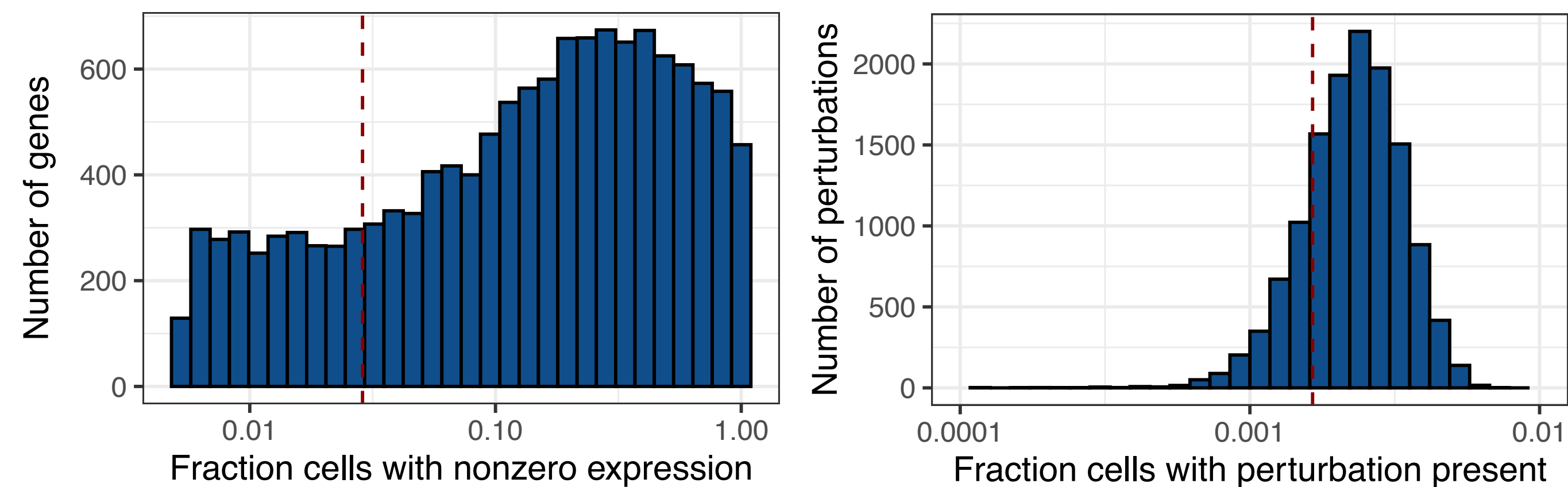


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Computational challenge: directly applying computationally heavy is infeasible.

Challenges: scale and sparsity

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Goal

Efficient computation
under error-control
constraints

Challenges: scale and sparsity

Effective Sample size
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transcripts)

Multiplicity
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Computation efficiency
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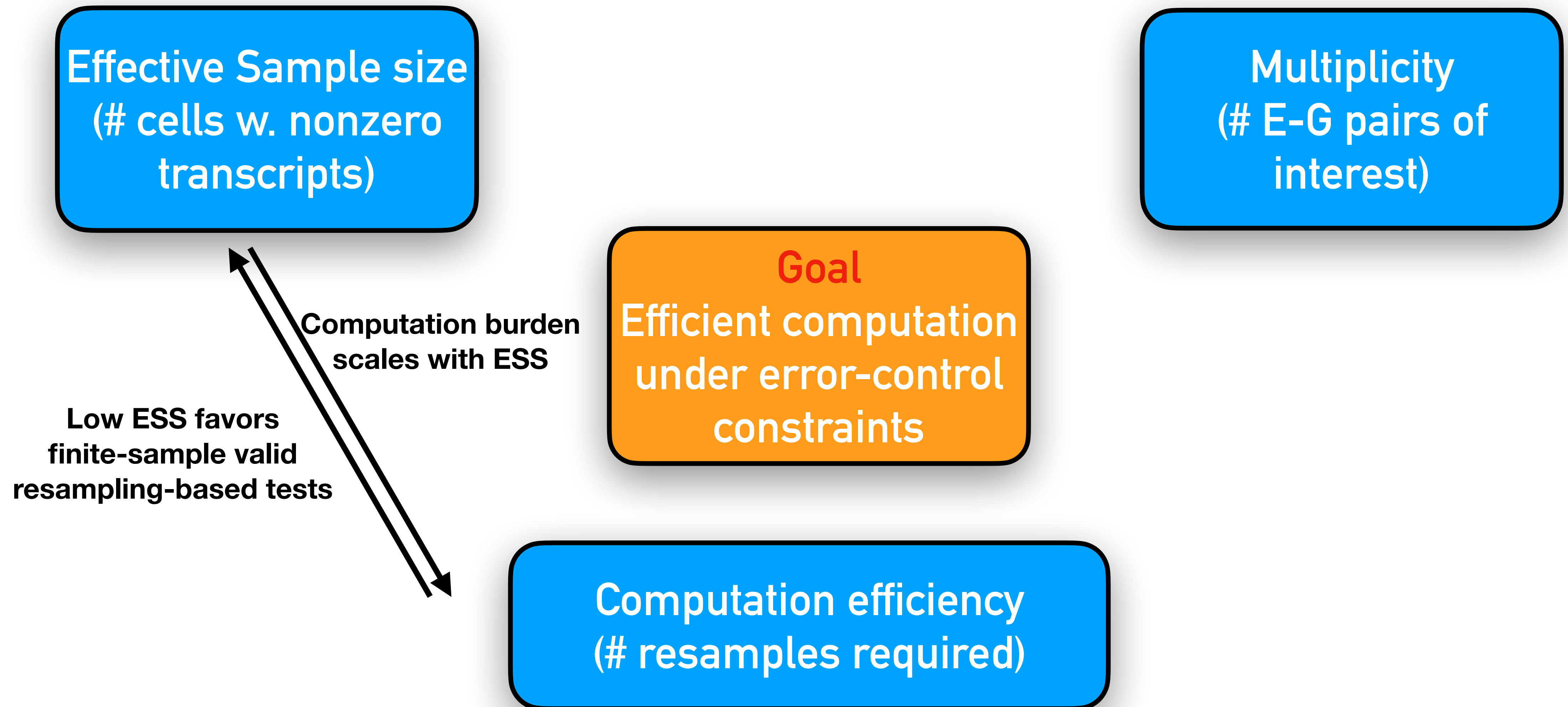
Computation burden
scales with ESS

Goal

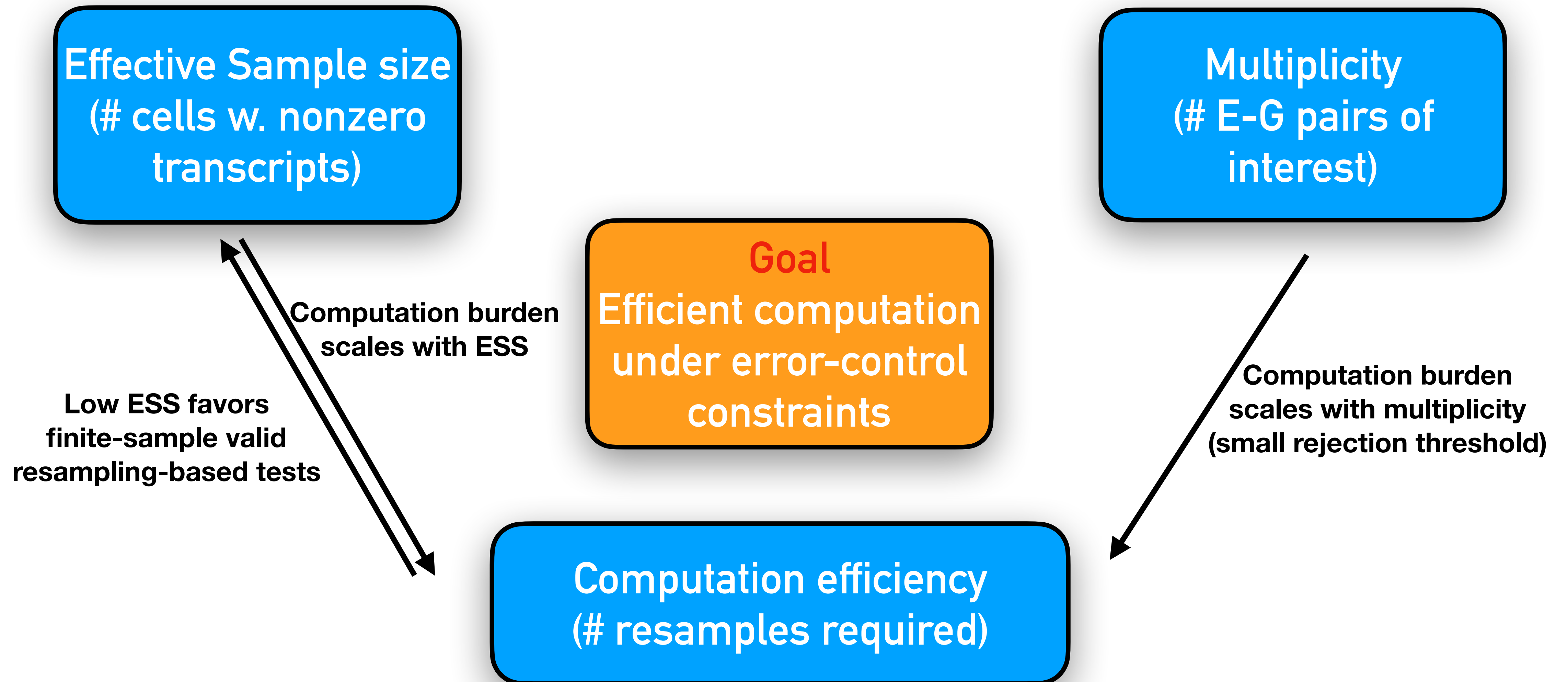
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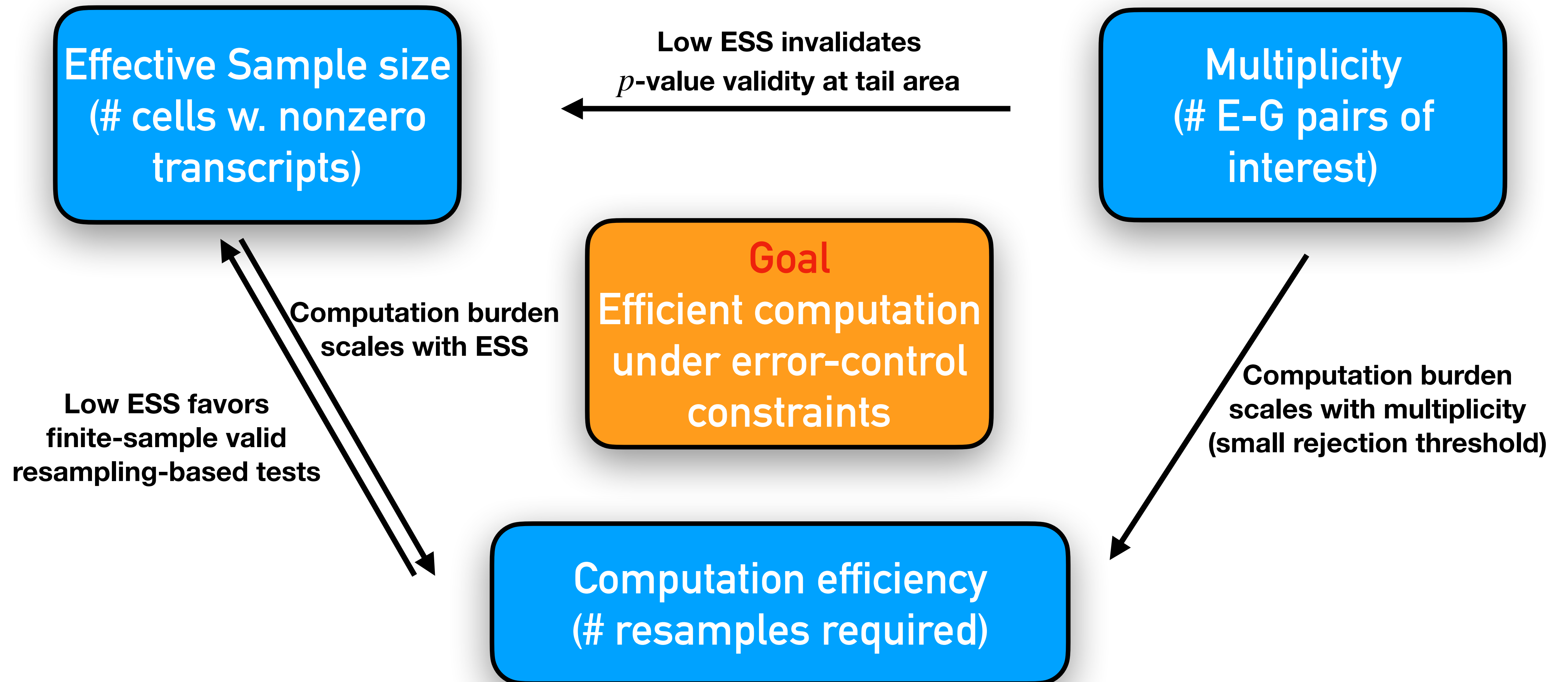
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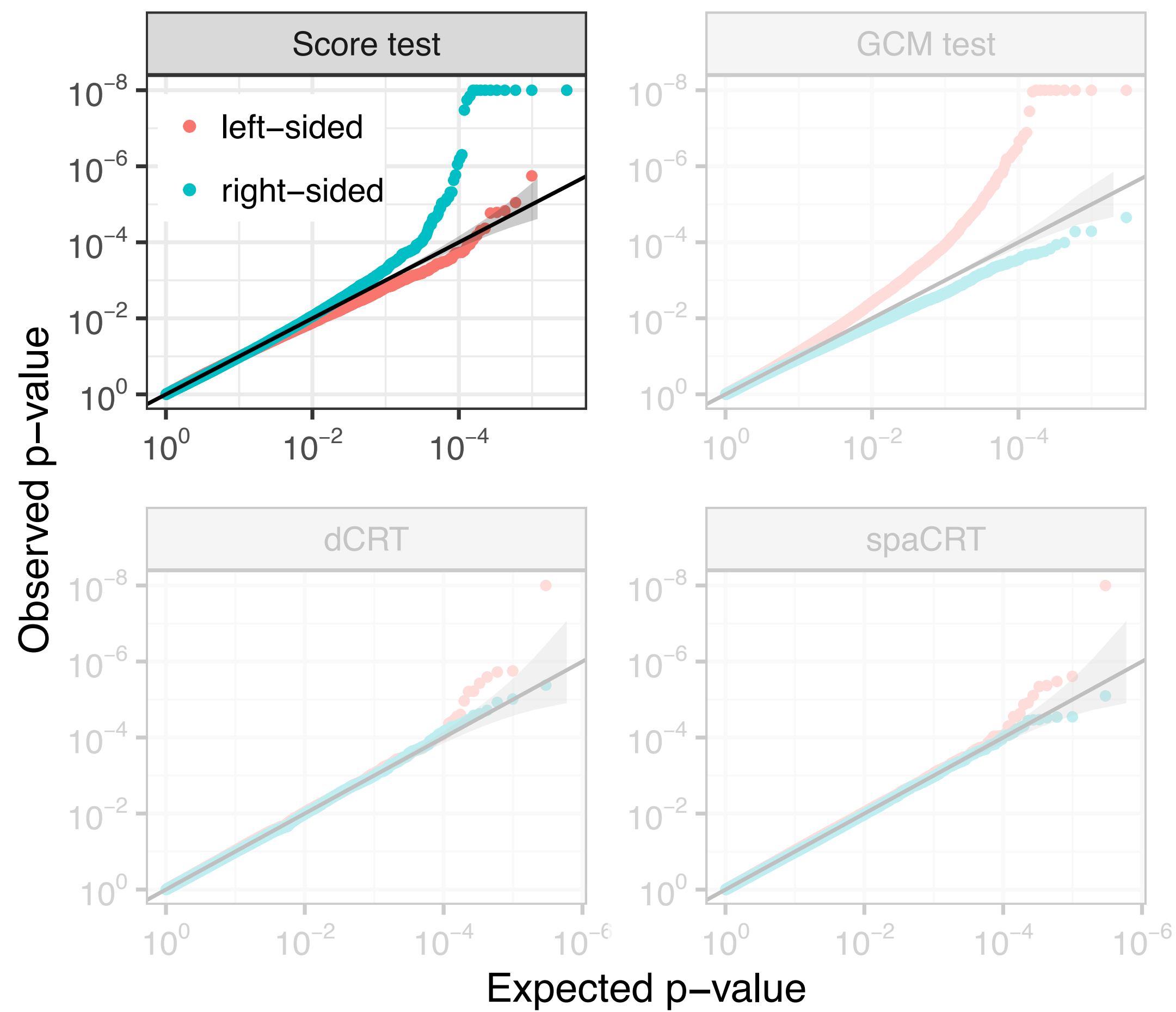


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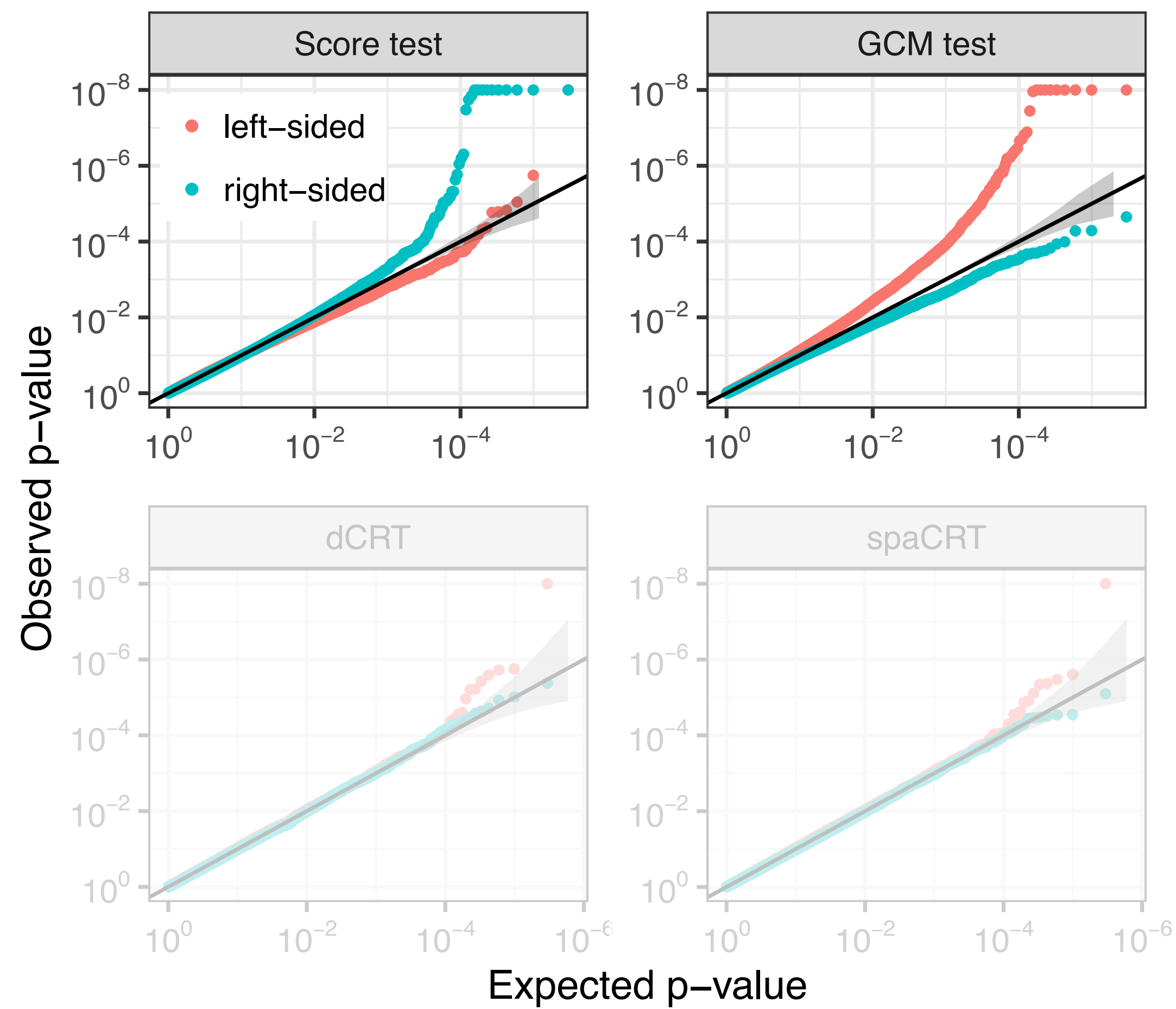


Benchmarking Type-I error on negative controls

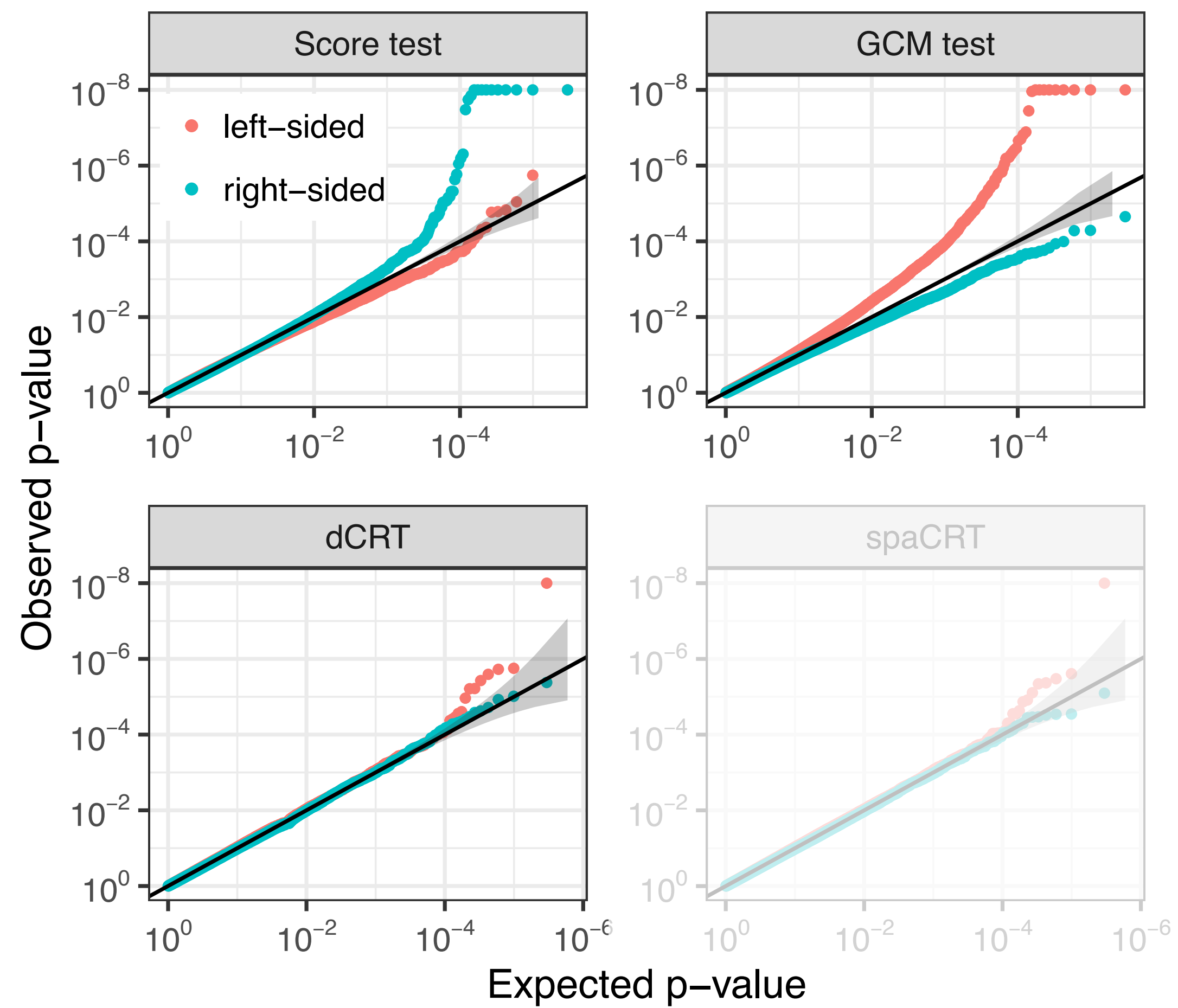
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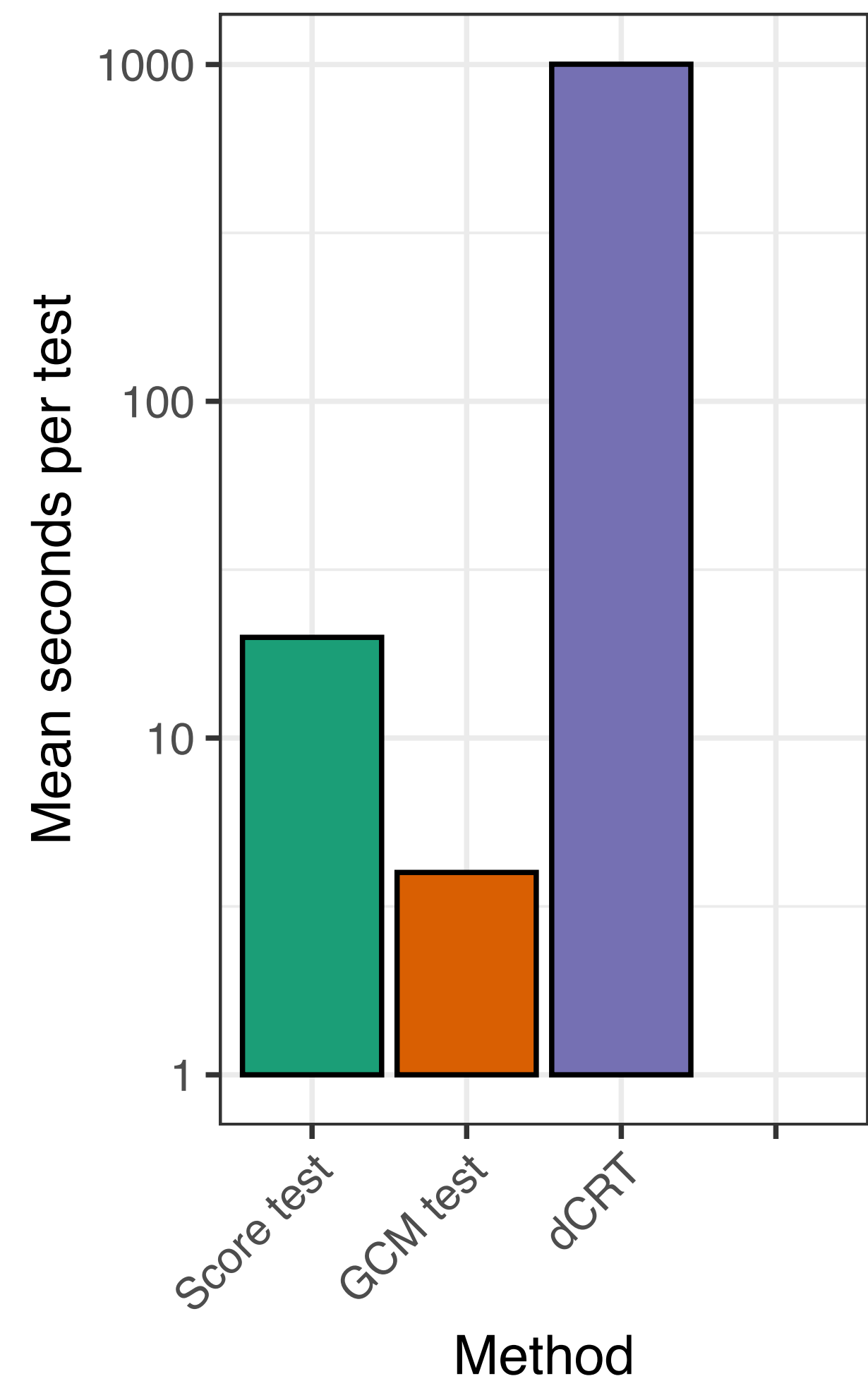
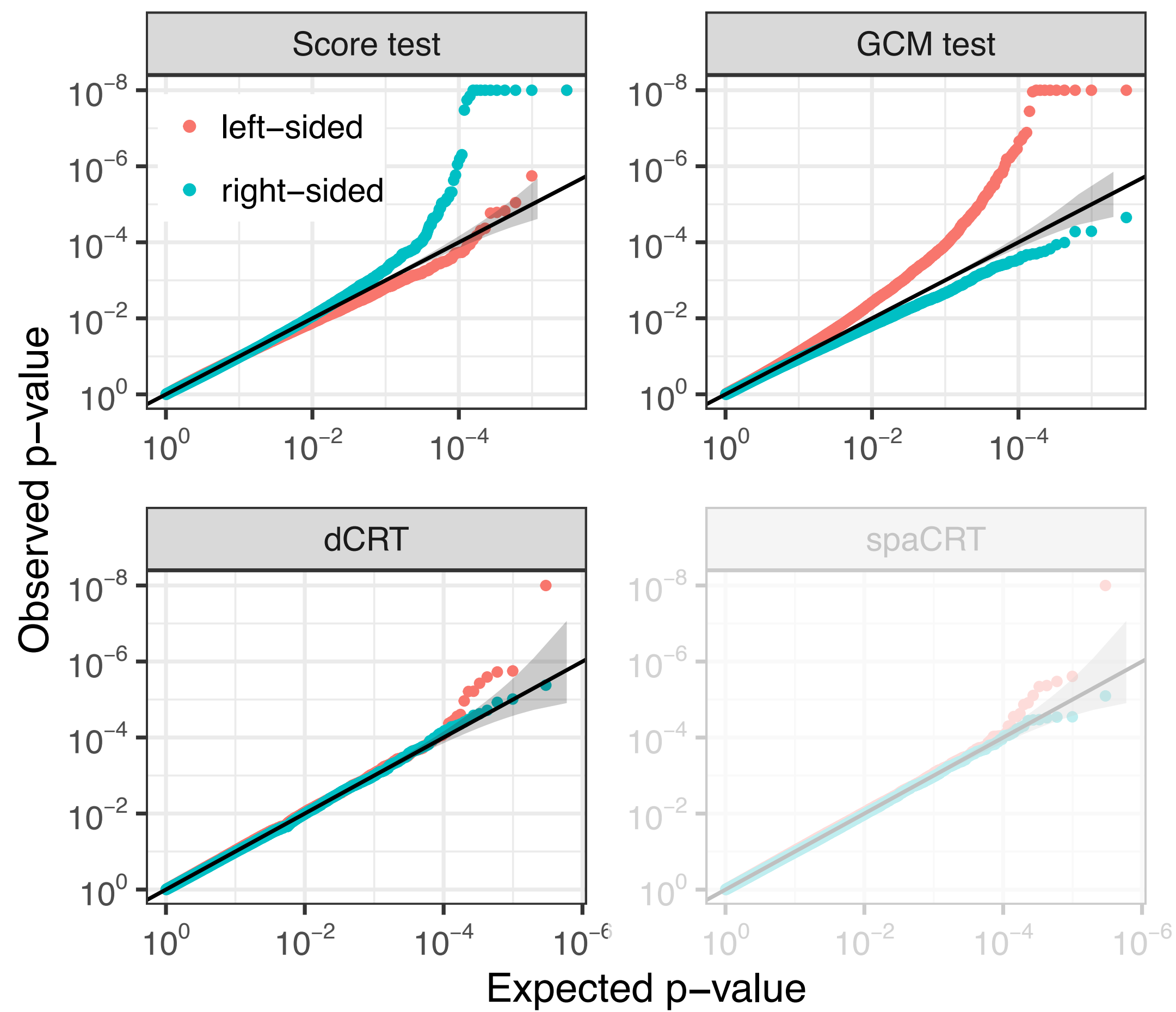
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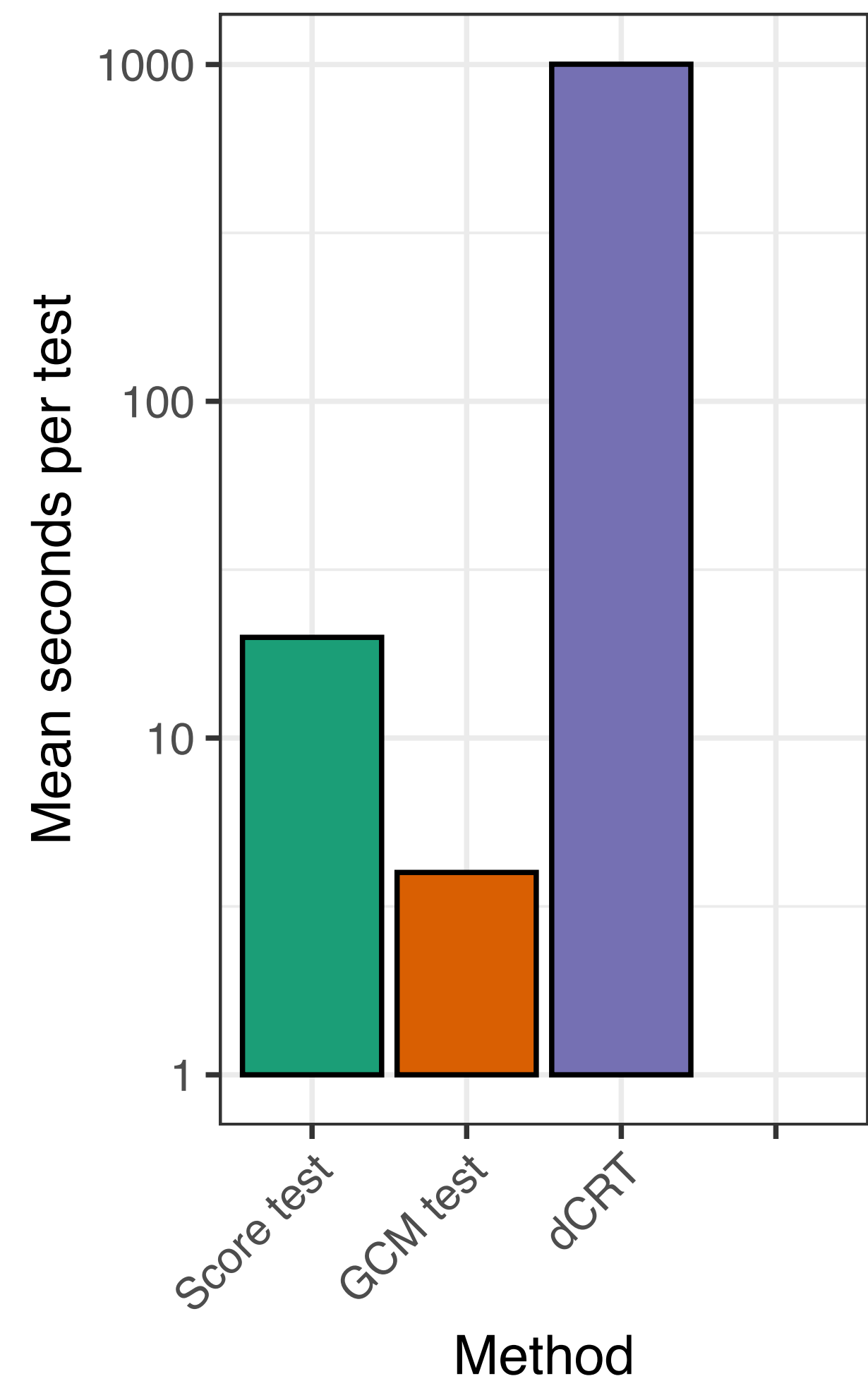
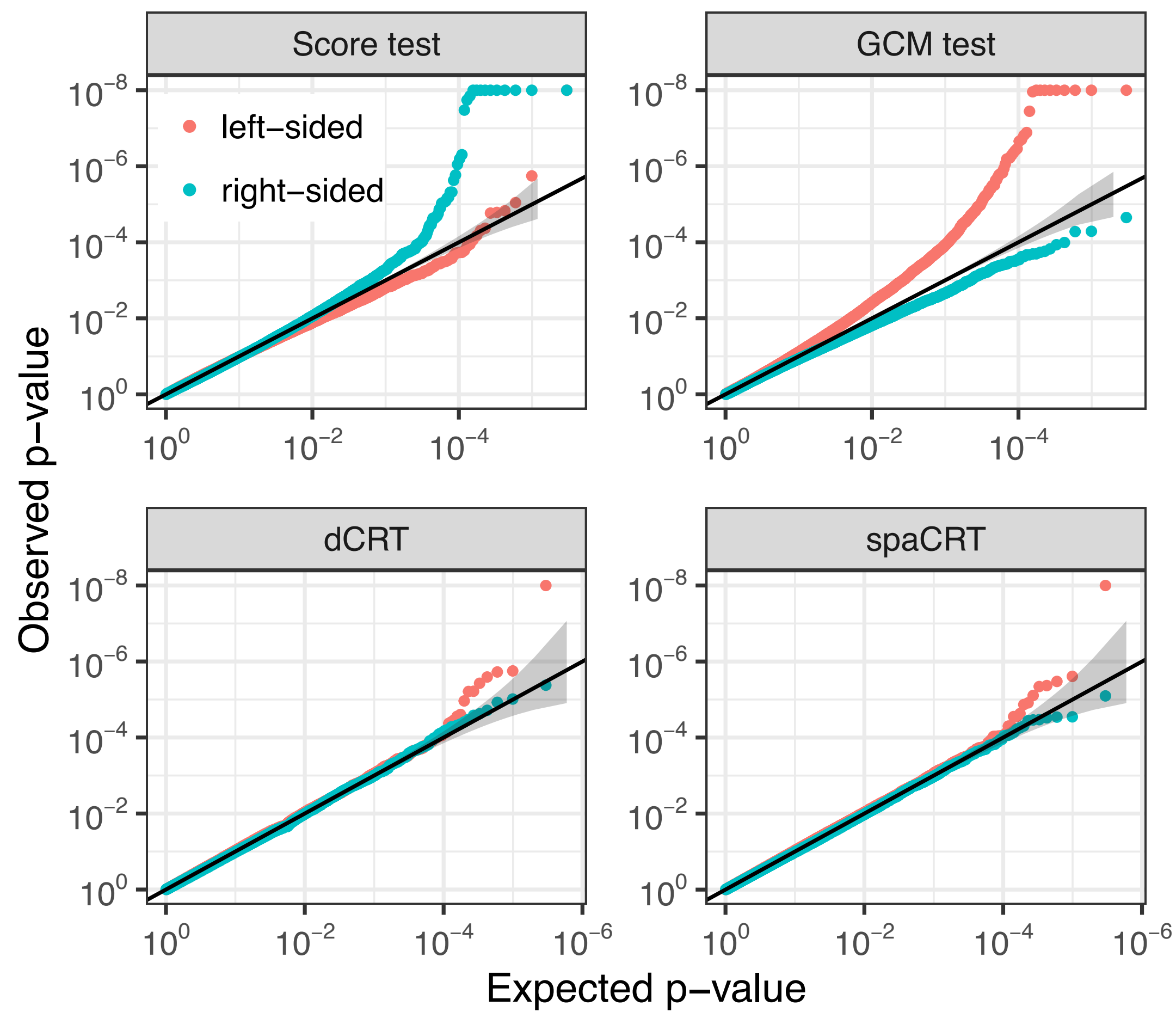
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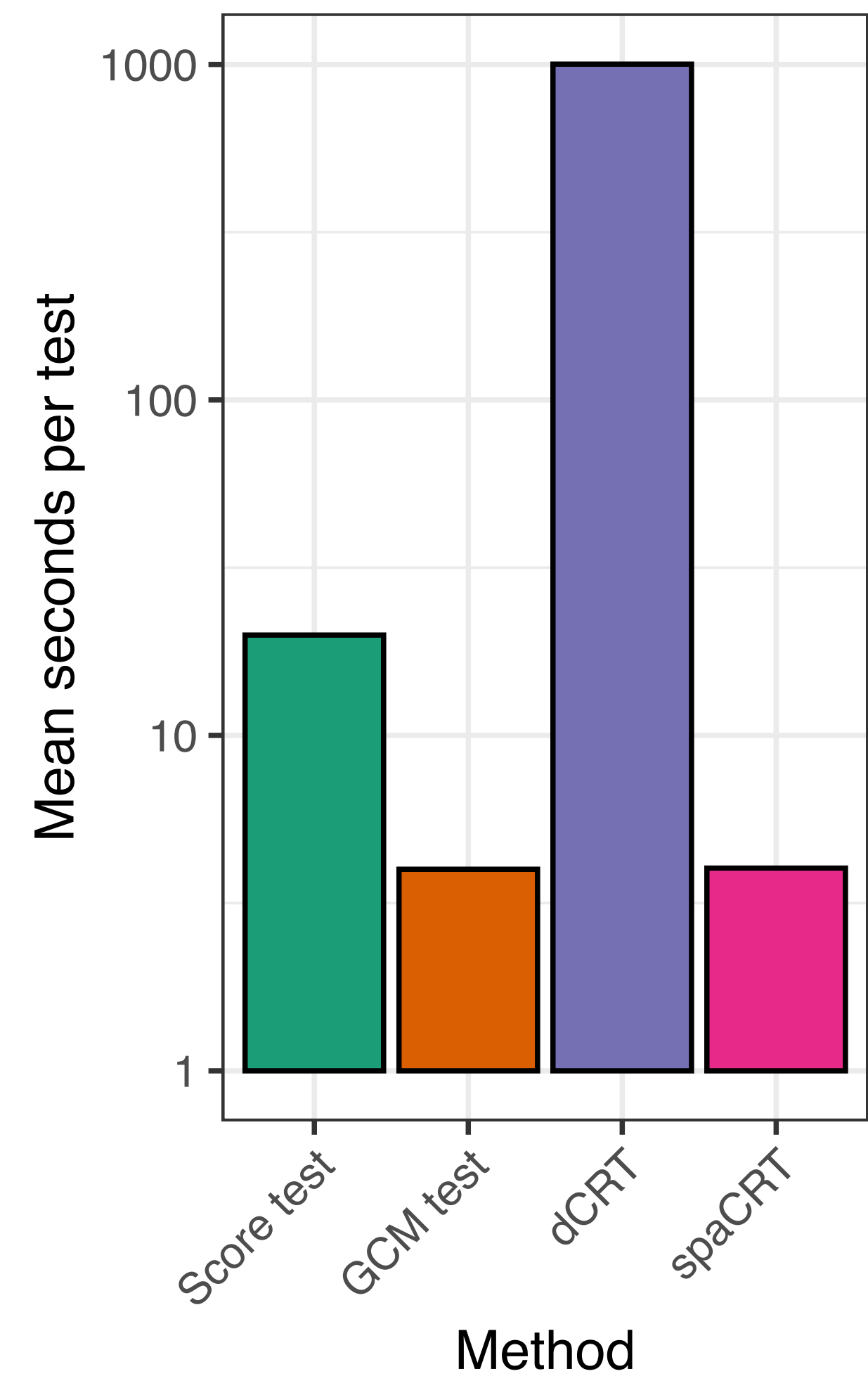
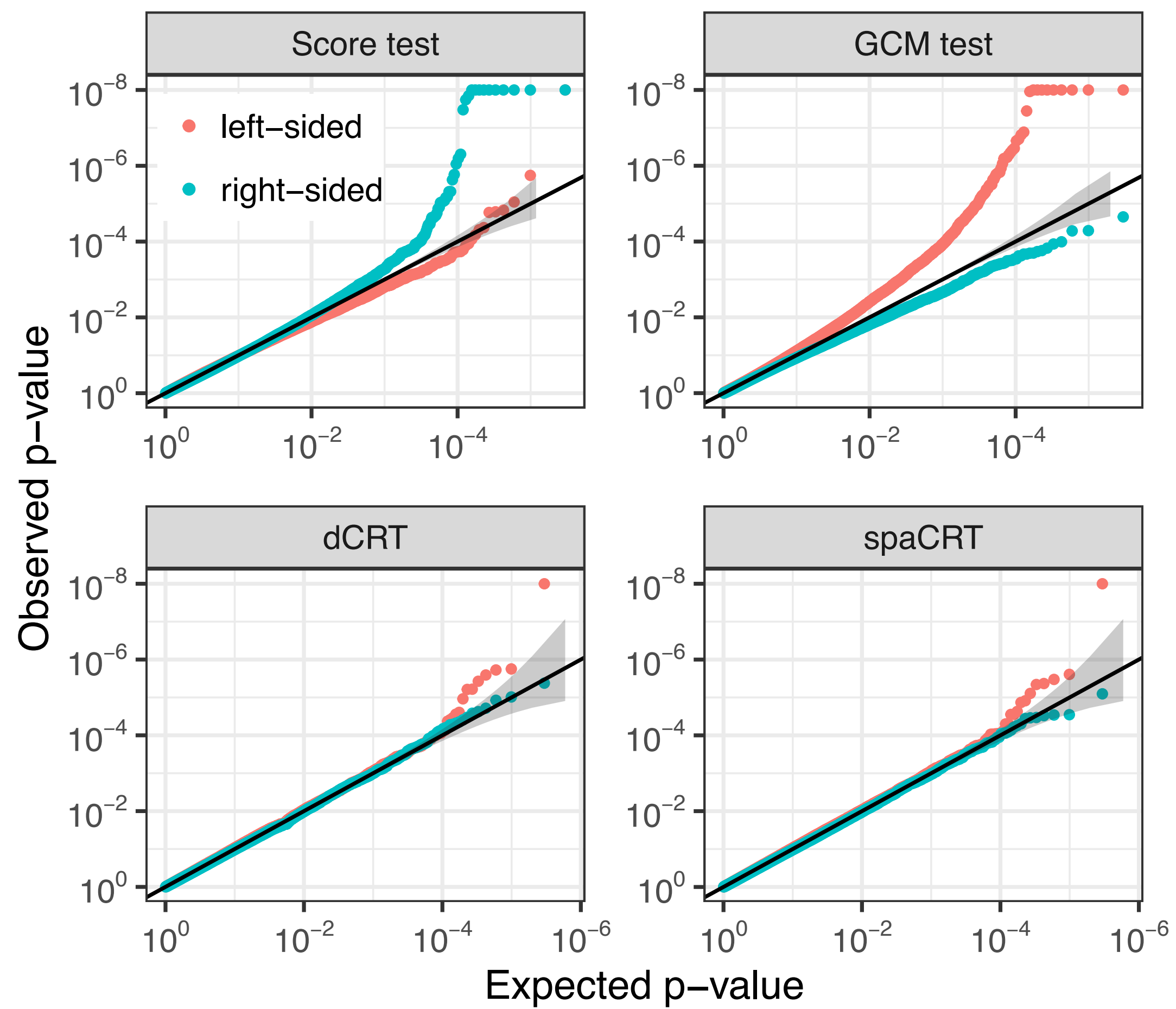
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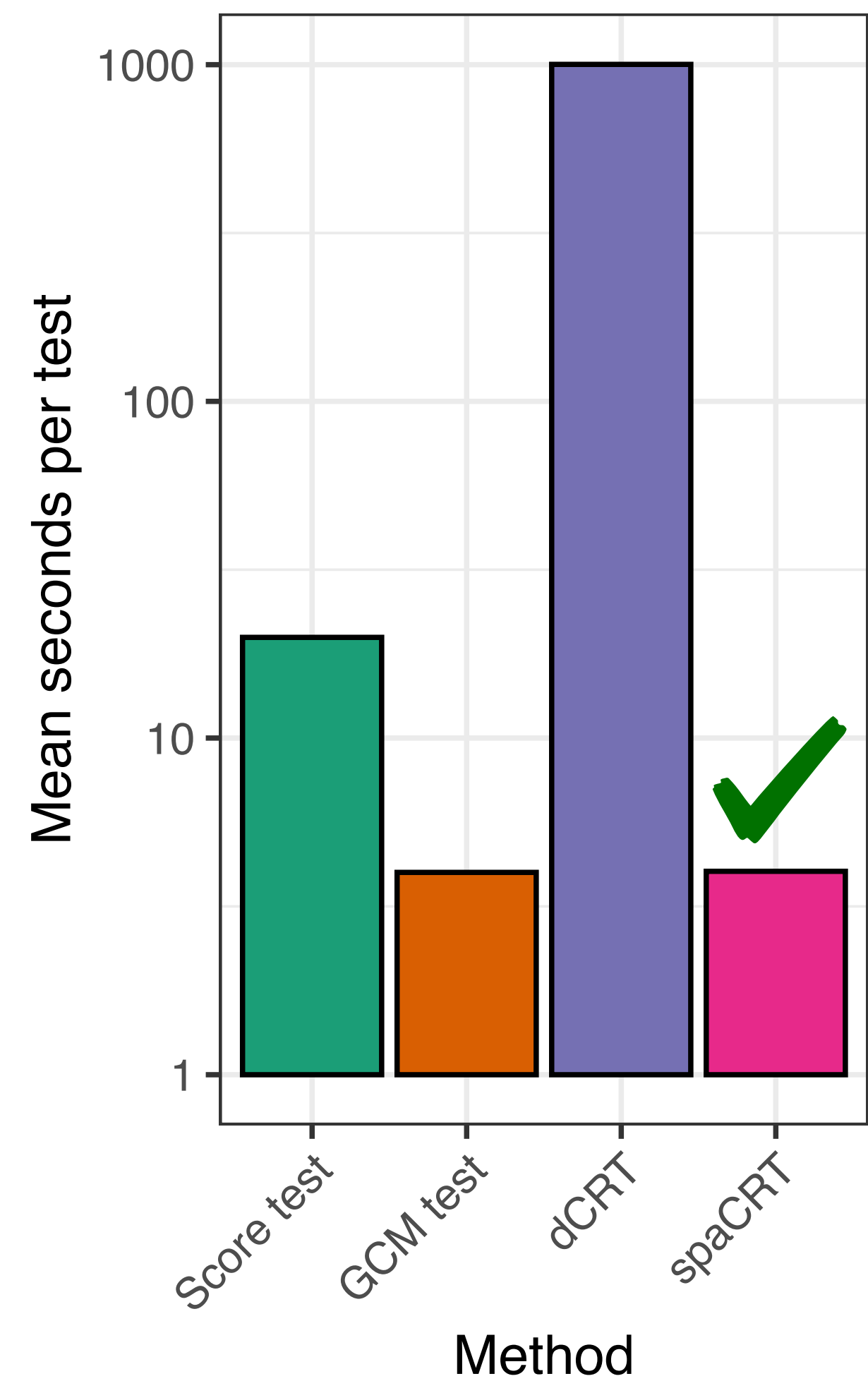
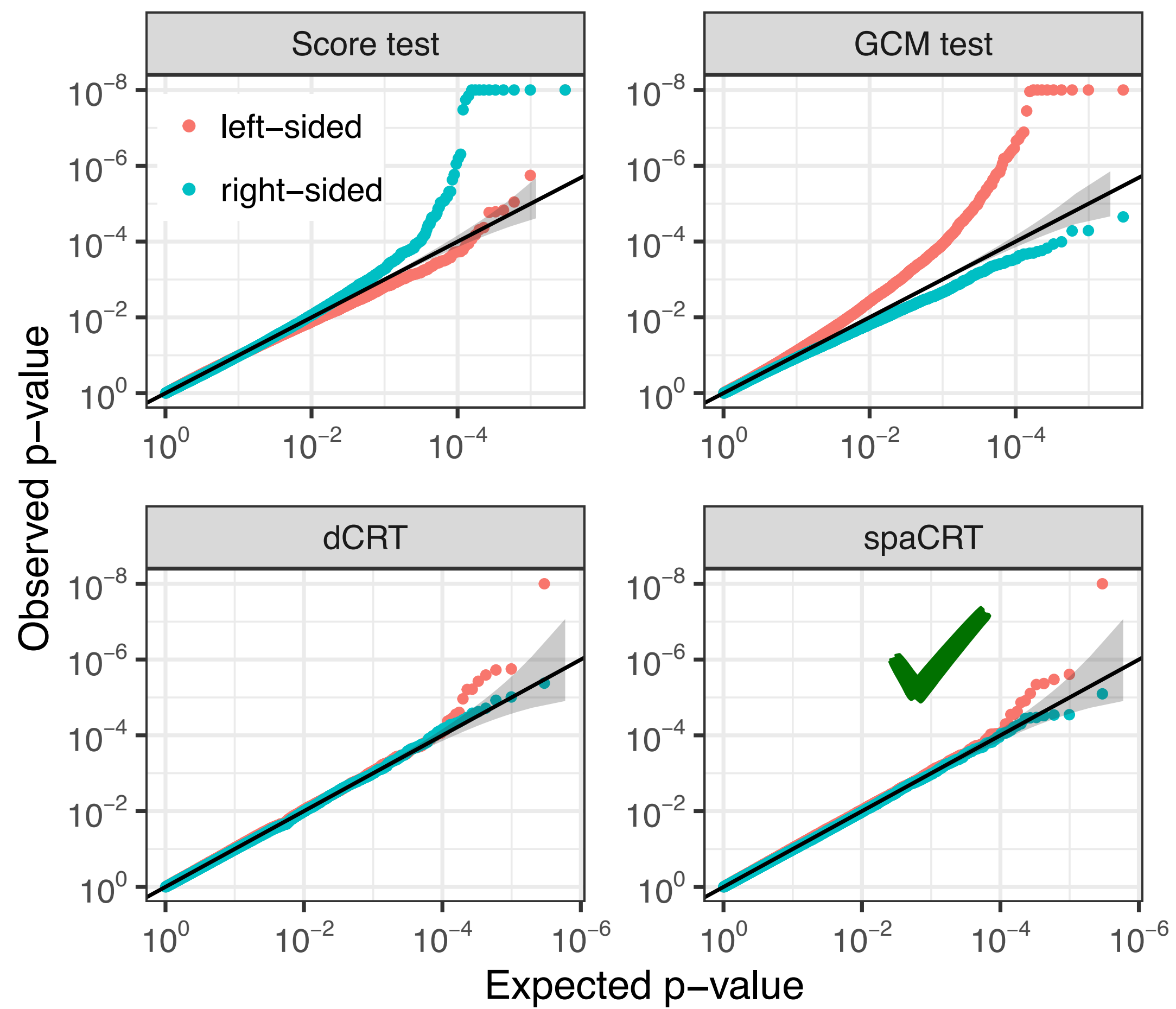
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The distilled conditional randomization test

dCRT algorithm

The distilled conditional randomization test

Suppose $X_i \stackrel{\text{ind}}{\sim} \text{Ber}(\pi(Z_i))$ for some function π .

Learn $\hat{\pi}(Z_i)$ by regressing X on Z and $\hat{\mu}(Z_i) \approx \mathbb{E}[Y_i | Z_i]$ by regressing Y on Z .

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3. dCRT p-value $p_n^{\text{dCRT}} = \frac{1}{B+1} \left(1 + \sum_{b=1}^B 1(\tilde{T}_n^{(b)} \geq T_n) \right)$.

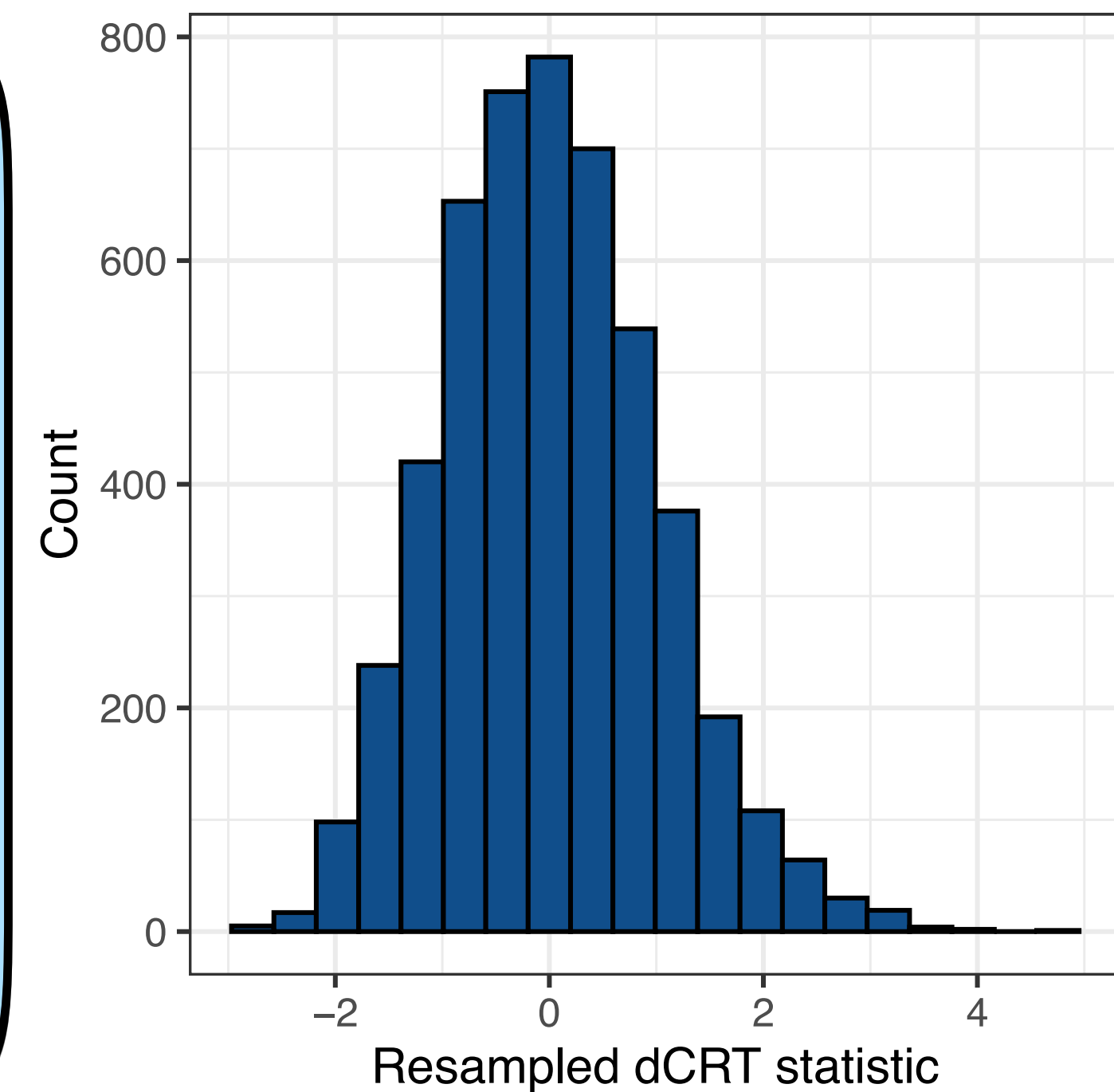
The distilled conditional randomization test

Suppose $X_i \stackrel{\text{ind}}{\sim} \text{Ber}(\pi(Z_i))$ for some function π .

Learn $\hat{\pi}(Z_i)$ by regressing X on Z and $\hat{\mu}(Z_i) \approx \mathbb{E}[Y_i | Z_i]$ by regressing Y on Z .

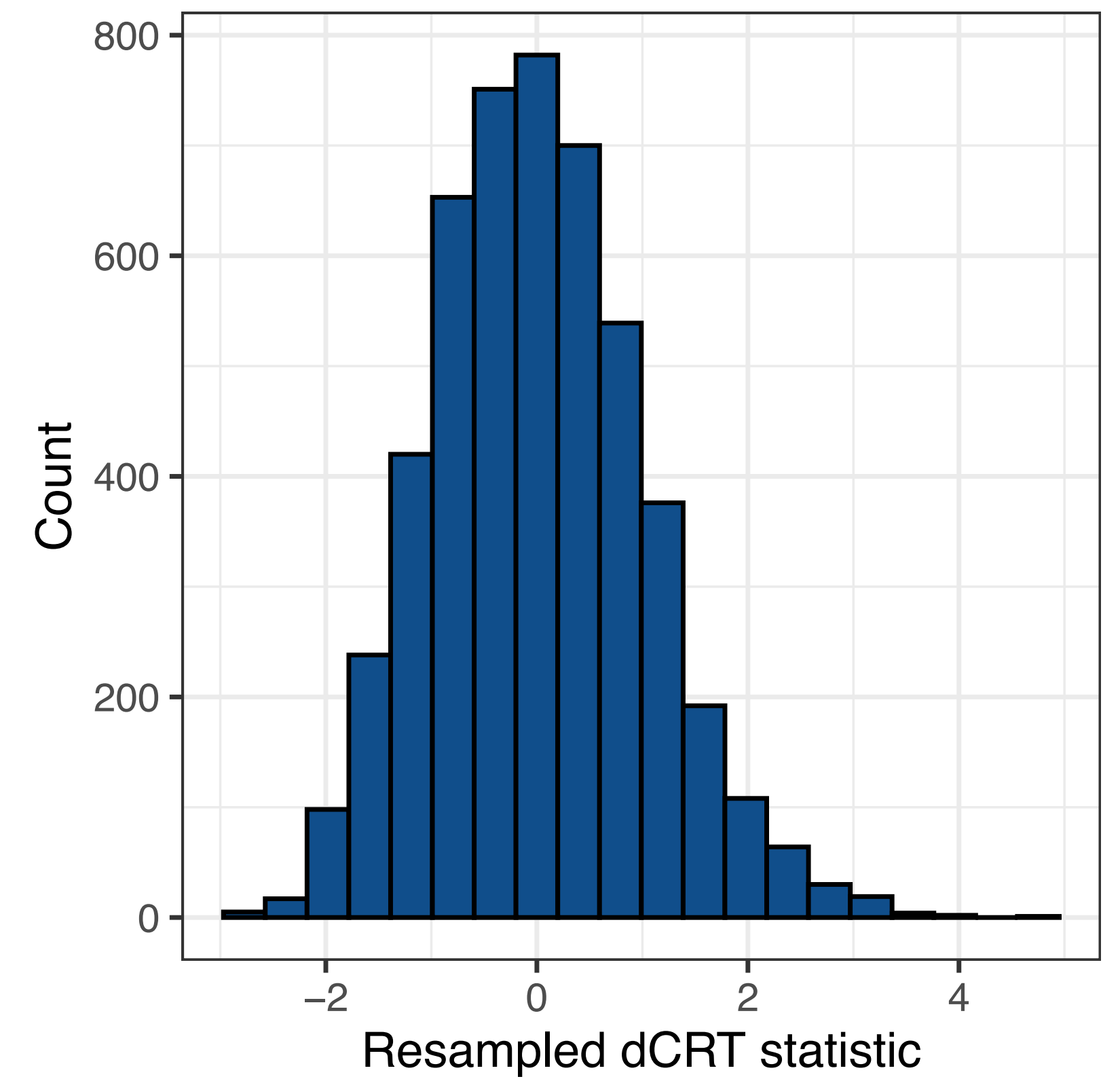
dCRT algorithm

1. Compute $T_n \equiv \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\pi}(Z_i))(Y_i - \hat{\mu}(Z_i))$.
2. for $b = 1, \dots, B$:
 - a. Sample $\tilde{X}_i^{(b)} \stackrel{\text{ind}}{\sim} \text{Ber}(\hat{\pi}(Z_i))$.
 - b. Compute $\tilde{T}_n^{(b)} \equiv \frac{1}{n} \sum_{i=1}^n (\tilde{X}_i^{(b)} - \hat{\pi}(Z_i))(Y_i - \hat{\mu}(Z_i))$.
3. dCRT p-value $p_n^{\text{dCRT}} = \frac{1}{B+1} \left(1 + \sum_{b=1}^B 1(\tilde{T}_n^{(b)} \geq T_n) \right)$.



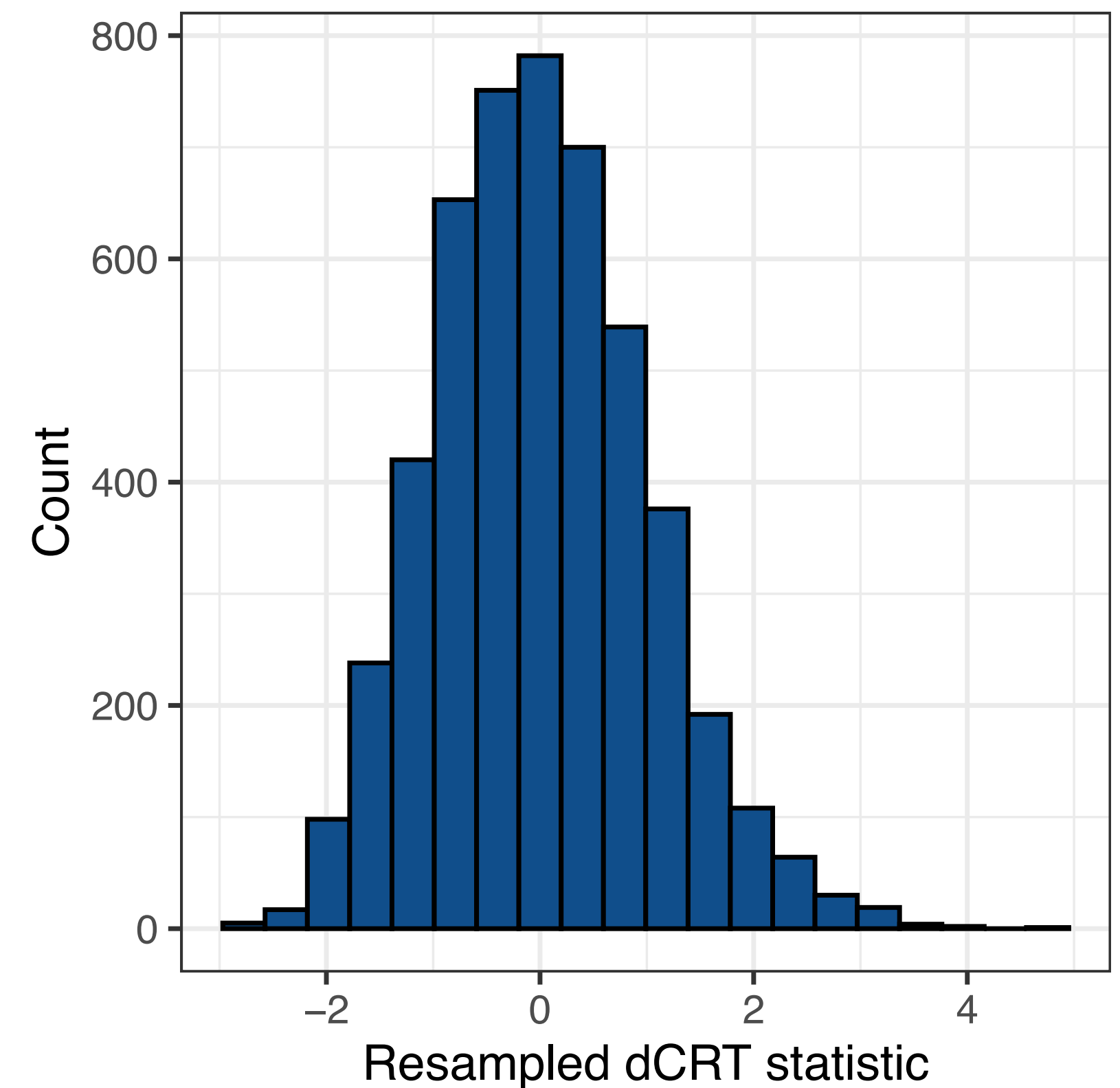
Approximating dCRT p -values

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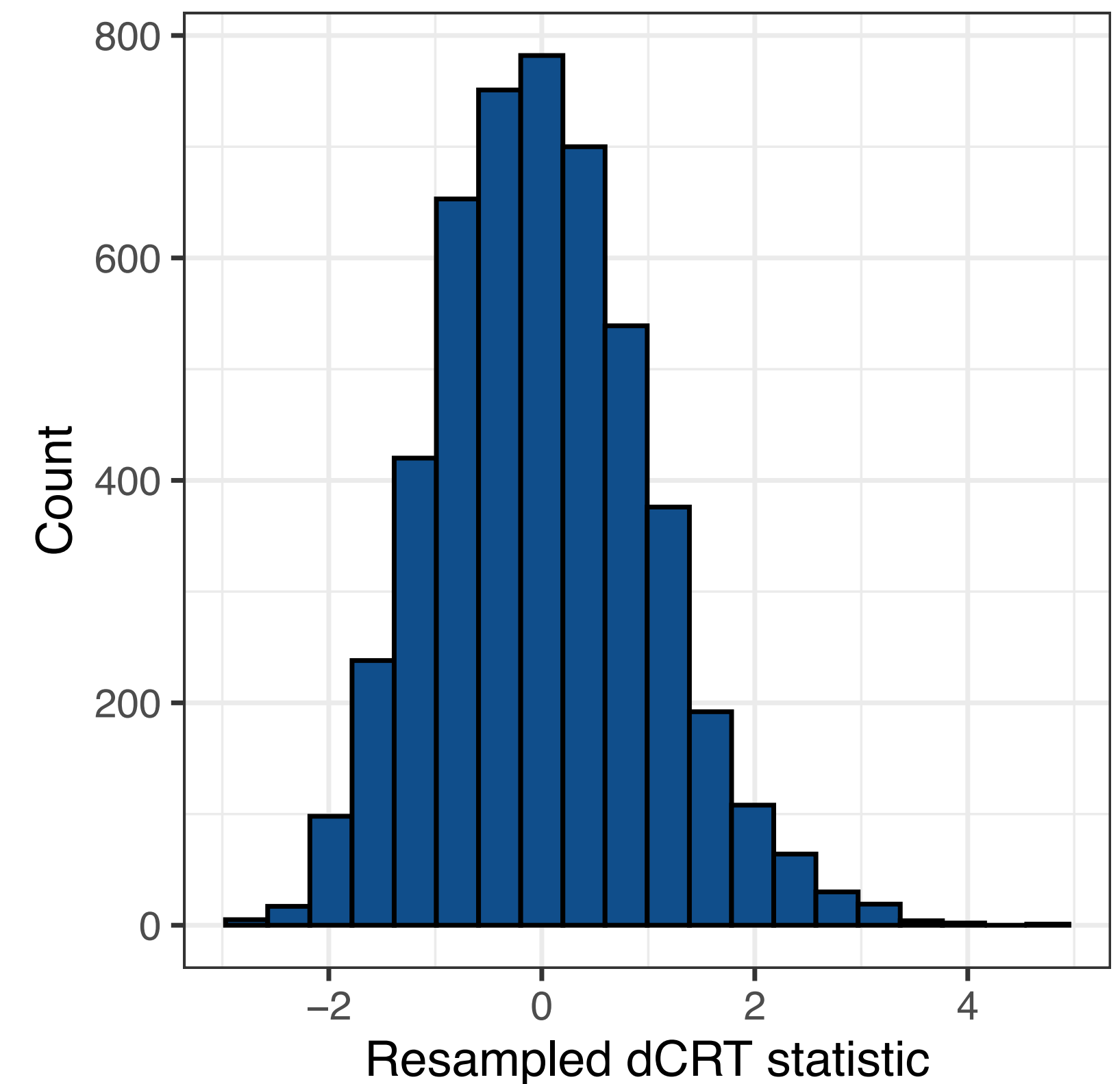
Approximating dCRT p -values

- Draw moderate number of resamples and fit parametric curve to distribution.



Approximating dCRT p -values

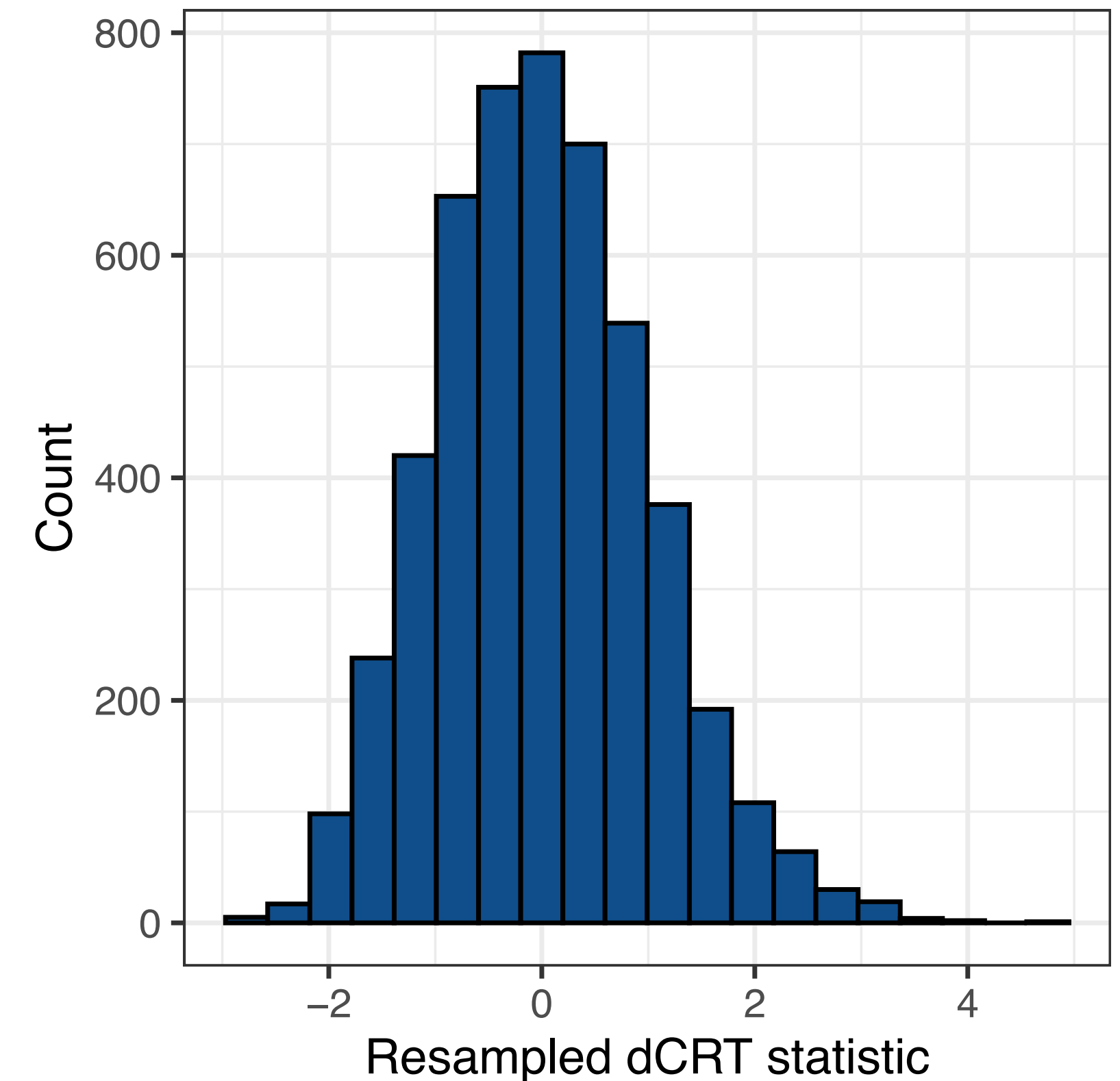
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¹Barry et al. (2021)

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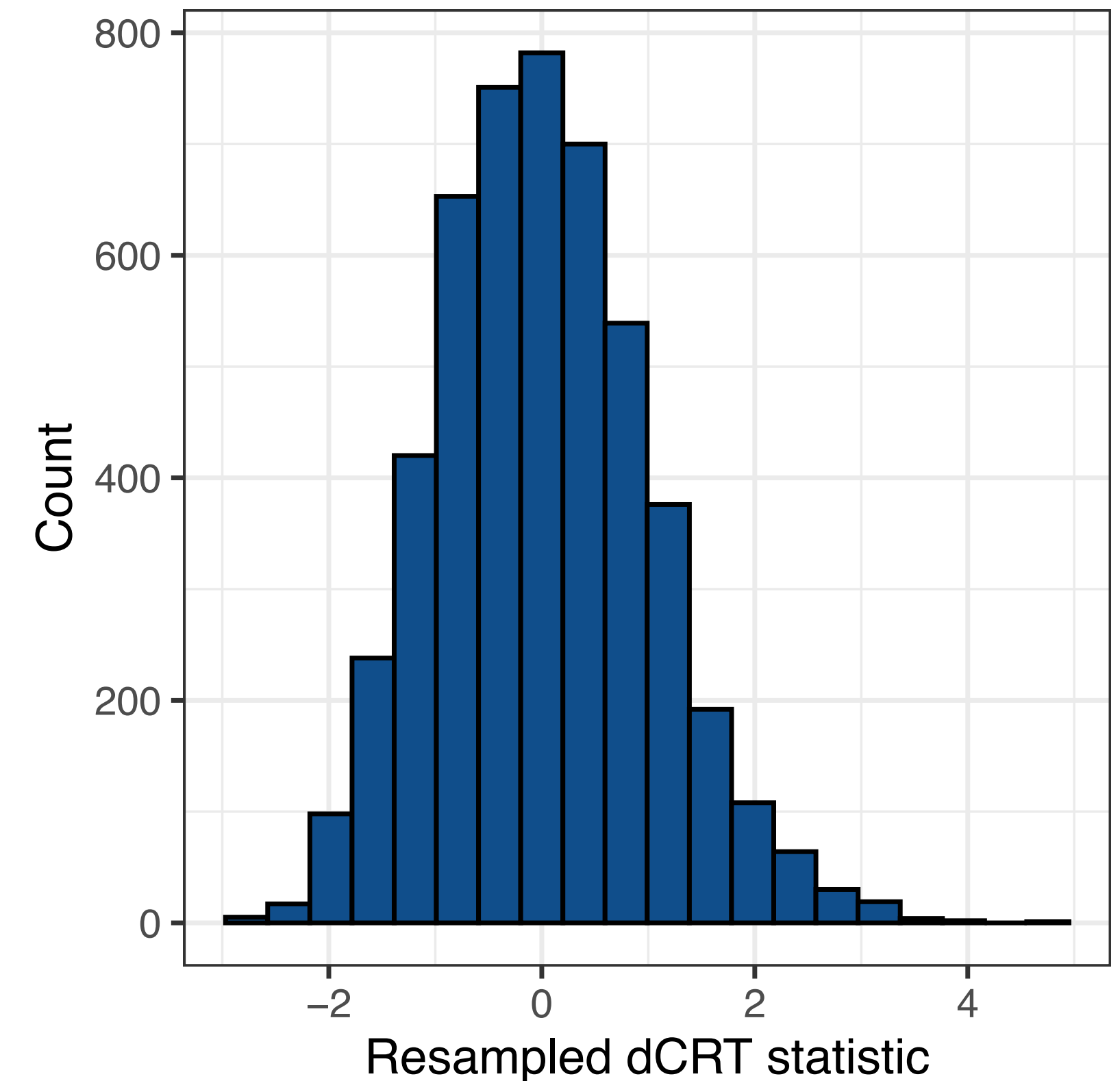
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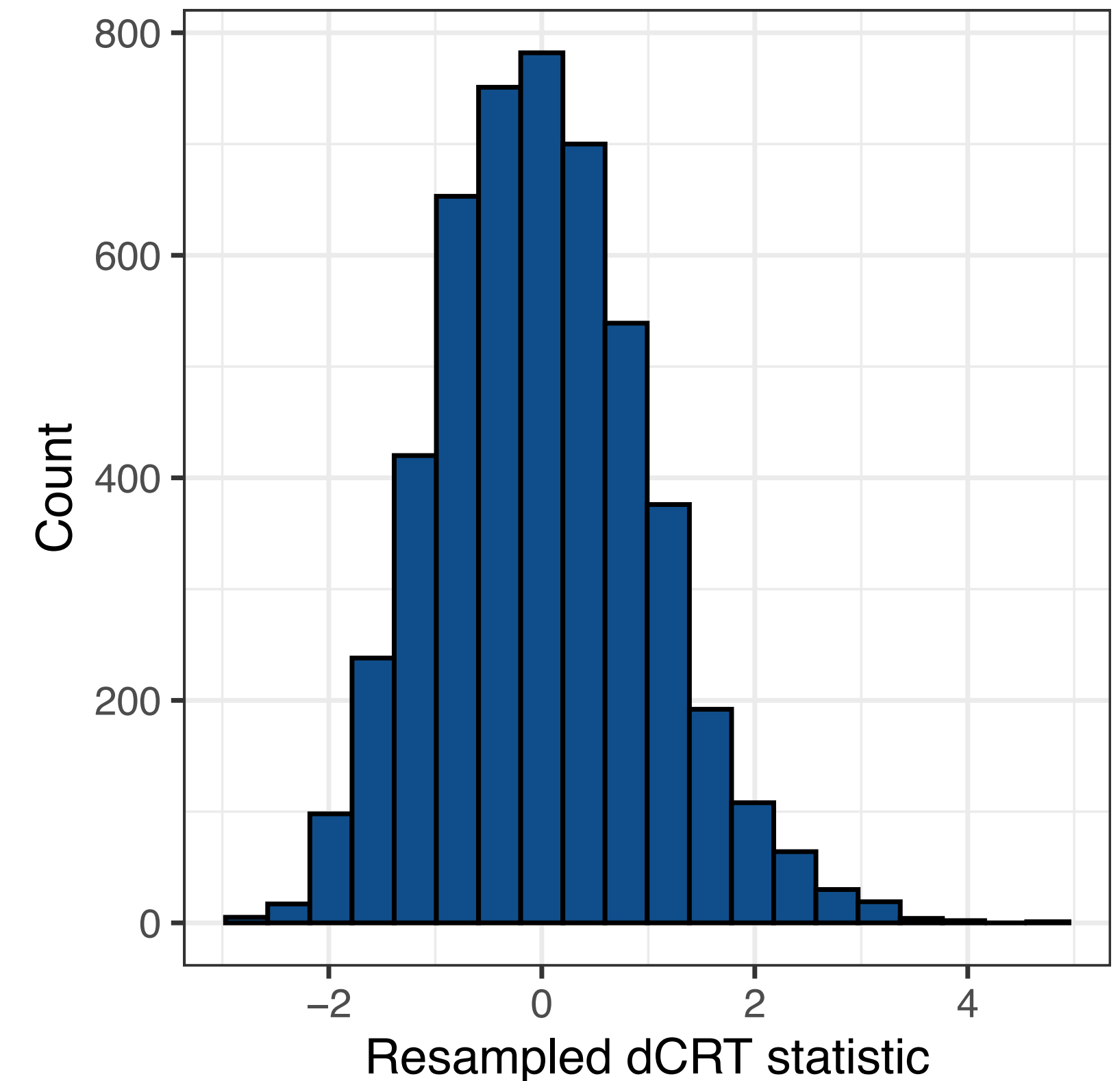


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- Draw moderate number of resamples and fit parametric curve to distribution.¹
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- Use the saddlepoint approximation for sums of independent random variables.

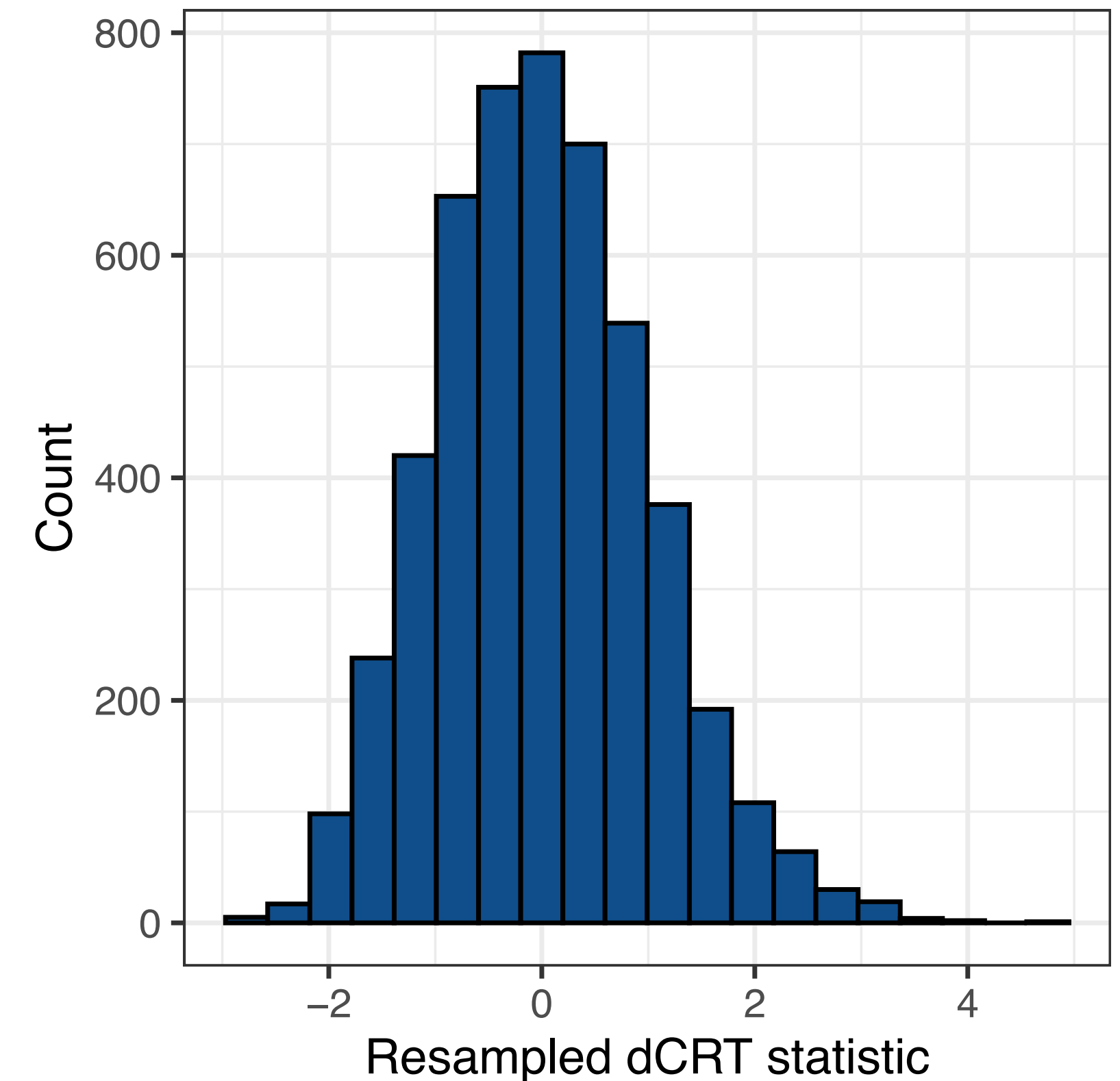


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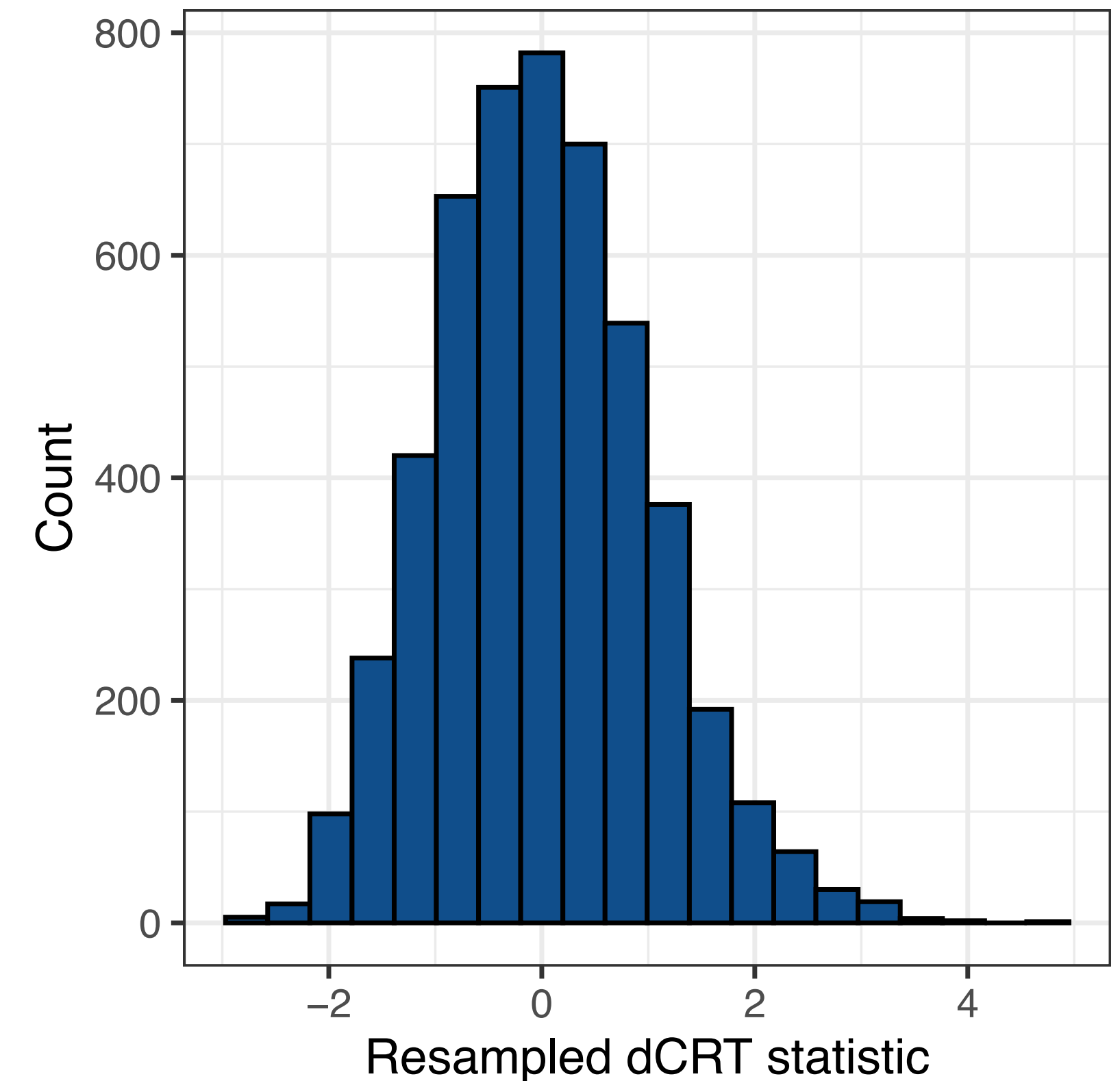
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1. Learn $\hat{\pi}(Z_i)$ by regressing X on Z and $\hat{\mu}(Z_i) \approx \mathbb{E}[Y_i | Z_i]$ by regressing Y on Z .
2. Let $\hat{s}_n$ be the solution of the saddlepoint equation in s :

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$$\lambda_n = \hat{s}_n \sqrt{\sum_{i=1}^n (Y_i - \hat{\mu}(Z_i))^2 \tilde{\pi}(1 - \tilde{\pi})}; \quad r_n = \text{sgn}(\hat{s}_n) \sqrt{2 \sum_{i=1}^n \left\{ X_i \log \frac{\tilde{\pi}}{\hat{\pi}} + (1 - X_i) \log \frac{1 - \tilde{\pi}}{1 - \hat{\pi}} \right\}}.$$

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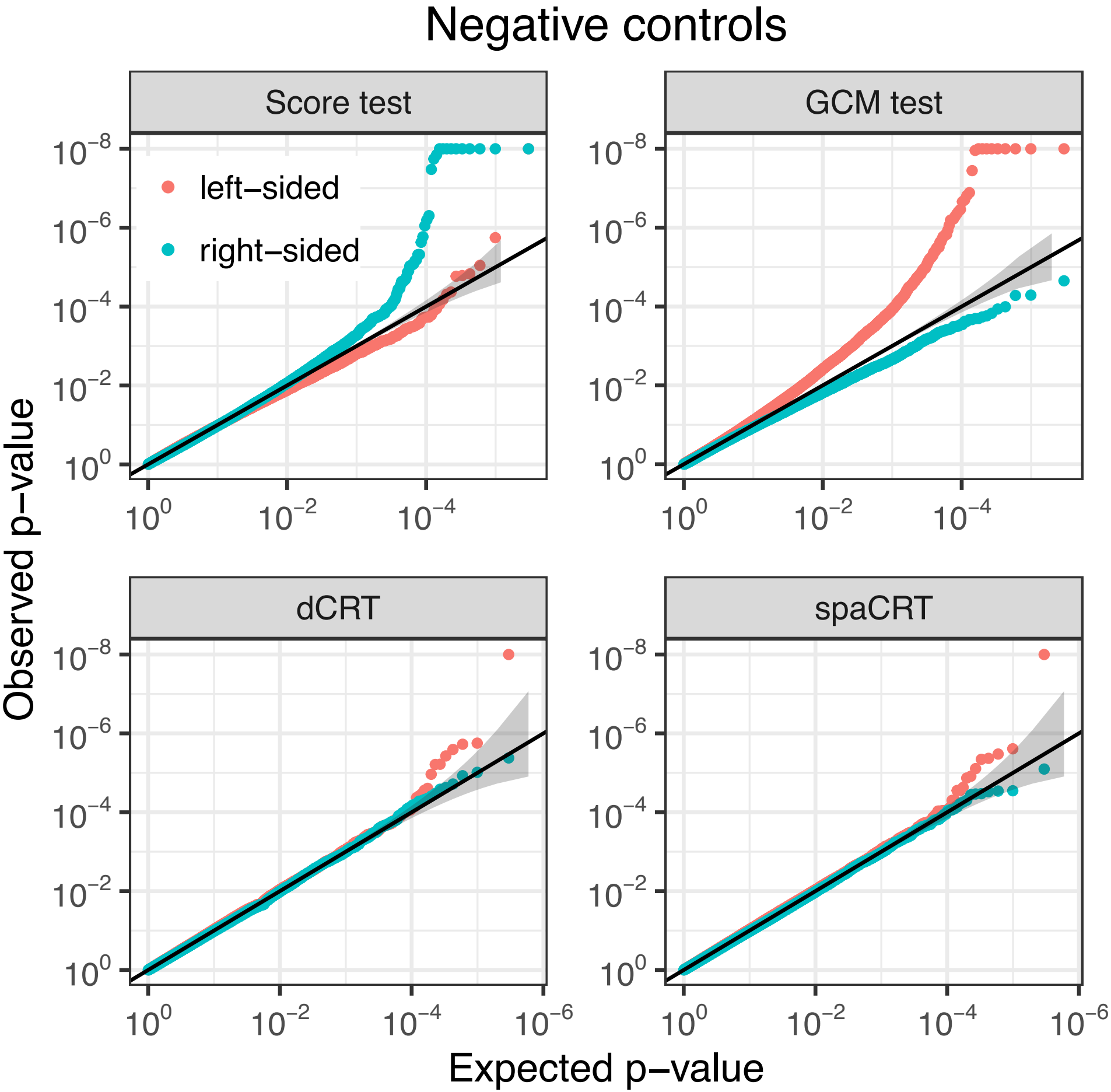
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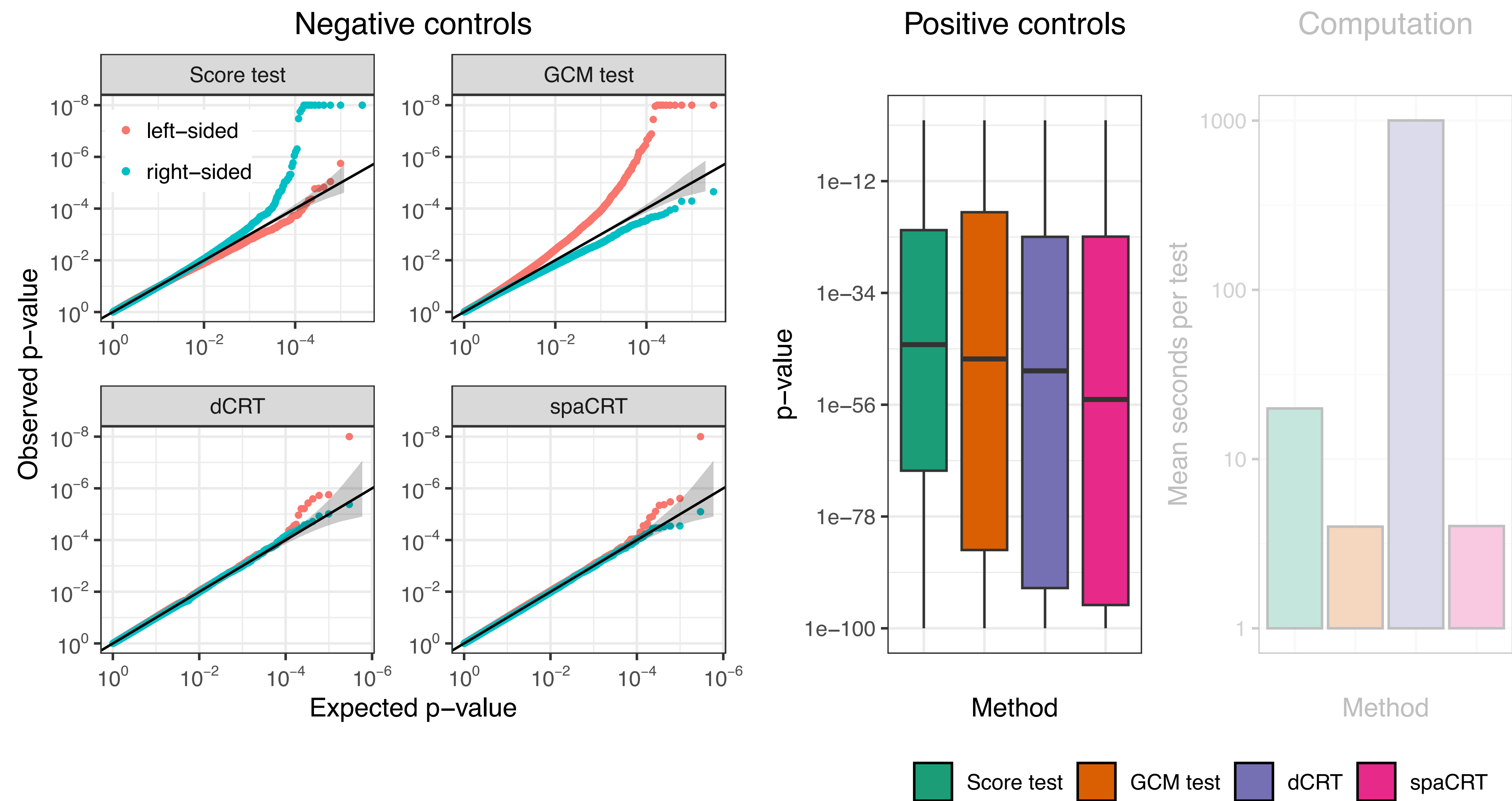
4. Return $p_n^{\text{spaCRT}} = 1 - \Phi(r_n) + \phi(r_n) \left(\frac{1}{\lambda_n} - \frac{1}{r_n} \right).$

Results on Gasperini data analysis

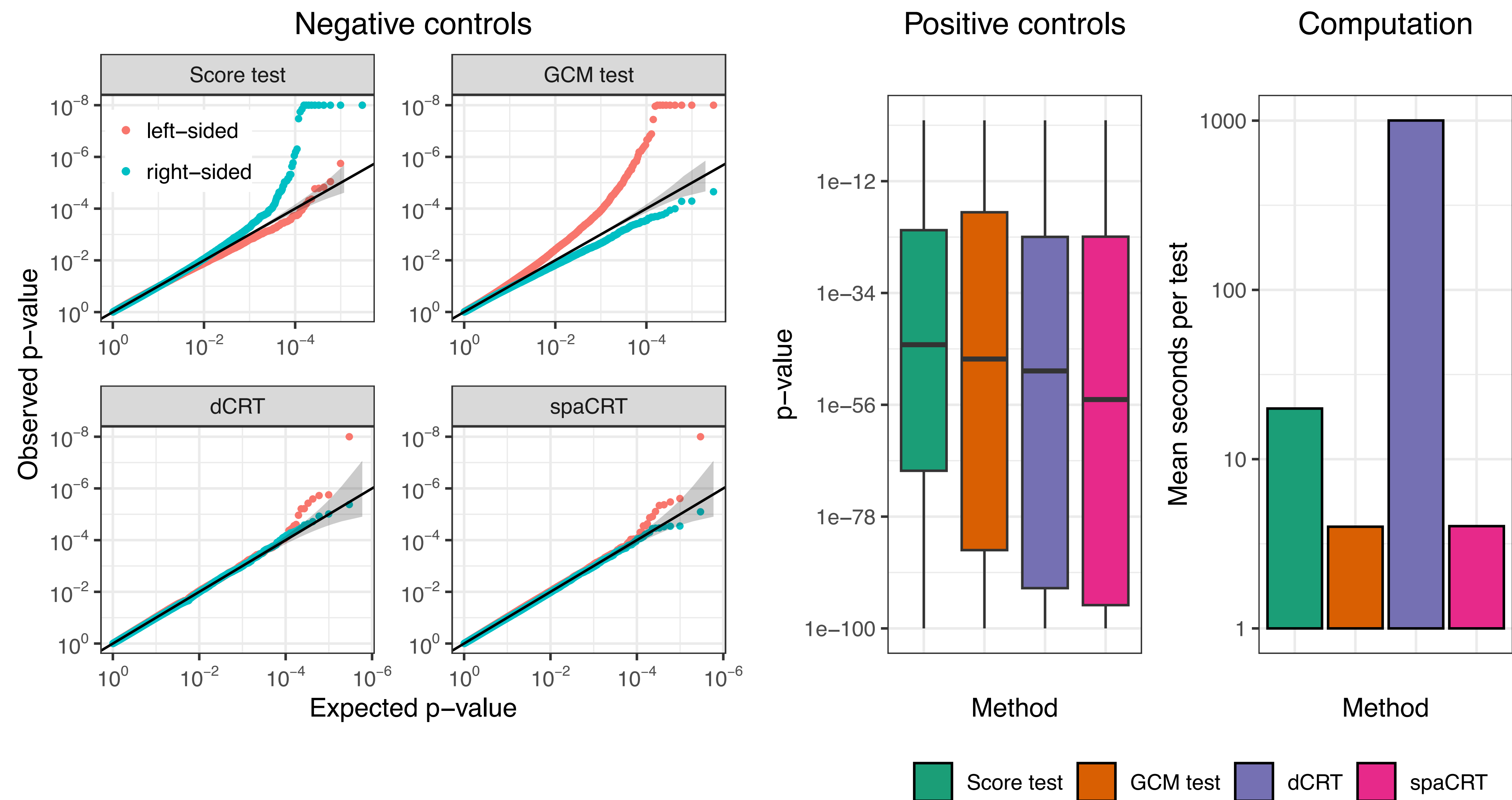
Results on Gasperini data analysis



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Conclusions



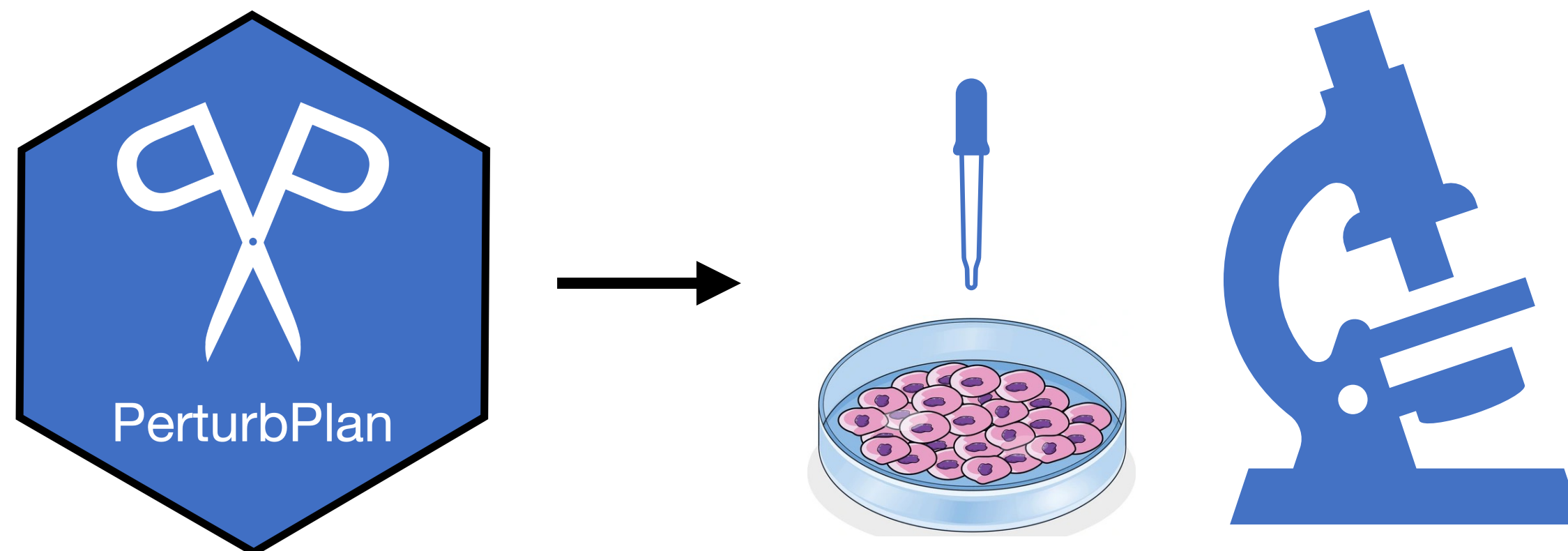
Conclusions

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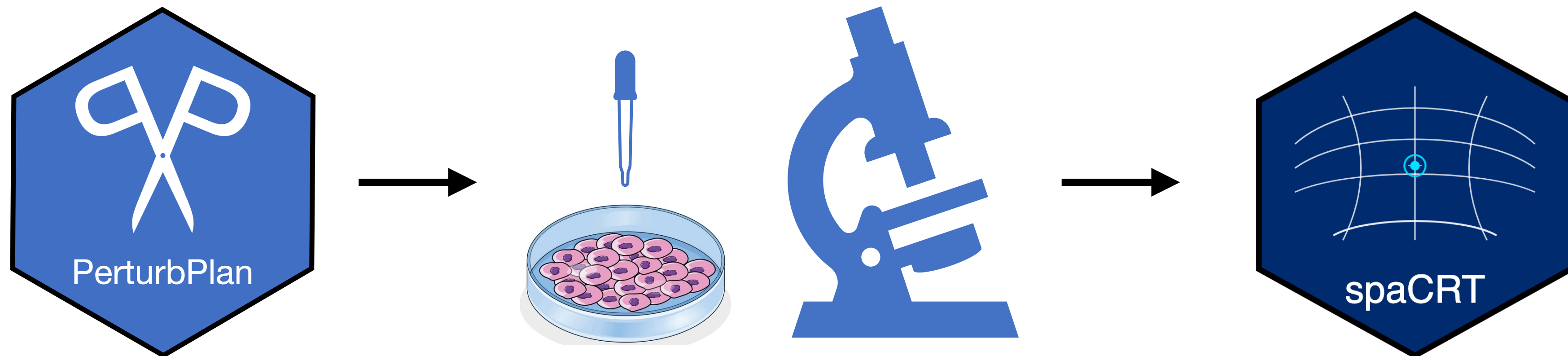
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Post-experimental analysis: spaCRT supports computationally efficient differential expression analysis, with strong statistical validity guarantees.

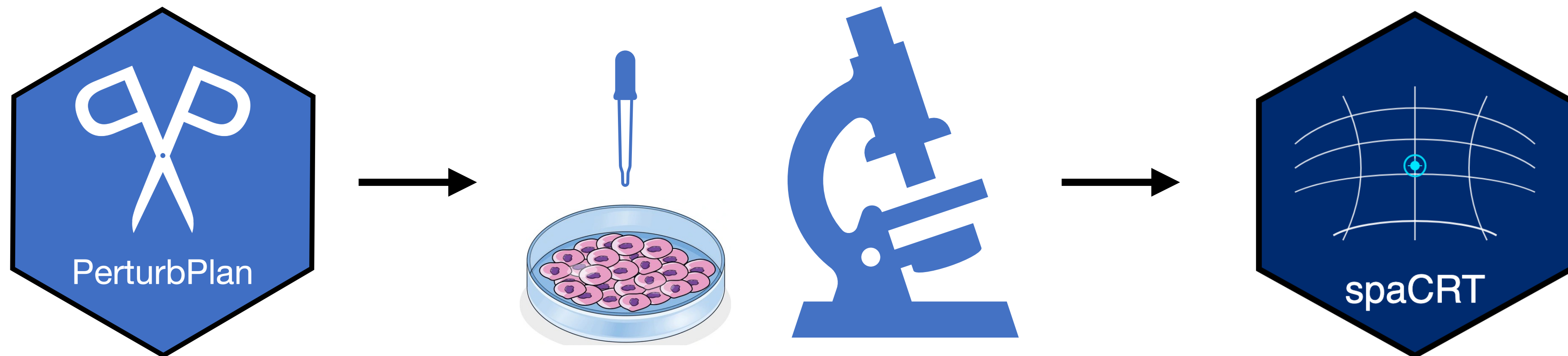


Conclusions

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Post-experimental analysis: spaCRT supports computationally efficient differential expression analysis, with strong statistical validity guarantees.

Both tools are open-source and available for use by the community.



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Thank you!
Questions?